Perhaps the obvious suspicion (hypothesis) is that she drowned but there are other possibilities too and the police will need to be certain – or as certain as possible – about the real cause of her death, so they will need evidence which rules out other possibilities and supports just one. For this reason they will need a pathologist's report; a pathologist is an expert who knows what evidence to look for in a case like this. Perhaps the woman simply fell into the lake and drowned, perhaps she had a heart attack first and fell into the lake, or perhaps she was murdered first and then dumped in the lake – and no doubt there are other possibilities. The expertise of the pathologist consists in knowing what sort of evidence will show the cause of death – what evidence suggests drowning, what evidence suggests a heart attack, and so on. Thus, after carefully inspecting the body and subjecting it to various tests, which are standard in his profession, he will write a report for the police saying what evidence he has found and what this leads him to conclude about the cause of death and how confident he is of his conclusion.

His report might be very straightforward. For example, if the body has not been in the water for long and all the evidence is clear and points in the same direction, he may be able to conclude with great confidence that the cause of death was X (drowning, a heart attack, a gunshot wound to the head, or whatever). On the other hand, it might be more complicated. If the body has been in the water for longer, some of the evidence might be lost or the evidence might point in different directions; in that case he may be unable to be sure of the cause of death and his report will again give the evidence, but will explain his degree of uncertainty.

In either case the pathologist will use a combination of observation and theory (including hypotheses). On the basis of what he sees initially (some initial observations) he will guess (hypothesise) some possible causes of death and will then look for further evidence which supports or excludes those possibilities. This is fundamentally the pattern in reasoning about all causal explanations where the cause is not immediately obvious; there will be many differences of detail in detective work, historical enquiries and scientific investigations, but this pattern is common: *Given what you want to explain, what are the possible explanations and what evidence will support/rebut each of these? Then, given the evidence you have or can find, which explanation is rendered most likely and fits best with everything else we know and believe? And how confident can we be that it is the right explanation?*

Question 10.1

Scientists are not sure how some species of birds are able to navigate with such accuracy when migrating thousands of miles. One theory is that the birds' brains contain molecules – called magnetite – which contain iron and which somehow enable the bird to detect the Earth's magnetic field in much the same way that the needle of a compass does. What other *possible* explanations can you think of?

10.2 An example of causal explanation

Following on from our example of how the pathologist works, let us imagine that a historian has produced the following argument:

> Napoleon must have died of arsenic poisoning whilst in exile on St Helena. Arsenic can be administered in small doses which will not be noticed but will eventually kill the victim. Arsenic poisoning leaves traces of arsenic in human hair, and reliable tests recently showed that Napoleon's hair contained abnormally large amounts of arsenic. It had been thought that he died of cancer, but his symptoms included nausea, chills, weakness and increasing corpulence, which cancer specialists say are not symptoms of cancer. However, these are typical symptoms of arsenic poisoning according to specialists.

This is typical of the kind of reasoning which occurs in causal explanations. The occurrence of the word 'must' shows the intended conclusion and 'died of arsenic poisoning' is clearly a causal claim. The reasoning then presents evidence for this claim. And, in this case, it also considers a possible alternative explanation (that cancer caused Napoleon's death) and argues that the evidence conflicts with this alternative hypothesis but supports the favoured one.

Using the notation we introduced earlier, plus a new symbol AH for 'alternative hypothesis', we could make the structure of the reasoning explicit as follows:

> C[Napoleon **must** have died of arsenic poisoning whilst in exile on St Helena] **for the following reasons**. R1<Arsenic can be administered in small doses which will not be noticed but will

eventually kill the victim> and R2<Arsenic poisoning leaves traces of arsenic in human hair> and R3<reliable tests recently showed that Napoleon's hair contained abnormally large amounts of arsenic>. It had been thought AH[that he died of cancer], but R4<his symptoms included nausea, chills, weakness and increasing corpulence>, and R5<cancer specialists say these are not symptoms of cancer>. However, R6<these are typical symptoms of arsenic poisoning according to specialists>.

Then R4 and R5 are the evidence against the alternative hypothesis and R1, R2, R3, R4 and R6 provide support for C.

We shall shortly explain how to judge whether this is good and compelling reasoning, but before we do let us point out some characteristic weaknesses we tend to display when thinking about causes.

10.3 Characteristic weaknesses in our thinking about causes

When the cause or causes of an event are not immediately obvious we may need to work things out by doing some detective work, a historical study, a scientific experiment or some other kind of investigation. Unfortunately, we often do nothing of the kind. We do not proceed scientifically, historically or whatever; in fact we do not do anything remotely systematic. Instead, we *jump to a conclusion* and accept the first explanation which comes into our heads; we do not even entertain any alternative explanations. Most of us do this sometimes; for example, we jump to a conclusion about why the car will not start, instead of considering possible alternatives – could it be out of petrol? Could it be an electrical fault? and so on. Or we hear a TV news report of a car bomb explosion in the heart of London and we immediately jump to a conclusion about the likely perpetrators – without even considering possible alternatives. Whether jumping to a conclusion matters depends on the case.

The second weakness we sometimes display when considering the causes of events is that we fail to consider all the relevant evidence; we take the bits of evidence which support our favoured explanation and ignore the others – or we do not even look for conflicting evidence. For example, cigarette smokers commonly pay more attention to the cases of smokers who have lived long and healthy lives (like Winston Churchill)

than to the much larger percentage of smokers when compared with non-smokers who have suffered ill-health or died early from smoking-related diseases.

It is worth spelling out these characteristic faults in a way which is easy to remember, because these are the faults we will be inclined to slip into and we want to guard against:

Common faults in thinking about causes

1 We consider only one possible cause and accept it without considering other possibilities.
2 We attend to only some of the relevant evidence in determining what causes or caused something.

The answer is clearly to consider several possible causes of the event or phenomenon in which you are interested and to ask yourself what evidence would favour these various alternatives and then look for the evidence.

Remembering the basketball analogy we used in chapter 1 to explain how to change your ways of thinking, we now need a model of good thinking about causes. The key is to ask the right questions and in the light of what we have said earlier it is fairly obvious what these should be.

10.4 The basic questions for skilful causal explanations

It follows from what we have been saying that the basic questions we need to ask ourselves when handling causal explanations are these:

Thinking map

Skilful causal explanation

1 What are the **possibilities** in this case?
2 What **evidence would** count for or against the likelihood of these possibilities (if you could find it)?

3 What **evidence** do you have already, or can you gather, that is relevant to determining what causes what?

4 Which possibility is rendered **most likely** by the evidence? (What explanation fits best with everything else we know and believe?)

The first question, 'What are the **possibilities** in this case?', may require us to be quite imaginative about other possible explanations. How imaginative depends on the case. For example, if the question is relatively unimportant, like what caused your current headache, we might be content to think of very few possibilities, but if the question is what caused a serious epidemic of some disease we might need to be much more imaginative about the possibilities. Notice that this is one of those points where critical thinking has to be 'creative' and imaginative to be good thinking.

Often, in deciding which possibilities need to be considered, we need to take into account whatever else is known about the subject. For example, the pathologist investigating the death of the woman found at the bottom of the lake needs to bring his experience and expertise to bear to decide whether it is necessary to consider the possibility that she killed herself somehow. Our answer to this initial question may well build in various assumptions so, depending on the importance of the case, we may need to note these too.

With the second question, 'What **evidence would** count for or against the likelihood of these possibilities (if you could find it)?', again, this may require you to be quite imaginative or may require you to draw on expertise in the area. Again, how much care you should give to thinking about the evidence which tells for or against the various possibilities will also depend on the seriousness or importance of the case and again you may need to be alive to the assumptions you build into your thinking. In short, if the case matters, we must not only think of alternative possible explanations, but we must also list some of the things that would support or tell against these alternatives. This second question may prompt us to look for information we would otherwise not have considered and this will help us to avoid making causal judgements on the basis of too little evidence (one of the characteristic weaknesses we want to guard against in thinking about causes).

Question 10.2

Consider the 'magnetite hypothesis' described in question 10.1. What evidence could you look for to support or rebut this hypothesis?

The third question, 'What **evidence** do you have already, or can you gather, that is relevant to determining what causes what?', is crucially different from the second. To answer the second question, you should so-to-speak 'look away' from the rest of the argument which is presented to ask yourself what on earth **would** prove the claim you are interested in (see Fisher, 2004, on the Assertibility question). However, once you know what you are looking for, you have to actually look for it – and of course the evidence may be very hard to find. Detectives know what would prove that Jones shot Smith (forensic evidence about the bullet linked to Jones's fingerprints on the murder weapon, and so on) but it may be very difficult to find the evidence. Similarly, it is commonly very difficult to find the evidence we need in science; it was one thing to say what **evidence would** support Einstein's theory of relativity but it was another thing for the Michelson–Morley experiment actually to provide the evidence.

The fourth question, 'Which possibility is rendered **most likely** by the evidence?', requires you to pull together what evidence you have, given the alternatives you are considering, to come to the best judgement you can. This will often mean that you have to come to a tentative conclusion, signalled by saying something like, 'This seems most likely but the evidence is not at all strong.' Sometimes, of course, you will be able to come to a confident conclusion; for example, the pathologist might report, 'I am absolutely confident that she died from a heart attack before falling into the lake.' In the pathologist's case, he will be able to bring all kinds of expertise to bear, and often the confidence you can place in a judgement about what causes what will depend on how much else you know about the subject matter and how well the evidence fits together with everything else you know or believe. Judging causal connections with any degree of confidence often requires bringing together a lot of related knowledge and belief, which is why the last question is supplemented with the question 'What explanation fits best with everything else we know and believe?' to remind us to consider the evidence in that context.

To summarise, for explanatory reasoning to be successful it must:

(i) *consider reasonable alternatives,*
(ii) *find evidence which (a) rules out other possible explanations and (b) supports the favoured explanation,* and
(iii) *fit well with everything else we know.*

These are the tests to apply when judging the reasoning used in justifying causal explanations. In this chapter we shall discuss only relatively simple and easily accessible examples but the same principles apply if you are considering much more complex or arcane examples.

Question 10.3

In the light of the previous guidance, look again at the example about Napoleon's death (section 10.2) and write a brief evaluation of the argument, saying whether or not it is convincing (or what its strengths and weaknesses are). You may search the internet if you think it will help you.

To reinforce the points we have just been making, let us look at another example:

> It is widely believed by experts in the field that the dinosaurs died out quite suddenly about 65 million years ago. The most likely explanation for their destruction seems to be that a large meteor hit the Earth at about that time. The resulting explosion would have sent a huge cloud of dust into the upper atmosphere which would have circulated the Earth for some years. This would have shut out the Sun, thus stopping photosynthesis in plants and destroying the dinosaurs' food supply. It is also believed that the dinosaurs were cold-blooded so they needed the Sun to keep warm. Either they froze to death or they starved. The best evidence for this is that an exceptionally large amount of iridium is found in layers of the sea thought to have been formed 65 million years ago and iridium is found in meteors in much higher concentration than on the Earth's surface.

If we consider this reasoning with a view to evaluating it skilfully, we have to ask ourselves whether there are other possible explanations for the extinction of the dinosaurs and what sorts of considerations

would favour or tell against the various options. We then have to draw on whatever evidence we have to make the best judgement we can.

Question 10.4

10.4.1 What other possible explanations can you think of for the extinction of the dinosaurs?

10.4.2 What **evidence would** support or refute the possibilities you envisaged in 10.4.1?

10.4.3 Assume that what the passage says is true; how strongly does it support the explanation it offers?

10.4.4 Assume that what the passage says is true and bring in anything else you know about the death of the dinosaurs to produce the best argument you can to explain their extinction.

10.4.5 Suppose we discovered in the fossil record that some species of plants which require very little light did *not* die out at the same time as the dinosaurs. What does that tell us?

10.4.6 Suppose we discover that in fact different species of dinosaurs died out at different times separated by intervals of hundreds of thousands of years. What does that tell us?

10.4.7 Suppose we discover that there have been other mass extinctions of species at other periods in the Earth's history. What does that suggest?

This question (10.4.1–10.4.7) should help to emphasise the importance of being *imaginative* in evaluating the reasoning provided in support of causal explanations. As we have said before, good critical thinking often requires creativity and imagination. You might find browsing the internet helps you with this – both in this example and with the others below.

Question 10.5

Using the thinking map for skilful causal explanation (see the start of this section) to guide you, write a brief evaluative response to the following passage:

> An underwater survey of the Witch Ground, in the North Sea off Aberdeen [Scotland], has discovered a trawler which was

probably sunk by a sudden burst of methane gas escaping from a vent in the sea floor – known as the Witch's Hole. When methane – or natural gas – bubbles through the sea in big enough volumes, it lowers the density of the water around it to a point at which objects, including ships, will no longer float. 'Any ship caught above [such a blow-out] would sink as if it were in a lift shaft,' said Alan Judd, a marine geologist from the University of Sunderland who led the Witch Ground survey expedition. In this case the trawler had sunk 'flat' with its hull sitting horizontally on the sea bed exactly over the Witch's Hole vent. This was consistent with the vessel being sunk by a methane blow-out; if she had been holed she would have sunk with the holed end lowest. Even sailors who jumped overboard wearing life-jackets would sink like stones. The Witch Ground, and the Witch's Hole in particular, have long been known among fishermen as treacherous waters. Methane blow-outs are thought to have destroyed about 40 oil platforms around the world; the Witch Ground is just 22 miles from the Forties oil field. (Adapted from an article in *The Times*, 30 November 2000, by their science correspondent Mark Henderson)

Question 10.6

Using the thinking map for skilful causal explanation (see the start of this section) to guide you, briefly evaluate the following reasoning:

Studies on individuals have attempted to determine whether raising the consumption of cholesterol affects the level of blood cholesterol. For example, volunteers drank four pints of milk a day for a period: their blood cholesterol levels were unaffected. Even more damaging to the hypothesis that ingested cholesterol can damage the heart are the results of two recent and independent studies funded by the British Medical Research Council. One found

> that men who drink no milk have ten times as many heart attacks as men who drink more than a pint a day. The other discovered that men who eat margarine have twice as many heart attacks as those who eat butter. There are in fact good reasons why one would not expect blood cholesterol to vary with diet. First, the liver manufactures three or four times as much cholesterol as is normally ingested. Secondly, the body itself regulates the amount of cholesterol in the blood: its level is normally kept constant regardless of what is eaten, though some unfortunate people have too high a setting and are likely to die young through heart attacks. The true causes of high blood cholesterol are not known. What is known is that although reducing blood cholesterol levels by drugs reduces heart disease, it does not increase longevity: people die of cancer instead. There is no solid evidence that cholesterol intake affects blood cholesterol levels, yet jumping to conclusions on the basis of insufficient evidence has caused a considerable scare. (Sutherland, 1992, p. 184)

For more recent evidence on this topic see www.ravnskov.nu/myth3.htm where Professor Uffe Ravnskov discusses several 'cholesterol myths' and argues that diet has little to do with blood cholesterol levels.

10.5 The language of causal explanations

At this point you should probably look back at section 3.7 to remind yourself again of the points made there in some detail. Remember, it is easy to confuse arguments and causal explanations because of the ambiguity of certain words and phrases like 'because' and 'that's why'. These words sometimes signal that a reason is being given for a conclusion but sometimes signal that the author is giving a causal explanation and we provided a test to help you decide which is the case. With a little practice, however, explanations leap to the eye and you will quickly learn to ask the right questions.

In general, the language we use in reasoning about causal explanations is very similar to the language we use in other kinds of reasoning (cf. sections 2.3 and 2.5); thus, we use the same language

mentioned there to signal the confidence we have in claims, to criticise reasoning and so on. But in explanatory contexts, there is special emphasis on 'hypothesis', 'explains', 'causes' and 'evidence', so the language we use tends to include these and related words, like 'the theory says', 'the facts/data are/suggest/imply', 'is/is not consistent with', and so on. It is also common to use verbs or phrases which imply a causal relationship without using the word 'cause', like 'drown', 'died from arsenic poisoning', 'died out', 'hit', 'sunk', 'affects the level of blood cholesterol' to give some examples we have encountered in this chapter. It is also quite common to reason 'by analogy' and to refer to what experts in the field believe when reasoning about causal explanations. Often, of course, such language is absent, but it is often implied and it could be helpful to write it in as we did with the 'therefore' test.
Consider the language of the following passage:

> The distinctive yellow-green colours of Van Gogh's paintings of landscapes and domestic interiors may be due to the side-effects of his taking the drug digitalis. Although there are few medical records to prove the point, it is generally known that digitalis was commonly prescribed during the time Van Gogh lived. Also, Van Gogh's portraits of the doctor who treated him show the doctor holding a stem of foxglove, the very plant from which digitalis is made.

Once you think about it you can see that the author is offering a possible causal explanation for Van Gogh's distinctive use of colours ('due to the side-effects'). It is tentatively put forward ('may be . . .'). We could call this claim her hypothesis or her conclusion – either way it is very tentatively advanced and the reasons (or pieces of evidence) offered in its support are presented as only weak evidence for her theory ('Although there are few medical records to prove the point . . .'), so overall it is a very tentative argument in support of a thesis.

However, it is an interesting idea; perhaps a painter's way of painting could be influenced by taking certain drugs and perhaps this was the case with Van Gogh. Of course, as it stands, the case is extremely weak, but if we wanted to consider the hypothesis seriously it is clear from our thinking map what questions we should ask. We would need to consider whether there are other possible explanations for Van Gogh's use of those colours (he mixed them himself from unusual ingredients; he was partly colour blind), and we would need to consider what evidence would support or conflict with these ideas and then we would need

to look for it. For example, did Van Gogh suffer from any illnesses for which digitalis was commonly prescribed? Is there any other evidence that he took digitalis? Does taking digitalis have a similar effect on painters' choice of colours now? Do we understand the mechanism at work?

Though the language of the passage is not completely transparent, a little thought shows what is being suggested. ('Perhaps Van Gogh used those distinctive yellow-green colours in his paintings of landscapes and domestic interiors because he took the drug digitalis.') By the criterion we explained in section 3.7 it is clear that this is an explanation (since it is known that Van Gogh used distinctive colours – that is clear for all to see – but it is not known whether he took digitalis). Furthermore, though it takes a bit of imagination, it is also reasonably clear what questions we should ask if we want to determine whether it is a good explanation (and again, if you want to know more, the internet might help).

Question 10.7

Bearing these remarks in mind, decide what is being explained and what is being argued for in the following passage. Is the argument convincing to you?

> Since 1980 [the number of human twin births has] been increasing – they are up by 42 per cent since that date throughout the developed world. For every 1,000 births in the developed world 29 are now twin births. This figure is likely to continue to increase because many of the extra twin births are caused by fertility treatments which result in the release of more than one egg during ovulation. This is also why the whole increase in twin births is taken up by fraternal twins who come from separate eggs. The rate of identical twins, from division of the same egg, has remained constant at 4 for every 1,000 births.
>
> So, while the number of twin births is set to increase even further in the short term as more women seek fertility treatments, there will be a limit to the number of extra twin births because, first, not all will seek this treatment and, second, not all women who do will have twins.

> At what point the figure will stabilise we do not yet know, but we can rest assured that the majority of births will remain as single births. ('Twin sets', *New Scientist*, 18 November 2000, p. 121)

This neat example illustrates the difference between explaining and giving reasons for a conclusion; the first paragraph explains a phenomenon, the second gives reasons for a conclusion. It arose from a question about the increase in twin births and whether this would continue until all human births were twin births.

10.6 Making things fit together

The thinking map for skilful causal explanation (section 10.4) included the question 'What explanation fits best with everything else we know and believe?' In general scientists are trying to construct a 'web' of beliefs which best explain how things happen in the world, which make the best possible sense of our experience, so everything is intended to fit with everything else. The implication is that the more science you know, the better you are able to judge whether a given explanation is plausible or not. But you can practise this skill with whatever level of knowledge you happen to have. Let us look at an example to illustrate this, taken from an article called 'When North Flies South' by Lou Bergeron, published in *New Scientist*, 30 March 1996 (pp. 24–8):

> Compasses will not always point north. Every 500,000 years or so the earth's magnetic field flips, swapping the positions of the magnetic poles. The last time this happened was 780,000 years ago so we are long overdue for another flip. When it comes it could be faster than anyone expected.
>
> Geophysical processes tend to be slow, and until recent times most geophysicists believed that a complete flip would take around 5,000 years. But last year a team of scientists announced they had found evidence in the magnetism of some rocks that parts of the flip might happen so quickly you could almost see the compass needle move. So fast indeed that long-haul flights could be thrown completely off course, and migrating birds, used to trusting their inbuilt compasses, could be left perplexed . . .

Question 10.8

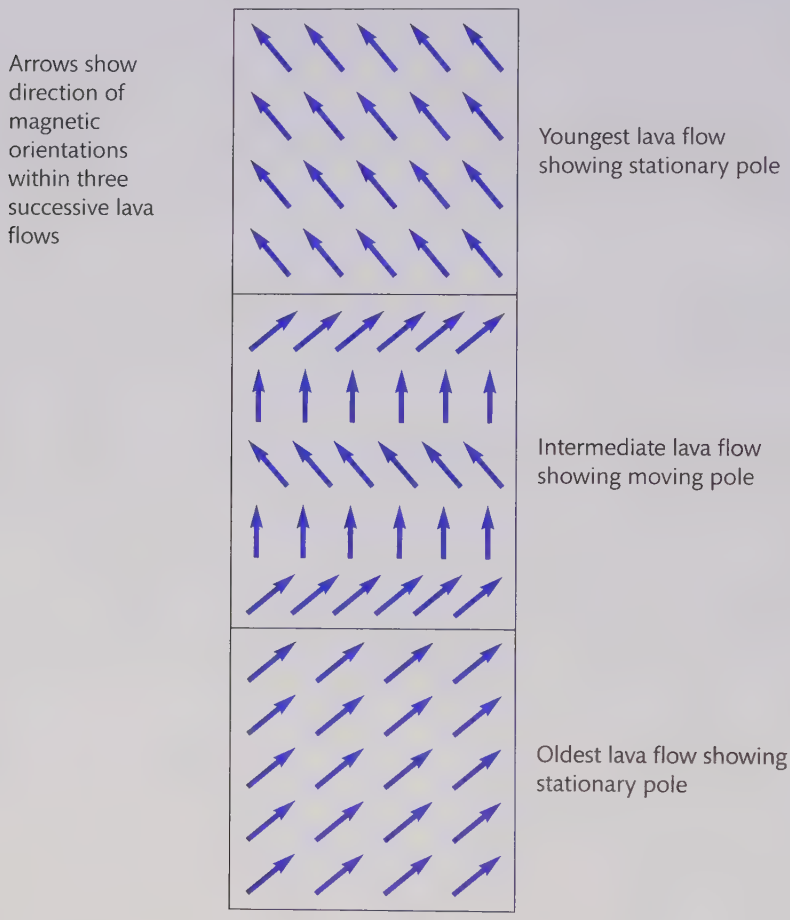
Before reading on, note what evidence you can think of which might support/refute what is claimed here (which incidentally is not a straightforward causal claim but a more general claim about a scientific phenomenon).

The most dramatic evidence for this claim comes from a pile of successive lava flows almost one kilometre thick (and 16.2 million years old) on Steens Mountain in eastern Oregon, USA. As Bergeron explains:

> Lava flows erupting on to the surface of the Earth contain particles of [iron] and when the lava cools below about 580 degrees C these particles become magnetised by the [Earth's magnetic] field. This freezes the direction of the prevailing geomagnetic field into the rock, and from this the scientists can work out the position of the geomagnetic pole at the time the rock cooled. Most lava flows take only a matter of weeks or months to cool – much shorter than the time taken for a complete flip – and from the flows that formed while a reversal was under way scientists have concluded that the reversing field moved fairly slowly. But Steens Mountain contains two flows that appear to have caught the moving field racing from one position to another. . . .
>
> When a lava flow is deposited, it cools first at the top and bottom, and only later towards the centre. If the [Earth's magnetic] field is stable while the lava is solidifying, the orientation within the magnetic particles is the same throughout. But the key flows at Steens Mountain look very different. The top and bottom of the flows have exactly the same orientation as the underlying lava. As you move towards the centre of the flows, the polarity slowly changes orientation until in the very centre it exactly matches that of the overlying flows. This is just what you would expect if the lava flows had caught the field in the act of moving.

The following diagram illustrates this. It is most easily grasped if you start from the bottom of the diagram, which shows the oldest lava flow – with a stable magnetic field throughout its cooling period, then

look at the top, which shows the youngest lava flow – again with a stable magnetic field throughout its cooling period (but with a quite different direction), then look at the intermediate lava flow, where the top and bottom of the flow have the same orientation as the oldest flow but the centre shows the orientation of the youngest flow:



The evidence from the Steens Mountain lava flows was published under the title 'New evidence for extraordinarily rapid change of the geomagnetic field during a reversal' by R.S. Coe, M. Prevot & P. Camps, in *Nature* Vol. 374, 20 April 1995. What they found in a lava flow like the central one in our diagram was that during the time the lava flow was solidifying, the geomagnetic pole moved as much as six degrees a day over a period of thirteen successive days.

Extrapolating from these figures the North and South magnetic poles could flip completely in two to three months – a dramatic contrast with the 5,000 years scientists formerly believed it took.

Bergeron reports that other geophysicists regard the scientists who published these results as very reputable, and their work was published in a reputable scientific journal, *Nature*. The problem for geophysicists is that it is very hard to reconcile this evidence, and the conclusion drawn, with other things they believe about the Earth's core and what causes the Earth's magnetism. But other possible explanations of the Steens Mountain evidence, for example the occurrence of one of the Sun's periodic magnetic storms, just don't seem to fit either. To make things even more puzzling, further similar evidence has now been found in 12-million-year-old rocks in New Mexico.

This is an interesting example, partly because it shows how instructive it can be to ask 'What would show X?', but also because subsequent reasoning by the scientists concerned has been largely about what 'fits' with everything else we know and believe.

Question 10.9

Consider the claim that 'working systematically through this text will improve your critical thinking ability'. What evidence do you think would show this to be true or false?

Question 10.10

Using the lessons of this chapter, write a brief response to the following passage:

> There is a small area in the middle of the right-hand side of the brain which produces a curious effect if an epileptic focus develops there. In such a focus the nerve cells from time to time all fire together: when they do so, they cause an epileptic seizure. A focus in this particular area can render the person highly religious, and cause him to avoid sex in any form and to give up all addictions such as smoking and alcohol. Remarkably, when the focus is removed and the person goes back to his previous existence, he may become an atheist, and return to cigarettes, alcohol and the pursuit of sex. It may be that the form the Christian religion has taken was in part caused by St Paul suffering an epileptic attack on the road to Damascus. (Sutherland, 1992, pp. 10, 11)

10.7 Summary

Much of our reasoning is concerned with causal explanations. Sometimes we just see what causes something, but mostly we have to work it out. In the latter case it is common for us to slip into one of two errors; either we consider only one possible cause or we attend to only some of the relevant evidence.

To remedy these faults, we need to ask the right questions when considering reasoning to a causal explanation. These are:

1. What are the **possibilities** in this case?
2. What **evidence would** count for or against the likelihood of these possibilities (if you could find it)?
3. What **evidence** do you have already, or can you gather, that is relevant to determining what causes what?
4. Which possibility is rendered **most likely** by the evidence? (What explanation fits best with everything else we know and believe?)

Examples of reasoning to causal explanations show how these questions can help us think better about such cases.

Of course scientists, detectives and many others have to ask themselves questions very like the ones we have presented here if their reasoning about causal explanations is to pass the scrutiny of others.

Further reading

Ennis (1996, chapters 8 and 9).
Swartz, Costa et al. (2010).
Gardner (2000).
Goldacre (2008).

1 Decision-making: options, consequences, values and risks

How often have you said, or heard someone else say, 'How could I have been so stupid?' after making a decision which turned out to be a disaster? As with causal explanations, the thinking involved in *deciding what to do* – or *recommending a course of action*, or *weighing someone else's recommendation* – needs special attention because it is very common, is often important, and has to be evaluated in a distinctive way. In earlier chapters we have already encountered numerous examples of reasoning which recommended a course of action, for example the argument that we should not permit prospective parents to choose the sex of their child, the argument that young people should not get married, and numerous others.

We are all used to making decisions (or recommending a course of action), but sometimes we do that less skilfully than we might and the purpose of this chapter is to explain how to do it better and to give you practice in doing so.

Of course, we make decisions and recommendations about many matters which are not very important, like what clothes to put on in the morning before lazing about at home, or what to have for breakfast on such a day, but in this chapter we are mainly concerned with decisions and recommendations about more substantial matters: personal issues such as which university to go to, which subjects to study, whether to take a job, whether to become vegetarian, and so on, and more 'policy' ones like whether some currently illegal drugs should be legalised, whether boxing should be banned, whether fox-hunting with hounds should be illegal, whether parents should be punished if their children break the law, and so on. In short, we are more concerned with decisions of a significant kind – which merit careful consideration but which we often decide rather hastily. Sometimes, of course, decisions have to be taken very quickly – perhaps to deal with an emergency – and then one just has to act. But often we do have time to think, and if the decision

matters, the question is how to make a good decision rather than a poor one. Most of us have never been taught how to take decisions well – in particular what questions to ask before deciding – but this is a skill which can be taught, as we shall show. Whether we are trying to make a decision ourselves or are wishing to evaluate someone else's decision or recommendation, we need to be clear about the kind of reasoning which is appropriate in such contexts, and that is what this chapter is about.

Question 11.1

Think of a poor decision you made (about a significant matter) and write down anything you did – *or failed to do* – in the process of coming to that decision which caused you to make a poor decision. (For example, perhaps you did not really think about the decision at all!)

11.1 Common flaws in our thinking about decisions

Most of us have made a poor decision at some time and can put our finger on some flaw in the process which led to that decision; either we simply did not think, or we did not find out some relevant information, or we listened to our heart and not to our head, and so on. I have often asked this question in workshops on decision-making, and participants usually come up with a list very like the following:

I did not give the matter enough thought.
I did the first thing I thought of.
I didn't think of possible alternatives.
I didn't consider the consequences of various courses of action.
I needed to get some more information.
I didn't really consider what mattered to me.
I was too hasty.
I was too emotionally involved.
I did what the 'boss' said.
I accepted what others recommended without thinking about it.

Maybe your answer to question 11.1 above is included in this list or maybe you mentioned some other problem. Either way, most people have no difficulty in recognising these problems where decision-making is concerned. The question is, 'What can be done to combat

these weaknesses?', and the answer is surprisingly simple: if you recognise problems in the process of making decisions then you are halfway to being able to remedy them. If you think back to our basketball analogy in chapter 1, you will realise that you then need a model of good decision-making to emulate. So let us present such a model now.

11.2 A model for good decision-making

The model we present here arises out of the list of weaknesses we identified earlier. There is nothing very controversial about this model; the ideas behind it have been around – in what is sometimes called 'decision theory' – for at least 50 years, but most people find that it is a great help to their own decision-making and to responding to other people's reasoned recommendations.

11.2.1 Be clear why the decision is necessary

Suppose you are considering whether to become a vegetarian. You might consider this issue for at least two quite different sets of reasons. On the one hand, you might feel this is essentially a moral issue – about the rights and wrongs of treating animals in a particular way and eating them. Or you might see it mainly as a health issue – about the benefits and health risks associated with eating meat. Of course, you might think both sets of considerations are relevant, but it is very common for people to adopt one perspective or the other. The point is that you need to be as clear as possible about the background to the decision, the assumptions which lie in the background, the context and so on if you are to weigh the case wisely. Although models of good decision-making often fail to mention this requirement explicitly, it is implicit, and getting clear about why a decision is necessary helps to focus one's thinking.

Sometimes being clear about why a decision is necessary means identifying a problem and sometimes it will require you to think carefully about *objectives* – what you or others want to achieve. So, for example, if you are considering which university to attend, you might realise you know very little about the institutions you could attend, so that is a problem. Alternatively someone might advise you to apply for university A 'because that is the most prestigious place for the subjects in which you are interested', but you may feel that this is not necessarily

right for you, so you may need to think carefully about what you want to get from your university education, what you are looking for – what your *objectives* are.

11.2.2 Make sure you consider alternative possible courses of action

One of the commonest mistakes we fall into when we are faced with making decisions or considering recommendations is that we fail to think of some reasonable alternatives. In fact, most people admit that they commonly make decisions by first deciding what they want to do and *then* rationalising that choice (if you reflect on your own decision-making, you may be surprised how often you do this). The solution is clear enough. When faced with a decision, you need to pause and ask yourself what the alternatives are. This sounds easy enough, but it is quite hard to change old habits, so some practice will be required to achieve this habit. Not only do most of us have the habit of leaping first and producing the justification afterwards, but most of us find it hard to be imaginative about what alternatives there are. How important this is depends on the case.

If the decision concerns something relatively straightforward (like which route to drive between A and B) it might be necessary to think of only one or two alternatives. But if the case is complicated and important it might be necessary to be very imaginative and inventive about possible alternatives (for example, whether Truman should have authorised the use of nuclear weapons on Hiroshima and Nagasaki). This should remind us of what we have said several times earlier: good critical thinking often requires creative thinking, imaginative thinking and 'thinking outside the box'. Sometimes you need just to think of a small number of reasonable alternatives, but sometimes you need to 'brainstorm' to think of unusual possibilities if the case is very important and knotty – like the case of Truman and the atom bomb. Anyway, when reasoning to a decision which matters, it is always important to think what other options there might be.

For an example which illustrates this, look again at an argument we have already seen (question 9.5):

> Young people in Britain should not get married. Current statistics show that 40 per cent of marriages end in divorce and one can safely assume that many of those couples who remain together are unhappily married. Therefore, it is more likely than not that young

people who marry will divorce or be unhappily married. These are daunting odds for any young couple.

This can seem a powerful argument until you ask, 'What are the alternatives (to getting married)?' Presumably these are things like living together on a long-term basis (without marrying), avoiding long-term relationships and having a series of short-term relationships, or avoiding close relationships altogether! But what are likely to be the consequences of these alternatives? Perhaps we could find statistics showing how individuals in these other relationship patterns fare; perhaps long-term relationships where the partners do not marry break up as often as marriages. Perhaps many of those in a series of short-term relationships suffer in ways they would not wish on others. And perhaps many of those who avoid close relationships suffer considerable loneliness. Until we know how well these alternatives work out statistically, this argument does not carry much weight. Perhaps it is just a fact that life is risky whatever you do!

It is not easy to say how many and what alternatives one should consider when faced with making or evaluating a decision or recommendation; it depends on the case, on how important it is, how much time is available, whether relevant information is available (or easy or expensive to obtain), whether you can 'go back' and change your mind if things go wrong, and so on. The important point is that enough alternatives need to be considered in the circumstances if a decision or recommendation is to be adequately justified; this might mean only a small number, or only 'reasonable' alternatives, or perhaps 'imaginative' options too – it depends on the case.

Question 11.2

11.2.1 Imagine you are the teacher who receives the complaint from Hans about the unfair critical thinking test (example 1, section 2.2). Why do you need to make a decision about what to do? And what alternative courses of action are open to you?

11.2.2 Suppose you are prime minister or president in your country and a terrorist threat is brought to your attention. Why is a decision necessary and what alternatives are open to you?

11.2.3 Imagine yourself in President Truman's position as he faced the decision about whether to drop the atom bomb; bearing in mind whatever you know about that event, list as many options as you can which you think need to be considered. (Brainstorming could help you.)

11.2.3 Consider the possible consequences of the various alternatives

It is very common for people to say of a decision they took, 'I just didn't think about what was likely to happen.' But it is often very important to avoid that mistake and to think about the possible consequences – both good and bad – of a course of action. For example, suppose you are considering which university to go to; if you do not check out the syllabus of the various courses which interest you and if you do not visit the institutions concerned you may well go somewhere where the course is not really what you want or where you do not really like the place or the people. If you had checked out these possible consequences in advance, it could have saved you a lot of disappointment, frustration and perhaps even failure.

Working out the possible consequences of alternatives may require some imagination in order to think of consequences which need to be considered. Some possible consequences will be very obvious and will immediately spring to mind – for example, before choosing a university, it is obvious that you must investigate the course syllabus. But it may require a bit of imagination to think of some possible consequences (for example, do you need to consider the sporting facilities available at the university where you are thinking of studying or don't these really matter to you?).

Question 11.3

11.3.1 Refer back to your answer to question 11.2.1 and write out the possible consequences of two of the alternative courses of action open to you.

11.3.2 Refer back to your answer to question 11.2.2 and set out the possible consequences of two of the alternative courses of action open to you. (Use whatever background knowledge you have.)

11.3.3 Refer back to your answer to question 11.2.3 and, using whatever you know about the bombing of Hiroshima and Nagasaki, think of 'possible consequences' of two important options which President Truman should have considered in deciding whether to drop the atom bomb. (Since this was an issue of enormous importance you should be as imaginative as you can about possible consequences and perhaps use the internet or do some reading on the subject if you know little about it.)

It is also clear that working out the consequences of possible alternatives will commonly require some investigation – as in the example we have been discussing about which university to go to. This certainly takes time and might cost money (for example, if you have to travel to universities to find out about them and their courses). How much investigation is appropriate depends on the case – on how important the decision is, how sure you want to be (some people like risks and uncertainty) and how difficult it is to find out what you need to know. In general a decision is reasonable if 'enough' of this investigation has been done in the context, but there is no general rule saying how much is 'enough'.

11.2.4 Consider how *likely/unlikely* and how *valuable/undesirable* the possible consequences are

It is very common for people to say things like, 'You should (or should not) do X because otherwise Y *might* happen.' For example, 'You should be vaccinated against tuberculosis because otherwise you *might* catch it.' Here they are talking about a *possible* consequence of an action but they are making **no** estimate of how *likely* it is, or how *serious* it would be – but both of these matter a great deal.

Consider, for example, the UK National Lottery; for some time this was advertised with a picture of someone pointing at the viewer; the advertisement resembled the First World War Lord Kitchener advertisement where the caption was 'Your country needs YOU'; in the National Lottery advertisement the caption was 'It could be YOU', meaning that *you* could win the National Lottery. Well, it is perfectly true that if you buy a National Lottery ticket you *could* win and this would

give you millions of pounds which would be *highly desirable* to most people. But what are the chances? How *likely* is it that you will win? If you buy one ticket the chance of winning the National Lottery is about 10,000,000 to 1 against! So, yes, you *could* win but it is very, very *unlikely* (you are far more likely to be killed in a car accident!). This example illustrates the difference between how *likely* and how *valuable* a possible consequence of an action is and how important it can be to distinguish between the two (since they are independent of each other – something may be very desirable and either likely or unlikely, for example). This is not a new discovery, as this song from Henry Fielding's farce *The Lottery*, first performed in London in 1732, illustrates:

A Lottery is a Taxation,
Upon all the Fools in Creation;
And Heav'n be praised,
It is easily raised,
Credulity's always in Fashion:

Of course, we often have to estimate how likely something is without being able to give figures like those in the National Lottery case. In such cases we use language like 'very likely', 'almost certain', 'extremely unlikely', 'as likely as not' and similar phrases. These are vague but have more content than 'might happen' alone. Note also that when something **undesirable** might happen it is very common to speak of 'risks'; for example, if you have some medical operation it is common to be told that there is such and such a risk of undesirable side-effects – including possibly death!

Question 11.4

11.4.1 What is your view about buying National Lottery tickets for yourself *with your own money*? Write a brief case for or against doing this which takes account of alternatives, possible consequences, and how likely and how valuable these are.

11.4.2 Refer back to your answer in 11.3.3 and, using what you know about dropping the atom bomb on Hiroshima and Nagasaki, estimate how likely and how valuable/undesirable the various possible consequences of your two options were.

11.2.5 Take proper account of moral or ethical commitments

You may have found question 11.4.1 (about buying National Lottery tickets for yourself) a bit problematic because you are morally opposed to gambling. This may be something you feel very strongly about, or perhaps something you believe but not very passionately. If this is a serious commitment, the only 'possible consequence' which will matter to you is that buying National Lottery tickets would violate your moral code and this is something you regard as highly undesirable, so you will instantly decide against on that basis alone. Other people, who believe that gambling is immoral but do not feel very strongly about it, might work out the consequences of buying tickets, giving some weight to doing an immoral act, but only some. It would depend on the strength of their commitment.

In general, you can take account of moral and ethical commitments in arriving at a decision by including them in the list of consequences and giving a measure of how undesirable it is for you to violate them. For example, if you were faced with the awful decision about whether you or your partner should have an abortion, this would involve moral considerations for most people (about the 'right to life' of the foetus or about a woman's 'right to choose') and these could be included in the list of consequences with values attached to them which would depend on how strong your moral commitments were. Similarly, perhaps President Truman considered the morality of killing many thousands of innocent civilians if he authorised the dropping of the atom bomb and the morality of subjecting many more US soldiers to injury and death if he did not.

Question 11.5

Consider what sort of moral considerations you think would be relevant to deciding the following:

11.5.1 A friend has confided in you, making you promise not to reveal her secret to anyone else. Other friends very much want you to tell them her secret. What should you do?

11.5.2 As in 11.5.1, but what she has told you could affect your other friends seriously. What should you do?

11.5.3 You know a school/college friend is getting deeply into serious drugs. What should you do?

11.2.6 Weigh up which alternative is best in the light of the consequences

Once you have considered a reasonable range of options and worked out their consequences – how likely they are and how valuable/undesirable – then you can make a decision about what ought to be done which is reasonably considered.

Here is an example where this technique was applied in my own case. Some time ago my daughter complained that our old piano was worn out and was preventing her from preparing well for her music exams. There seemed to be a problem, so what was to be done? Here is my reasoning about three options:

> *Option 1:* **Do nothing**. I watched and listened to her playing; there was a small chance that she would simply cope with the piano and continue to enjoy playing; however, it seemed much more likely that she would become increasingly frustrated and play less, perhaps losing interest. I didn't want to risk this; encouraging her interest in music was very important to me.
>
> *Option 2:* **Buy a new piano**. I took my daughter to play a wide range of new pianos; she really liked playing some, but these cost far more than I could afford. I felt sure she would love a good piano and would keep playing at least until she went to university (in two years); that was very important to me. But we couldn't afford the ones she liked.
>
> *Option 3:* **Have the old piano 'restored'**. Three companies looked at it and said it would be expensive and they could not guarantee success. A fourth said they could guarantee success; their price was reasonable and one I could afford; furthermore, they said they could sell it for twice the cost of restoring it if we didn't like the result of their restoration (because it was special and unusual). I checked with former customers and their credentials were excellent. Furthermore, my daughter preferred to keep her old piano if it could really be restored. Bingo!

It is clear that arriving at this decision involved considerable investigation; it took quite some time and cost some money (telephoning, visiting shops and getting estimates), but looking at the options and working out their likely consequences and the value of these enabled us to arrive at a well-reasoned decision. Not only was it well reasoned, it also turned out well, since the restoration was excellent and my daughter progressed musically as a result!

To conclude this section, if we ask the right questions we have a better chance of making good decisions than most of us do without the aid of such strategies.

11.3 A thinking map for handling decisions/ recommendations skilfully

If we put together the preceding considerations, we can produce a thinking map which helps us ask the right questions when faced with making a decision or evaluating someone else's case for a recommendation.

Thinking map

Handling decisions/recommendations skilfully

1 What makes this decision **necessary**? (Objectives?)

[**2** What is **recommended** and on what **grounds**?]

3 What are the **options/alternatives**? (Realistic or unusual?)

4 What are the **possible consequences** of the various options – and **how likely** are they? (On the basis of what evidence and how reliable is it?)

5 How **important** are these consequences – for all those affected?

6 When I **compare** the alternatives in the light of their consequences, which is best? Is the recommended course best?

[**7** How can I carry out this decision? (Contingency plans?)]

Let us consider how this thinking map works out with an example. Imagine you are an inspector of prisons, and a prison governor with special responsibility for drugs policy in prisons reports to you as follows:

> Random drug-testing of prisoners was introduced five years ago in order to solve the many problems associated with prisoners taking drugs. Since cannabis can be detected in the body up to a month after having been smoked, prisoners are tempted to switch to heroin, which stays in the system for only 48 hours. As a result, since drug-testing

was introduced, cannabis use has declined by a fifth, whereas heroin use has doubled. Heroin is not only a much more damaging drug than cannabis, it is also extremely addictive. There is evidence that heroin addiction encourages prisoners to intimidate others in order to pay for the drug.

Question 11.6

Before reading on, consider what *you* think should be done.

Following the thinking map, your thinking might go something like this:

> There are many problems associated with prisoners taking drugs; the policy we introduced to deal with this has had some unintended and undesirable consequences, so we need to think again. If we abandon random drug-testing the old problems will very probably return and we don't want that. If we continue with the present policy we shall probably retain a lower use of cannabis but we shall make some prisoners into heroin addicts and risk increased intimidation. So, what other alternatives are there? We could attempt to make it much harder to smuggle drugs into prisons, with more severe penalties; we could do more drug-testing of prisoners and make the penalties fiercer; we could relax our attempts to control drug-taking in prisons and try to contain the consequences. Perhaps there are other possibilities too. We probably need some experiments in different prisons to see what the likely effects of these are, but drug-taking is a problem for us and we can't just ignore it.

This thinking does not get very far, but it is on the right track so far as it goes. Next we need some evidence about likely consequences and so on, and this will require extensive investigation.

Question 11.7

Refer to your answer to question 11.1 and use the thinking map to see how following it would have changed your decision-making process and how (if at all) your decision would have been different.

Question 11.8

Briefly (in about 100 words) review some options and possible consequences in the case of the arguments contained in two of the following:

11.8.1 Questions appendix, passage 8.

11.8.2 Many foods which are marketed to children are regarded by health experts as 'unhealthy'; they are high in fat, salt and sugar and low in vitamins, minerals and other things important for good health. The result is that many children are overweight or even obese and are increasing their chances of getting heart disease, diabetes and various other diseases. Since governments prohibit the sale to children of other harmful products – like cigarettes and alcohol – why not treat harmful foods in the same way? Young people under 18 should only be allowed to buy foods which are healthy and nutritious.

11.8.3 Questions appendix, passage 25.

11.8.4 Questions appendix, passage 35.

These examples do not allow us to get very far in deciding what to do because we mostly need rather more information. If we look back to my piano example (section 11.2.6) it is clear that finding the relevant information can be time-consuming and costly. This would clearly be the case with the example above about drug-taking among prisoners. In general, the cost of finding the relevant information, in both time and money, may be a factor you need to take into account when trying to decide what to do, and sometimes it will be too high to be worthwhile. This is a judgement you will sometimes need to make and it will depend on how important the decision is, how serious it would be to 'get it wrong', how costly the extra information would be and whether there is enough time available. Leaving those problems on one side, let us now explain the connection between our model and making the 'right' decisions.

11.4 Decision procedures and making the 'right' decision

We pointed out earlier (chapters 6 and 7) that making judgements about which sources of information are reliable can be done more or less skilfully. But we also pointed out that making these judgements skilfully does not *guarantee* that we arrive at the *truth* (that deciding such issues rationally does not *necessarily* produce the truth). Asking the right questions helps us to avoid falling into some familiar errors and is more likely to produce the truth, but we can still make mistakes. The justification for making credibility judgements in the way we explained earlier is that in general this procedure will produce more reliable beliefs than not doing so; but this does not mean that in a particular case we won't make a mistake.

The same is true with decision-making. Most people have had the experience of making some decision carelessly, quickly and without much thought – and have been lucky because the decision has *turned out well*. Equally, it sometimes happens that people have the opposite experience – in which they consider very carefully what to do, following all the procedures we just explained – and their decision *turns out badly*. For example, sometimes people who have taken great care over deciding whether to buy a particular car have nothing but trouble with it and find that it is just a 'lemon'. Again, the point is that following good decision-making procedures will help us avoid familiar mistakes and generally produces better decisions than doing otherwise (acting on intuition or whatever), but it will not *guarantee* a 'good' decision (since the best-laid plans can go wrong). For a dramatic real-life example which illustrates the point, consider the case of Monsieur Raffray and Madame Calment.

> Jeanne Calment, the world's oldest person, died on August 4th 1997 aged over 122. In 1965, as a 90-year-old widow, Madame Calment agreed to bequeath her flat to Monsieur Raffray, her family lawyer, in exchange for a monthly pension of 2,500 francs, that was equivalent to about £250. He died in 1995 having paid out £120,000, more than three times the flat's value. On the face of it, Monsieur Raffray made a very reasonable deal in 1965, since no one could have foreseen that Madame Calment would live so long, but in the event it turned out to be a disastrously expensive deal. (*Daily Telegraph*, 5 August 1997)

Clearly then, there are two ways of assessing the quality of a decision. One is simply in terms of its *out-turn*. The other is in terms of the *quality of the procedures* used in making it. Everyone hopes their decisions will turn out well, but most people also recognise that things

can go wrong even with the most rational procedures. The justification for using the procedures we have recommended is not that they will always guarantee a 'good' out-turn, but that they help to save us from error and generally tend to produce better decisions in the sense of better out-turns than any other procedure.

Question 11.9

There are two parts to this question:

11.9.1 Suppose someone offers you the following bet – they will toss a coin and if it comes down heads they will pay you £1,000 but if it comes down tails you will have to pay them £100 (and this will be done only once). Should you take the bet – assuming it is a standard coin, that you can afford to lose £100, would like to win £1,000 and are not opposed in principle to gambling?

11.9.2 With the same scenario as 11.9.1, what should you do if they agree to do it five times?

Question 11.10

Bearing in mind how decision-making can go wrong, try to put yourself in Admiral Kimmel's shoes (without the benefit of hindsight) and say what questions he should have asked at various key points:

> In the summer of 1941 Admiral Kimmel, Commander in Chief of the American Pacific Fleet, received many warnings from Washington about the possibility of war with Japan. [His men were not fully ready so he instituted a training programme, but he did not suspend peace-time shore leave. As a result,] at weekends there were sixty American warships anchored in Pearl Harbor and the airports on Hawaii contained lines of planes wing-tip to wing-tip . . .
>
> On 24 November he was warned by naval headquarters that a 'surprise aggressive movement in any direction including attack on Philippines or Guam is a possibility'. Despite a meeting with his staff he decided not to change his orders. One of [his] staff pointed out that Pearl Harbor had not been mentioned in the message from Washington

and was therefore not at risk. Although this was clearly not implied by the message, which referred to an 'attack in any direction', the meeting came to the conclusion that there was no further need for action . . . If [Kimmel] thought the message was ambiguous, he should have asked Washington to clarify it. Moreover, he assumed, wrongly, that the army, which manned the anti-aircraft guns, was on full alert. He had only to pick up the telephone to check his assumption, but he failed to do so . . .

Further warnings of war were received on 27 November and 3 December. The latter reported that American cryptographers had decoded a message from Japan ordering their embassies throughout the world to destroy 'most of their secret codes' . . . Kimmel and his staff seized on the word 'most': surely if Japan were going to war with America they would have instructed their embassies to destroy *all* secret codes. On 6 December, the day before the battle of Pearl Harbor, there was more evidence of an impending attack. Kimmel was given orders to burn all confidential documents on outlying Pacific islands. Moreover, his chief intelligence officer reported that the location of Japan's aircraft carriers was unknown, since for several days it had been impossible to intercept their radio signals. This information convinced him that they were about to attack: the question was where . . . his staff officers reassured him, arguing that the Japanese had not sufficient strength left over from their operations in the Asiatic area to attack Pearl Harbor.

Five hours before the Japanese attack, two American mine sweepers saw a submarine which they assumed to be Japanese just outside Pearl Harbor. Because there was no full alert, this was not reported, but one hour before the attack, a Japanese submarine was sunk near the harbour entrance. The officer of the watch reported it to all the relevant naval officers he could contact and the message reached Admiral Kimmel. Instead of taking immediate action, he decided to wait for confirmation that the submarine really had been Japanese. The destruction of the American fleet followed. As for Admiral Kimmel, he was court-martialled and demoted. (Sutherland, 1992, pp. 131–3)

11.5 To summarise

The thinking involved in making decisions or in recommending a course of action deserves special attention. We often do this poorly because we jump to a decision and then rationalise it. If we are to make decisions well, we need to avoid the common weaknesses in thinking of this kind, so we need to be clear what the problem is, we need to consider a reasonable range of options and their possible consequences and so on *before* coming to a decision or recommendation.

In getting clear what the problem is, this may necessitate formulating objectives, though not necessarily. In thinking of possible consequences we need to be as imaginative as possible, but we must then judge carefully *how likely* they are, and *how valuable,* to come to a rational decision. In doing this we may also need to undertake some investigations and take account of moral considerations. We may then be in a position to come to a well-reasoned overall decision.

These are simple ideas but they have proved their value over many years and are widely used. Charles Darwin says, in his *Autobiography,* that he used the method of spelling out pros and cons in deciding whether to marry! I'm not sure how well the method applies in such a case but it *is* very useful in many decision-making situations.

Question 11.11

Use the ideas you have studied here to help you decide what should be done in response to two of the following:

11.11.1 Questions appendix, passage 12.
11.11.2 Questions appendix, passage 48.
11.11.3 Questions appendix, passage 49.
11.11.4 Questions appendix, passage 52.

Further reading

Swartz, Costa et al. (2010).

12 How to get reliable information from the internet (Critical thinking about the internet)

We often use the internet to get information. For example, if we want to check the weather forecast for London we simply type 'weather forecast London' into the search box of our preferred search engine, press 'Enter' and it's likely the first few sites will provide us with what we need. Or if we want to read reviews about a digital camera we are thinking of buying, we enter the name of the camera followed by the word 'reviews' and, again, we expect to find what we need among the first few sites listed. Indeed, most people search the internet in this straightforward and simple way.

It is well known that most search engines (Google, Yahoo and so on) aim to organise their searches so that the first few websites which come up in response to a query are likely to be the ones which interest most people. However, if the first few sites *don't* give us the information we want, there are several things we can do and we shall explain some of these here.

The obvious first step is to learn how your preferred search engine works and to read its own guidance on how to maximise the chance of success. This will help up to a point, but there will still be a significant problem – which is our main concern in this chapter. Since *anyone* can put *anything* on the internet, it can be very difficult to distinguish good sources of information from poor ones – to separate the wheat from the chaff – to find *credible, reliable* and *authoritative* sources of information, and to do this reasonably quickly. In what follows we explain how to think critically about material found on the internet; this is done in a way which is intended to be of interest to the general reader, but also with the particular aim of helping school, college and university students with their academic course work or research.

12.1 An outline of our approach

To find reliable information on the internet – and to do so as quickly as possible – will usually require thinking in the ways we advocated in chapters 6 and 7. For example, suppose you want to find out whether there is still a ban on hunting whales, then you put something like 'hunting whales ban' into Google and trawl through the resulting list until you see something which looks interesting, scan that for names of organisations or individuals you take to be authoritative, and look at their websites or publications by them on the subject. If you do this, you will quickly find your way to the International Whaling Commission (IWC) site. You will find that the IWC is an intergovernmental organisation established in 1946; it has a scientific committee; it reports that hunting whales was banned in 1986 and reports recent meetings of the Commission; it also reports what has happened under the ban and whether the ban has been (partially) lifted. Furthermore, the IWC ought to be a reliable source to answer your initial question!

Of course the key thing about any search, as seen in chapters 6 and 7, is to find people/sources who ought to know or who have the relevant expertise, and a good way to check this out quickly is to follow up on the home page or 'about us' page, or even using the 'contact us' link. These are sometimes missing – so in that case you know very little about the source of the 'information' and you should move on. But sometimes the source will tell you a lot about itself and you will see what sort of organisation or person you are dealing with. You might see that it is a non-profit organisation which campaigns in a particular way – which you might think is biased or is relevant to your search; alternatively you might find it is a commercial organisation which has a vested interest, and so on. In the case of our previous example, the IWC site includes a lot of information about its history and purpose, so you can quickly see that it is likely to be a reliable source for your question.

An alternative way of finding out about the ban on whale-hunting is to choose a news site you think is likely to be reasonably reliable, say the *New York Times* (or the *Guardian* if you prefer). You then go to the *New York Times* site, enter 'whale hunting ban' into its search facility and look through some of its recent reports for the names of experts or organisations whom you think should be reliable. For example, such news reports might quote academics who research the subject or lobby groups who know a lot about it – and you can then check their

websites directly, to see what they say on the subject (and to check their credentials). If you find your way to an academic expert, their website may have recent papers available and these will have a bibliography which could also be helpful.

Another alternative is to use Wikipedia. If you search this on the whale-hunting ban, you can again hope to find references to organisations or experts you can then check out. Even if there are questions about Wikipedia's reliability, it often cites sources which are worth investigating and may well be reliable (see below) – and, of course, you can verify these sources in the way described above.

So, in short, first choose suitable 'key' words and do an initial search, possibly in a source you think should be reasonably reliable, and see what leads it produces – what organisations or experts it cites whose web pages or publications you can investigate in the way we have just described. If it matters enough, you could even email the authors of material which looks interesting.

Sometimes, when trawling through material looking for reliable sources, you may find some claim which looks interesting and possibly relevant to your enquiry but which has no source. Then, one way to check out *this* claim is to type it into a search engine such as Google – with or without quotation marks (see below, p. 197) – and use this search to track down a source whose reliability you can then check out. For example, in reading about the ban on whale-hunting, you might read of 'declining whale numbers' so you could put that phrase into Google to search for a reliable source (confirming or denying the claim).

In general, a site is reliable as a source of information if it meets the conditions we discussed in chapters 6 and 7. So we need to ask whether the source has the relevant expertise, what the source's reputation is, whether the authors are biased or have a vested interest, what the nature of the claim is, how well justified it is and whether there is corroboration from independent sources.

For example, a college professor, writing on his subject, is very likely to be a reliable source; he will have the relevant expertise, will be able to justify his claims, and so on. (If you find a site which is an academic's personal page, it should be reliable because their reputation is on the line!) Some news organisations are regarded as reasonably reliable sources – like the *New York Times* or the BBC. We cannot guarantee that what they say is true, but they check their sources and have a good reputation – generally speaking. Other sources, like encyclopaedias, dictionaries and

other reference works, can be reliable sources – and even Wikipedia is a surprisingly good source if you know how to use it (despite what many academics have said in the past). Commercial organisations may also be reliable sources of information; it depends on the case. For example, the BBC is a commercial organisation (as indicated by its website address, bbc.co.uk). And if you want to know Exxon Mobil's declared policies on measuring oil reserves, Exxon Mobil must be a good source – but if you want to know whether these lobbying practices are ethically sound you may need to look elsewhere.

So much for our outline; now let us explain in detail how to do these things.

12.2 A few basics about searching the internet

Since so many people use Google as their search engine we shall begin with some details about Google. (For other search engines, see their own guidance.)

As we have said, if you want to find out something, think of words which capture your interest, type them into the search box and press Enter. Google will then produce a list of websites, the first few of which are likely to be the most relevant/useful (though see below). Suppose, for example, that you want to find out about oil production in Russia, then you type 'oil production in Russia' into your search box and press Enter. Google then uses some 200 factors in deciding how to rank search items as responses to your query – and how it does so is instructive.

12.2.1 How does Google order its search results?

Google ranks pages by what it calls 'relevance' and 'importance'. To decide if a page is relevant, Google checks how your key words occur on the page. For example, if a web page includes the words 'oil', 'production' and 'Russia' in its title, Google reckons that it is more likely to be relevant than one with the title 'Oil Production in the USA' which mentions Russia elsewhere in the article. Again, if your key words occur right next to each other on a given page, then Google calculates that the page is more likely to be relevant – i.e. to be about 'oil production in Russia' – than to be an article about oil production which just happens to mention Russia. If your key words occur several times in the web page, Google expects this to be more relevant than a page where they appear only once. There are lots of considerations determine if a web page is

relevant to your query. Google then uses its PageRank algorithm to rank them by importance. This process checks how many other websites are linked to a given page (the 'quantity' of links) and what their 'quality' is, taking the view that several links from high-quality websites like the BBC or the *New York Times* count more heavily than more links from less reputable or less well-established sites (though it isn't very clear what makes a site 'reputable'). The implication here seems to be that if other 'quality' websites think a given site is a good one for their purpose, then it probably is.

These are only some of the factors Google takes into account when ranking items in response to a query, but they give you the flavour of the way Google works and are enough for our purposes, except for one important note. Most users believe that Google orders its search results so that the first few are likely to be the most relevant/useful. However, this isn't quite true, because Google sometimes heads its lists with items which are 'sponsored links', i.e. these sites have *paid* to be at the top of this list. Some of Google's sponsored links will be at the side of the page – but it is very important to remember that some may simply be placed at the top of the list – looking almost as though they got there like all the subsequent items. Given that we are particularly interested in reliable information, these initial items – with a faintly coloured background and the words 'Ad' or 'Ads' in the corner in 2010 – may well have precisely the kind of vested interest which will undermine their credibility! This is not always the case, so you just have to inspect them carefully (for example, a current search on 'global warming' produces a list headed by a sponsored link from the Royal Society – a very reputable source of scientific information).

12.2.2 How to read search results

Suppose you have typed 'weather forecast London' into the search box; then one of the first sites to be listed will look something like this:

> BBC **Weather | London**
> **Weather forecast** for. United Kingdom; > Greater **London**; > **London**. Help for Location ... Wed Day weather. White Cloud with light rain forecast for 18:00 ... Weather: London forecast - Info for Sunday's weather - Info for Saturday's weather
> news.bbc.co.uk/**weather/forecast**/8 - Cached - Similar

The first line, BBC **Weather**|**London**, is called the *title* of the web page and it is what the *author of the page* chose as the best short description – or title – for the page. The next four lines, '**Weather forecast** for. United Kingdom; > Greater **London**; > **London**' etc. is called the *snippet*: it is a concise description of – or excerpt from – the web page. This is *not written by the author of the page* but is what Google describes as their 'algorithmic attempt to extract just the part of the page most relevant to your query'. Being 'algorithmic' means it is done electronically – by part of Google's software – not by a human being, and this can have odd consequences. On the next line is the URL (uniform resource locator), which is the *address* of the web page on the World Wide Web; in this case, it's the BBC in the UK. The word 'Cached' is a link to an earlier version of this page (which can be very useful as we'll explain later), and the word 'Similar' will direct you to other sites with similar content.

(Notice that if *you* try this example you may find it looks rather different, since web content can change quite rapidly, but our explanation here should enable you to understand what you read and to judge the usefulness of the site. This is equally true for all subsequent examples.)

All of these features are helpful in determining whether a page is likely to contain what you want; if you want a *reliable* weather forecast for London, this page looks promising; its title and snippet are very near to what you want and the BBC is generally regarded as a reliable source, so it is likely to be useful.

Staying with the example of London's weather, another site among the first few will look something like this:

> **London** & South East England **weather forecast** - Met Office
> Met Office 5-day **weather forecast** for **London** & South East England, 15-day and 30-day forecast for the UK.
> Tunbridge Wells weather forecast - Temperature - Croydon: Forecast
> www.metoffice.gov.uk/**weather**/uk/.../se_**forecast_weather**.html - Cached - Similar

This time, the title looks exactly right; it suggests a weather forecast for London from the Met Office (the official, government weather forecaster in the UK). The snippet looks OK – if a bit odd. And the address looks right.

12.2.3 Clues from URLs

You can get clues about the usefulness of a website from its internet address – its URL. Here's how to read a URL, using our previous example:

www.metoffice.gov.uk/**weather**/uk/.../se_**forecast_weather**.html

The initial 'www' stands for the 'World Wide Web'; 'metoffice' tells us the computer (or 'server', as it is often called) where these pages are hosted (and you can check whether it is what it appears to be by googling 'metoffice'); and the expression 'gov' is called an 'organisational code' – it tells us what *kind* of organisation is responsible for this site. The most common organisational codes are the following:

.co or *.com*	commercial/company
.org	non-profit organisation
.gov	government
.ac or *.edu*	academic/educational

There are others too, like *.mil* for military sites, *.net* for networks (internet service providers) and *.me* for personal blogs, but the six above are the most important ones for our purposes. Clearly, the organisational code of a site could be very useful in deciding whether it contains what you are looking for and in evaluating its contents.

The expression 'uk' is called a 'country code'; it tells us the server is based in the UK. Other country codes are *.fr* (France), *.de* (Germany), *.eu* (European Union), *.sg* (Singapore) and so on. Note that the URL does not always contain a country code; for example, many sites based in the United States do not include the country code (*.us*) in their URL (but they might in order to avoid confusion – for instance *.ca* means Canada, whilst *.ca.us* means California).

Question 12.1

Explain what the following search responses tell you about the content of the website, who wrote it, who published it, where and when it was published, and so on.

12.1.1 Obesity - Causes
25 Feb 2010 ... Information on **Obesity** from NHS Choices including **causes**, symptoms, diagnosis, risks and treatment and with links to other useful resources. www.nhs.uk/Conditions/**Obesity**/Pages/**Causes**.aspx - Cached

12.1.2 **Obesity** - Wikipedia, the free encyclopedia
Obesity is a leading preventable **cause** of death worldwide, with increasing prevalence in adults and children, and authorities view it as one of the most ... en.wikipedia.org/wiki/**Obesity** - Cached - Similar

12.1.3 **Ozone layer** most fragile on record | Environment | The Guardian
27 Apr 2005 ... The UN environment programme estimates that for every 1% thinning of the **ozone layer** there is a 2% to 3% rise in **skin cancer** ... www.guardian.co.uk/science/2005/apr/.../environment.research - Cached - Similar

---

12.2.4 Refining your search: key words and how they function

You may find the information you want straight away if you make the kind of basic query we have been discussing, but if you don't, the first step is often to think again about how you have phrased the query.

Most people know that Google considers only *'key' words* and uses their *order* as an important clue in conducting its search; it also takes the key words in your query *together* – as though they had the word 'and' between them. Thus, if you type 'oil production Russia' in the search box, Google will list pages which are about oil production in Russia, not about castor *oil*, not about the *production* of beef, and not about art in *Russia*. Google will also ignore many common words like *and, the, a, how, is, for, of* (and many others); also, there are words like *document, website, company, information* and so on which are usually unnecessary (because it's the specific content of the document etc. which interests you). So the first step in maximising the chance of a successful search is to identify the key words relevant to your enquiry – the words people are *most likely to have used* in writing relevant web pages. Remember, a search engine is not human. It can only match the words you provide to pages on the web, so in your search use the words that are most likely

to appear on such pages – they are the key words (and, of course, it is wise to have some alternatives in mind).

One quick way to check whether you have chosen the right key words is to click on the word 'Cached' which is often found after the URL. This will bring up a version of the listed web page with the words you have chosen highlighted throughout the page, so you can quickly skim the page to see if the occurrences of those words really are talking about what interests you.

It can be quite tricky to choose the right key words to find what you want; if, for example, you are interested in medical information about broken bones, you will need to use the term 'fracture' not 'broken' because that is the term doctors will have used. For another example, if you want to find out more about how prospective parents might choose the sex of their children (our example in section 3.5) you will not get far with the phrase 'choosing the sex of children', but if you trawl around a little you will discover that 'preconception sex determination' will give the kind of results you want.

Sometimes you will need to think of alternative key words to the obvious ones. Suppose you are considering the question 'Does violence on television affect teenagers?'; then the obvious key words are 'violence', 'television' and 'teenagers'. However, it is easy to imagine that there might be many relevant pages on the internet which use different key terms, for example (i) violence, aggression, brutality, (ii) television, TV, media, and (iii) teenagers, adolescents, juveniles. So how do you also find those pages? The answer is to combine the alternatives using OR (which must be typed upper case). In this example, you would put in your search box 'violence OR aggression OR brutality and television OR TV OR media and teenagers OR adolescents OR juveniles'.

Brainstorming may be enough to think of alternative key words, but sometimes it might be helpful to use a thesaurus. Remember also that North America and the UK often use different words for the same thing.

Question 12.2

Consider the following questions and say what you think the key words should be for a successful internet enquiry (and in what order). Suggest alternatives if necessary.

12.2.1 In what country are bats considered an omen of good luck?
12.2.2 Could digitalis explain Van Gogh's use of distinctive colours?
12.2.3 Is there evidence to think that Napoleon was poisoned with arsenic?
12.2.4 Is global warming caused by human activity?
12.2.5 How can you find reliable information on the internet?
12.2.6 Can you find good guidance on 'critical thinking about the internet'?

Another way of finding suitable alternative key words is to put the tilde sign '~' in front of what you have identified as key words, to see what comes up. For example, you could search 'Does ~violence on ~television affect ~teenagers' and the results may suggest alternative key words to those mentioned above (cf. section 12.4.1). This may help you focus on what really interests you.

12.3 Which websites are reliable?

It's time to say more about the question which really interests us in this chapter – how can you get reliable information from the internet? In general, it is true that the more you already know about a subject, the easier it is to frame a useful search. But, as we said earlier, so far as the reliability of sources goes, exactly the same principles apply as those discussed earlier in chapters 6 and 7; you want to know who the source (author and publisher) of the information is, whether they have the relevant expertise, whether they are genuinely in a position to know what they claim, whether they have a vested interest and what their reputation is, what the context is in which the claim is made, what its justification is, what its nature is and whether there is corroboration from other independent sources.

One of the problems about judging the credibility of 'information' on a website is that there is often no obvious author. Of course, if you want to know details of flights to a particular destination, the websites of companies which fly that route will reliably tell you what you want to know, but if you want to find reliable information about the extent to which the Earth's ice caps are melting, it is not so obvious how to find a reliable source who is in a position to know, has the relevant expertise and does not have a prejudicial vested interest – and this is especially because the authors of websites are so often unknown. However, you

can tell a certain amount from the snippets which come up in response to a query, in particular from their URLs, so it is worth taking a moment to explain these.

12.3.1 What you can tell from a URL

As we saw above (section 12.2.3), a URL can tell you whether a website is commercial (*.co* or *.com*), a government organisation (*.gov*), an educational establishment (*.edu* or *.ac*) or a non-profit body (*.org*). Though these organisation codes have to be interpreted with caution, they can be helpful in locating credible sites.

For example, URLs which end in *.co* or *.com* are commercial and will usually be trying to sell things, so might have the sort of vested interest which reduces their credibility. This is not always the case; for example, the BBC's organisational code is *.co* and the BBC will be regarded by most people as a reasonably reliable source on many things. It is also true that an airline's website (for example) will generally be reliable about details of its flights. There are campaigning groups which use the *.com* code, such as www.rethinkalberta.com which campaigns against exploiting the tar sands in Alberta. Furthermore many companies carry out respectable research and publicise the results – so we have to be careful about discounting these sites when looking for reliable information, but equally we need to watch for vested interests which raise questions about credibility.

The *.gov* code denotes a government organisation – and these tend to be reliable sources of information though in only narrow areas.

The *.org* code means that the organisation is non-profit or charitable. Again one has to be careful because these often have a particular bias; for example, if you look for information about global warming, you will find numerous websites with the *.org* code devoted to claiming that global warming is just a communist conspiracy! On the other hand, websites with the organisational code *.org* can be reliable sources of information, for example the Royal Horticultural Society, which is *the* organisation for gardeners in the UK, and whose website rhs.org.uk is a very reliable source of horticultural information.

Given our particular interest in finding reliable information about the kinds of things school and college students have to research, it is clear that educational/university sites (*.edu* or *.ac*) are likely to be useful and authoritative. They are likely to have the relevant expertise, they are unlikely to have a vested interest – and so on. Of course, you have to be

careful because some *.edu* or *.ac* sites are student sites, but some are the personal sites of academics and their reputation is on the line – so they will tend to be very careful about what they publish on the internet.

As we said earlier, a good way to check the credibility of a person or organisation will be to look at the home page or the 'About us' page. If there is nothing, move on, but often you will be able to find out quite a lot about the person or organisation behind the site – and this should tell you a lot about their credibility.

Question 12.3

Say what you can about the probable reliability of the sites mentioned in questions 12.1.1–12.1.3.

12.3.2 Searching in reliable websites

Depending on what you want to find out, you may know of a source/organisation you regard (or who is generally regarded) as reasonably reliable. For example, you may want to know about the 'weapons of mass destruction' claims which were used to justify the 2003 Iraq war or the so-called 'war on terror' – and you may feel that the BBC or the *New York Times* would be reasonably reliable sources. Then, the obvious way to find information about these topics is to go to the website of either the BBC or the *New York Times* and put your search terms into its search box. Alternatively, you can type into Google's search box 'war on terror New York Times' or 'war on terror site:nytimes.com'. However you do it, you can then also select a particular time period – using Google's Timeline – to research the subject in the last year or at some other time.

12.3.3 Searching by reliable authors

As we have said, knowing who is responsible for material on the web is key to knowing whether it is reliable. Sometimes you will know of an author you regard as a reliable source of information on the subject you wish to research and you can search for material he or she has produced on the subject by simply typing his or her name into the search box with the topic in which you are interested. For example, you might wish to learn what Albert Einstein thought about nuclear weapons, and if you put 'nuclear weapons albert einstein' in the Google search box, you will have a good chance of finding out. Alternatively, you can use

the 'author operator', putting 'nuclear weapons author: albert einstein' into your search box, and again you should find some good sources.

Question 12.4

Search for information on hunting whales, published during the past year, from the BBC, the *New York Times* and the International Whaling Commission. Say how reliable you think it is likely to be and why.

12.3.4 Finding web pages with similar content to a given website

Suppose you have found a website which seems to be near what you are looking for, or to be particularly helpful. In that case, you can search for web pages that have similar content either by clicking on the word 'Similar' which comes after the URL in a Google results list or by typing 'related:' followed by the website address in the Google search box (with no space between 'related:' and the address, as in 'related:metoffice.gov.uk'). This could be a way to get near the 'corroboration by independent sources' which was so important when we discussed credibility earlier; however, you have to be careful because so many websites 'derive' some of their content from other websites (and therefore are not independent in the way required for corroboration) – but still it can be worth a try.

Question 12.5

Try finding similar content with the following examples:

12.5.1 Ozone and skin cancer
12.5.2 www.library.jhu.edu/researchhelp/general/evaluating
12.5.3 Any subject which particularly interests you

12.4 Some special searches

12.4.1 Finding meanings/definitions of words

One simple task which often crops up, especially in the course of academic work, is finding out what some term or phrase means – or

how it is defined. To do this, type in the word 'define', then a space, then the word(s) you want defined. Google will work just as well if you write something similar, like 'definition of' followed by the terms you are enquiring about, or 'meaning of ...' Incidentally, Google advises that 'if you want to search not only for your search term but also for its synonyms, place the tilde sign (~) immediately in front of your search term.'

Question 12.6

Use the internet to find the meaning of three of the following (some were discussed in chapter 5):

12.6.1 Organic foods
12.6.2 Baroque architecture
12.6.3 Proved beyond a reasonable doubt
12.6.4 Polygon (to explain its meaning to a child)
12.6.5 Circumstantial evidence
12.6.6 Greenhouse gases

12.4.2 Phrase search: using double quotes to search for an exact phrase

If a basic search doesn't yield quite what you want, you may feel it could help to focus your search if Google looks for some particular phrase – particular words in a particular order. To do this you put double quotation marks around the words you wish to search for, as in "to be or not to be".

Question 12.7

Try this with two of the following:

12.7.1 "Global warming caused by sunspots"
12.7.2 "I have a dream"
12.7.3 "Never have so many owed so much to so few...."
12.7.4 "Let them eat cake"
12.7.5 "The myth of global warming"

12.4.3 Time limits: finding material from a particular time

One way to focus the responses to an enquiry is to limit the time period Google should consider. For example, you might just want to know the latest material on the subject of your search, or what is available from the last year, or what is available from the period 2001–03. To limit your search in this way, look to the left of any Google search to find the Timeline device, or click on one of the time periods listed under 'Any time' (use the 'More search tools' option to find these links if necessary). In the case of Timeline, Google will display a graph which shows how many items were published at different times on the subject you are researching. You can then select a period from which you want to see material. This is obviously particularly helpful if you are researching historical subjects, but is also helpful in limiting searches.

Question 12.8

Try to find reliable material published on the following during the given time periods:

12.8.1 Hunting whales (in the past year)
12.8.2 Melting ice caps in the Arctic and Antarctic (during the past year)
12.8.3 The 'war on terror' (during the past six months)
12.8.4 The causes of obesity (during the past five years)

12.4.4 . . . And a more general point

Sometimes you may wish to search for books, videos or images which relate to some particular enquiry and Google enables you to do this very easily. If you look on the left-hand side of a Google search page, beneath the word 'Everything' you will see options for 'Images', 'Videos' and 'Books' and you can search these specifically (click on the word 'More' if you don't at first see all of these options). So if you wish to find books on global warming which have been published in the past year, you select 'Books' and enter or select the appropriate dates in the 'Any time' section. You can similarly search images or videos on the subject.

12.5 Wikipedia

Clearly, a good way to find reliable information about a wide range of topics is to use a good, authoritative encyclopaedia, like *Encyclopaedia Britannica*. Of course, *Encyclopaedia Britannica* is available on the internet to subscribers, so it is worth checking to see if a library to which you have access does subscribe. But for our purposes, the important thing to note is why *Encyclopaedia Britannica* is widely accepted as a reliable source of information. Essentially, articles in *Encyclopaedia Britannica* are commissioned from *experts*, whose names are normally attached to the articles they write (in the *Macropaedia*) and who are subject to extensive editorial scrutiny by people who also know about the subject. *Encyclopaedia Britannica* also has a good reputation for reliability; its entries are updated quite frequently, again by experts, and with careful editorial control – which keeps up its good reputation. Its reputation is clearly vital to its success, so it has editorial processes in place to safeguard that reputation. It avoids writers who have a vested interest and its articles can generally be corroborated from other authoritative sources – to which it often makes reference. So, all in all, it meets the criteria for being an authoritative and credible source of information.

Given that we are focusing here on using the internet to find reliable information, the obvious question to ask is, 'What is the position with Wikipedia? Is it OK to use Wikipedia?'

Wikipedia is a really interesting phenomenon. Although its coverage is now enormous (over 3,000,000 articles in 2010), it was only founded in 2001, and it has grown and continues to grow at a phenomenal rate. It is distinctive in many interesting ways. For example, although it is an encyclopaedia, it is entirely *online*; there are no paper versions of it. Furthermore, unlike *Britannica* and other encyclopaedias, it is *free*; all you need is access to the internet and you can research almost anything via Wikipedia – and it really is free. Indeed, if you set out to research almost any topic, the Wikipedia entry will often be one of the first to be listed. So, to return to the key question, should one rely on its contents?

Well, perhaps Wikipedia's most distinctive feature is that *anyone can write an entry* for Wikipedia – and thousands of people from all over the world do just that. This might make Wikipedia sound like the rest of the internet – where anyone can write anything – but Wikipedia is importantly different, in that contributors have to follow certain rules – and Wikipedia editors or indeed anyone else can edit and correct an

item which doesn't follow those rules. Wikipedia is what its founder Jimmy Wales calls an 'open community', in that *anyone* can contribute to it provided they follow the rules – but the fact that anyone else can edit the page whenever they wish is what he calls a 'core guiding check on everything that we do' (cf. en.wikipedia.org/wiki/User:Jimbo_Wales/Statement_of_principles). Wikipedia, then, has a very distinctive system of editorial control which is well designed but can be slow to implement.

So what should we expect from a Wikipedia article? Wikipedia describes its 'three core content policies' as follows:

(i) Wikipedia articles adopt a *neutral point of view (NPOV)*.
(ii) The material in a Wikipedia article should be *verifiable*.
(iii) Wikipedia articles do *not* express *'original research'*.

By far the most important of the three rules from our point of view is the second, the *verifiability* rule. This requires that all material published in Wikipedia must either be well known (for example, that Paris is the capital of France) or be *attributable to a reliable, published source*. This does not mean that everything needs to actually be *attributed*, but that if it is 'challenged or likely to be challenged' – or if it is a quotation – then it needs to be attributed to a reliable, published source in the form of an inline citation and the source must directly support the material in question (cf. en.wikipedia.org/wiki/Wikipedia:Verifiability).

Here is an example of how inline citations look on a page from Wikipedia (accessed December 2010).

> The ***Encyclopædia Britannica*** (Latin for "British Encyclopaedia") is a general English-language encyclopaedia published by Encyclopædia Britannica, Inc., a privately held company. Articles are aimed at educated adults, and written by about 100 full-time editors and more than 4,000 expert contributors. It is regarded as the most scholarly of encyclopaedias.[1][2]
>
> The *Britannica* is the oldest English-language encyclopaedia still in print.[3] It was first published between 1768 and 1771 in Edinburgh, Scotland, and grew in popularity and size, its third edition (1797) and supplement (1801) reaching 20 volumes together.[4][5][6] Its rising stature helped recruit eminent contributors, and the 9th edition (1875–1889) and the 11th edition (1911) are landmark

encyclopaedias for scholarship and literary style.[5] Beginning with the 11th edition, the *Britannica* shortened and simplified articles to broaden its North American market.[5] In 1933, the *Britannica* became the first encyclopaedia to adopt "continuous revision", in which the encyclopaedia is continually reprinted and every article updated on a schedule.[6]

The current 15th edition has a unique three-part structure: a 12-volume *Micropædia* of short articles (generally fewer than 750 words), a 17-volume *Macropædia* of long articles (two to 310 pages) and a single *Propædia* volume to give a hierarchical outline of knowledge. The *Micropædia* is meant for quick fact-checking and as a guide to the *Macropædia*; readers are advised to study the *Propædia* outline to understand a subject's context and to find more detailed articles.[7] The size of the *Britannica* has remained roughly constant over 70 years, with about 40 million words on half a million topics.[8] Although publication has been based in the United States since 1901, the *Britannica* has maintained British spelling.[1] (see en.wikipedia.org/wiki/Encyclopædia_Britannica)

The references are listed at the end of the article, pointing you to some reliable, published sources which directly support the material in question.

12.5.1 What does 'attributable to a reliable, published source' mean?

Wikipedia explains at great length exactly what it means to say that material must be 'attributable to a reliable, published source', so let us look at their explanation.

The key idea behind this phrase is that any material published in Wikipedia must have *already* been published by some person or some organisation with a *reputation for fact-checking and accuracy* – for being reasonably reliable. For example, the BBC and the *New York Times* are generally thought of as reasonably reliable organisations, so material they present as true may be cited in Wikipedia articles. Certain people are regarded as authorities in certain areas and they may therefore be cited in Wikipedia articles provided they have written or said something which supports the material in question. For example, Albert Einstein was an expert on many branches of physics, so it would be proper to cite him in such contexts, if his remarks supported the Wikipedia material.

Wikipedia advises the reader to consider the cited material itself, its original author and its original publisher in trying to decide its reliability – in line with what we advised in chapters 6 and 7 (about acceptability, the nature of the claim, the expertise of the author and the context of publication). Wikipedia elaborates as follows:

> The appropriateness of any source depends on the context. In general, the best sources have a professional structure in place for checking or analyzing facts, . . . evidence, and arguments; as a rule of thumb, the greater the degree of scrutiny given to these issues, the more reliable the source. Academic and peer-reviewed publications are usually the most reliable sources where available, such as in history, medicine, and science, but material from reliable non-academic sources may also be used in these areas, particularly if it appears in respected mainstream publications. Other reliable sources include university-level textbooks, books published by respected publishing houses, magazines, journals, and mainstream newspapers. Electronic media may also be used, subject to the same criteria. (en.wikipedia.org/wiki/Wikipedia:Verifiability, accessed December 2010)

Newspaper and magazine blogs may also count as reliable sources:

> Some newspapers host interactive columns that they call blogs; these are acceptable as sources so long as the writers are professionals and the blog is subject to the newspaper's full editorial control. In March 2010, the Press Complaints Commission in the UK ruled that journalists' blogs hosted on the websites of newspapers or magazines are subject to the same standards expected of comment pieces in that organization's print editions.[3] Where a news organization publishes an opinion piece, the writer should be attributed (e.g. "Jane Smith has suggested . . ."). Posts left by readers may never be used as sources. (ibid.)

Wikipedia also explains what it sees as 'questionable' sources, as follows:

> Questionable sources are those with a poor reputation for checking the facts, or with no editorial oversight. Such sources include websites and publications expressing views that are widely

acknowledged as extremist, or promotional, or which rely heavily on rumor and personal opinion. Questionable sources should be used only as sources of material on themselves, especially in articles about themselves. . . .

Anyone can create a website or pay to have a book published, then claim to be an expert in a certain field. For that reason self-published media – including but not limited to books, newsletters, personal websites, open wikis, personal or group blogs, Internet forum postings, and tweets – are largely not acceptable as sources.

Self-published material may in some circumstances be acceptable when produced by an established expert on the topic of the article whose work **in the relevant field** has previously been published by reliable third-party publications. Caution should be exercised when using such sources: if the information in question is really worth reporting, someone else is likely to have done so. Self-published sources should *never* be used as third-party sources about living persons . . . (ibid.)

It is interesting to note how closely this guidance on verifiability corresponds to what we said earlier in chapters 6 and 7 about acceptability of claims and reliability of sources. This suggests that if writers actually follow this principle in writing a Wikipedia article, it could be a very reliable source of information. Let us look briefly now at Wikipedia's other core content policies.

12.5.2 No original research and neutral point of view (NPOV)

The *no original research* principle is closely related to the verifiability principle. By 'original research' Wikipedia means material which has not already been published by reliable sources. So it gives the following guidance on how to write a Wikipedia article:

The best practice is to write articles by researching the most reliable sources on the topic and summarizing what they say in your own words, with each statement in the article attributable to a source that makes that statement explicitly. Source material should be carefully summarized or rephrased without changing its meaning or implication. Take care not to go beyond what is expressed in the sources or to use them in ways inconsistent with the intent of the

> source, such as using material out of context. In short, **stick to the sources**. (See en.wikipedia.org/wiki/Wikipedia:No_original_research)

The *neutral point of view* principle means that anyone who writes an article in Wikipedia is required to write it from a neutral point of view, that is:

> representing fairly, proportionately, and as far as possible without bias, all significant views that have been published by reliable sources,

so authors have to explain who believes what and why, which points of view are most common, and so on (see en.wikipedia.org/wiki/Wikipedia:Neutral_point_of_view).

These three principles, 'verifiability', 'no original research' and 'neutral point of view' – Wikipedia's 'three core content policies' – together determine what is acceptable in a Wikipedia article.

12.5.3 A study on Wikipedia's reliability

An article in *Nature* entitled 'Internet encyclopaedias go head to head' (issue 438, 15 December 2005, pp. 900–1) reported how scientists rated Wikipedia by comparison with its major peer, *Encyclopaedia Britannica*. The study found that 'Wikipedia comes close to Britannica in terms of the accuracy of its science entries'. Jimmy Wales, the founder of Wikipedia, commented on the study:

> Our goal is Britannica-or-better quality, but we're not completely there yet, . . . We've been working on Wikipedia for less than five years, and it's a testament to the strength of our community that we should come so close to them at this point. As we get more people to contribute their expertise, I know that the clarity, readability, and accuracy of Wikipedia content will continue to improve.

Question 12.9

Find the source of the above quote. (The source cites the *Nature* article and a project to correct mistakes based on the study.)

12.5.4 So can we use Wikipedia as a reliable source in our research?

Many college and university departments will advise their students that it is not acceptable to cite Wikipedia as a source. However, you can see from the sections above that it could be a good and reliable source if the rules are followed in a given article. In particular, if reliable sources are referenced by means of inline citations, such Wikipedia material could be a good source at least to start one's research. Such an article would enable the reader to *track* its source to reliable published sources.

Of course, anyone who has read Wikipedia articles will know that this is often *not* the case; a lot of the material is unattributed. This is fine if the material is unattributed simply because it is non-controversial; on the other hand, material is often unattributed simply because the author has failed to meet Wikipedia's verifiability requirement. In this case, there will often be a note at the beginning of the article along the following lines:

> This article needs additional citations for verification. Please help improve this article by adding reliable references. Unsourced material may be challenged and removed.

In summary, then, although you don't usually know who has written the material in Wikipedia or who has edited it, *where there are inline citations to reliable published sources, Wikipedia could be a good and reliable source of information.*

Question 12.10

Find information on five of the following in Wikipedia and say how reliable it looks and why:

12.10.1 The death of Napoleon
12.10.2 The extinction of the dinosaurs
12.10.3 Whether global warming is man-made
12.10.4 Truman's decision to use the atomic bomb on Hiroshima and Nagasaki
12.10.5 Admiral Kimmel's decisions prior to the Pearl Harbor attack

12.10.6 Hunting whales
12.10.7 Melting ice caps in the Arctic and Antarctic
12.10.8 The 'war on terror'
12.10.9 The causes of obesity
12.10.10 Decision-making

12.6 Academic and scholarly internet searches

Although anyone can put anything on the web, whether they are expert or amateur, well meaning or malicious, whether their interest is commercial or scholarly, whether they wish to inform or to spread disinformation, whether what they write is accurate or inaccurate, we are beginning to see that careful scrutiny can help you find reliable information on many subjects.

Of course, if you are studying a course in school, college or university, your teachers will know the field well and will normally supply you with a list of key materials to read or view and these should be your starting point. Alternatively, you can consult *web services designed specifically for college and university work* such as your library website.

But, if you are not in that position, or wish to go further than the set material, or wish to research a subject outside your course work, how should you find good sources of information which will help you? Remember, it is easy to miss key sources of information if you just search online. A huge body of research literature is still available only in print form in books or journals. Furthermore, search engines don't cover everything that is online – some databases of research literature may not appear in your search results because they require a password or subscription to gain access (true of many modern electronically published academic journals).

Having said all that, the internet is often very useful, so how should we proceed to maximise its usefulness for academic and scholarly work? We've already explained a number of techniques for improving on the very simple search techniques most people use – but how can we improve still further on these?

12.6.1 'Peer-reviewed' sources

Academics are expected to publish much of their work only after '*peer review*', and in a *scholarly journal* or through a *scholarly press* (like a

university press). 'Peer review' means that other academics, who are independent of the author but have expertise in the area of his or her work (academic peers), read it to check that he or she (i) knows the literature in the field, (ii) has properly referenced it and taken it into account, and (iii) has something new to contribute. In essence, new publications should build on previous work, should contribute to an ongoing academic discussion – and are quality assured by (i) having been subject to the peer review process, and (ii) being published in a reputable publication. This system has been the quality control system in the world of scholarship for a long time. It is hard to imagine a publication process which is more different from what generally happens on the internet, where anyone can publish anything!

So these are the sorts of sources which are most reliable for many fields and which you will be expected to cite in your academic work. Of course, many scholarly journals are now available on the internet – as what are called 'eJournals'. Access to them is not usually free, but your library will have paid a subscription to those thought most worthwhile and your teachers or librarians will be able to tell you how to find and use the best ones for your purposes (for example, which passwords, databases and other sources to use).

Your library website will therefore be key to gaining access to good internet-based material for your studies – eJournals, bibliographic databases, archives and so on – so ensure your teachers or librarians advise you on which to use and how to access them.

12.6.2 Google Scholar

Ordinary Google searches will sometimes direct you to Google Scholar. If you enter search terms into ordinary Google you will sometimes see at the top of Google's list the heading:

Scholarly articles for **[your search terms]**

and if you click on this heading you will be taken straight into Google Scholar. Google Scholar helps you search for peer-reviewed literature from academic publishers, professional societies, university and other websites, whether in books, articles, abstracts, theses, or whatever. Indeed it can help you learn about key scholarly literature in any field.

To find your way straight into Google Scholar, either put 'Google Scholar' into Google or find the 'Scholar' option on Google's home page

(currently by clicking 'More' at the top of the page to access 'Scholar' in a drop-down menu). Once into Google Scholar, as with any other searches around a topic, you need to frame your search carefully, using the key words which people are most likely to have used in writing on the subject (see above on doing this skilfully).
One thing to note about Google Scholar is how it ranks documents:

> Google Scholar aims to rank documents the way researchers do, weighing the full text of each document, where it was published, who it was written by, as well as how often and how recently it has been cited in other scholarly literature. (scholar.google.co.uk/intl/en/scholar/about.html)

So, if this is true, Google Scholar ranks items quite differently from ordinary Google, and does so in a way which aims to ensure that the material is authoritative and reliable (or is regarded as being so by the author's peers). Notice also that for each reference produced by a Google Scholar search, Google's intention is that the 'Related articles' link lists articles in Google's index which are most closely related to it. You might experiment with this link to see how helpful you find it.

Question 12.11

Use Google Scholar to search for scholarly work on two of the following. (You will find a lot of scholarly material on all these topics; you might wish to experiment with using different key terms and restricting the time period over which you search, to find whatever looks most relevant to your interests.)

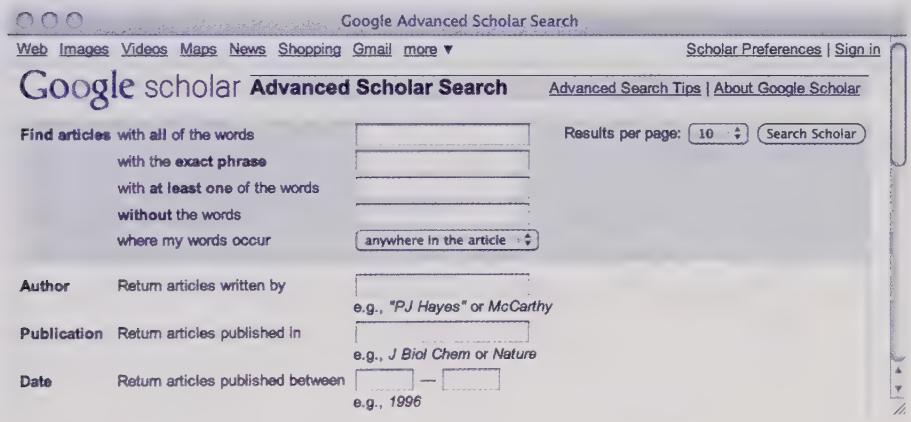
12.11.1 The death of Napoleon
12.11.2 The extinction of the dinosaurs
12.11.3 Truman's decision to use the atomic bomb on Hiroshima and Nagasaki
12.11.4 The causes of obesity (during the past five years)
12.11.5 Evidence about teaching thinking skills (during the past ten years)

Question 12.12

Having practised with some of the suggestions above, now choose any subject which interests you and research it using Google Scholar. Remember to select your search terms carefully and try different time periods to search.

12.6.3 Advanced Scholar Search

As we said earlier, the more you already know about a subject, the easier it is to frame a useful search. For example, you might know of distinguished and authoritative authors in the field in which you are interested or you might know of reliable publications which are likely to be helpful, such as *Nature*. In these cases, you can use Advanced Scholar Search. The top part of the page looks like this:



So, for example, you could put 'hunting whales' into the box which says 'with **all** of the words' and then 'Nature' into the box which says 'Return articles published in', and you might choose the past ten years in the box which says 'Return articles published between'. This way you are likely to get some reliable recent information on the subject.

Question 12.13

Use Advanced Scholar Search to help you with the following:

12.13.1 Search for material by Peter Facione on critical thinking.

12.13.2 Find the latest articles on global warming in the *Scientific American*.

12.13.3 Choose a topic which interests you and an author or scholarly source you would expect to be helpful, and see what Google Scholar produces.

12.6.4 To conclude on Google Scholar

Google Scholar can be very helpful *but* the crucial thing to remember about using it is that an enormous amount of good material is still not online. This is true and will remain so despite the Google Books project, which is in the process of making an enormous number of books and other documents available online. So the lesson is that teachers, librarians and other experts are still a vital resource for discovering good and reliable information.

12.7 Summary

As we have emphasised, anyone can put (almost) anything on the web. Much of what is presented as information is just plain false, is biased or is there simply to sell something, and there are many hoaxes, spoofs, urban legends, scams and plain frauds on the internet (for a good account of these see www.vts.intute.ac.uk/detective and read what it says about the 'ugly' on the internet).

There is some debate about whether it would be better to regulate what can be put on the web (so that one could know what was reliable information, for example). However, the present pattern is unlikely to change at all quickly. In the meantime we have explained how key words function, how to understand the information given about websites, how to find the work of particular authors or publications, when to use Wikipedia and how to use Google Scholar. All of these are steps in the right direction. They enable internet users to bring some 'critical thinking' skills to the web – and some crucial knowledge – to help them evaluate what they find on the web and to separate the wheat from the

chaff. It will be clear from what we said in chapter 1 that to improve these skills you have to practise them, so now it is over to you!

Further reading

www.internettutorials.net contains good material on 'Finding scholarly content on the web', on 'Subject directories and encyclopedias', and more.

library.wlu.ca/critical, from Wilfrid Laurier University Library, in Canada, contains extensive advice on critical thinking and the internet (though it was written some time ago).

Questions appendix

This appendix contains a large number of passages on which to practise your critical thinking skills. Many are taken or adapted from newspapers and other sources – in which case the source is shown. Some, which were used in the first edition, come from Critical Thinking (AS level) examinations produced by the Oxford, Cambridge and RSA Examinations Board; in these cases the OCR paper number and date are shown. Others are from the University of East Anglia *Test of Logical Reasoning* designed by Alec Fisher and Anne Thomson (shown as TLR).

Notice that many of the questions which are to be found in the text could be tried at different stages of your progress through this course. For example, you might like to write an answer to one or more of the questions in 7.11 *before* studying chapters 6 and 7 on making credibility judgements, and then, when you have worked through chapters 6 and 7 and without looking back at what you first wrote, write a second answer. You (or a teacher) could then compare the two answers to see if you wrote a really good answer first time or whether you improved as a result of studying these chapters.

Notice also that nearly all the passages in this appendix are rich enough to lend themselves to several uses/exercises in the context of studying critical thinking. Even though many of them are referenced in only one or two questions (and some not at all) they are all included here to provide a rich variety of examples so that a wide range of students will find material here to interest and challenge them as they sharpen up their skills.

Answers to most of the questions are to be found either in the text itself or in the 'Answers to questions' section which follows (p. 239).

All the passages used here are reproduced with the appropriate permissions.

Passages

1 Being an entrepreneur is a challenging and potentially very rewarding job. One has great freedom in working for oneself and it is not necessary to have a university education. Therefore, a suitable job for anyone who doesn't go to university is to become an entrepreneur.

2 In 2000 Dr Eleanor McGuire published a groundbreaking study of the brains of London taxi drivers. Brain scans showed that they had a much larger hippocampus (the part of the brain used for navigation) than most people. More importantly, it grew larger the longer they spent doing their job. So it is a mistake to think that we continually lose brain cells from an early age. (Cf. *Daily Telegraph*, 6 October 2008, p. 25)

3 If the world's climate is getting warmer, we should find that some of the ice at both the North and the South Pole is melting at an unusually high rate. If the ice is melting, we should see its effect in the raising of the level of the sea. There is evidence that this level is increasing, so the world's climate must be getting warmer. (*OCR 1999, Paper 1*)

4 Most people give very little thought to their health until they are unwell. The demand for health services increases every year, but what we really need is far better education on health issues. Many of the illnesses from which people suffer are due to their lifestyles – what they eat and drink, whether they smoke, how much exercise they get, whether they get enough sleep, and so on. The population could suffer far less ill health and we could save enormous sums on health costs if people adopted healthier lifestyles.

5 Many species are in danger of becoming extinct from the destruction of their habitats. Sometimes this is caused directly by human activity, as when forests are cleared for agriculture, but global warming is also having a devastating effect on many habitats. Of course, human activity is the main cause of this too. Human beings are the main threat to the existence of many species.

6 If you own and drive a car you pay numerous taxes for the privilege of doing this. Governments all over the world spend the income from road taxes on all sorts of things – from welfare to war. But this is unfair. Road taxes should only be spent on roads – on building them, maintaining them, policing them, and so on.

7 If people who claim to have been abducted by aliens really have been abducted, then we would need to take reports of UFO sightings very seriously. However, such claims are very suspect. There are many possible explanations for such apparent abductions ranging from obvious untruthfulness on the part of those claiming abduction to hallucinations and temporary paralysis. Given that we can explain these apparent abductions in ways that do not involve aliens, we do not need to take reports of UFO sightings seriously. (*OCR 1999, Paper 1*)

8 Bullfighting has a long tradition in Spain, but it is out of tune with modern thinking and campaigners there are fighting to get it banned. It is also legal in France, Portugal, Mexico, Colombia, Venezuela, Peru and Ecuador. Though some see it as a beautiful and noble sport, most people now regard it as a barbaric activity which should be banned.

9 A team from the Chinese University of Hong Kong has published results in the journal *Paediatrics* which suggests that though teenagers who lie in at the weekend may seem lazy, the extra time in bed may help them stay slim and healthy! The team studied some 5,000 children aged 5 to 15 and found that those who enjoy a lie-in were less likely to be overweight. (Cf. *Daily Telegraph*, 14 November 2009)

10 Human beings have only actually prospected for oil in a very small part of the world's surface. Some geologists have argued that if oil is as common in the untried areas of the world as it is in the areas where we have searched, then oil reserves are thousands of times present estimates. If they are right, there is no problem about providing all the oil the world needs for several centuries.

11 A group of European scientists looked at a number of studies of the link between passive smoking and lung cancer. The scientists said that these studies do not demonstrate that there is a significant risk of lung cancer for non-smokers who are exposed to environmental tobacco smoke at work or in the home, from either a smoking parent or spouse. But this analysis was commissioned by the tobacco industry. So findings of the analysis are likely to be incorrect. (*OCR 1999, Paper 1*)

12 Dr David Kessler, a former head of the US Food and Drug Administration, recently published a book called *The End of*

Overeating, in which he argued that thousands of foods have been engineered to be tasty in such a way that they also stimulate us to want more of them. As he puts it, 'It is time to stop blaming individuals for being overweight or obese. The real problem is we have created a world where food is always available and where that food is designed to make you want to eat more of it.'

13 The world produces enough food to feed everyone. The problem is what happens to the food after it is produced. For example, it takes four kilograms of grain to produce one kilogram of meat. If we all became vegetarian, there would be plenty of food to go round. Another problem is that people in rich countries price food out of the reach of people in poor countries. Also, people in rich countries eat more than they need. It is absurd to say, as Malthus did, that there must always be many people who starve.

14 Banning cigarette advertising could easily lead to an increase in smoking. If a government bans cigarette advertising within its territory then cigarette manufacturers save their advertising budgets in that country – and, in order to compete with each other, they are then likely to lower their prices. And what is the likely effect of that?

15 The famous frescoes on the ceiling of the Sistine Chapel were painted by Michelangelo between 1508 and 1512. In the following 500 years they must have lost their original colours. Every time there was a Mass, the candle flames sent up wax and soot; after some years, varnish was applied which gradually became yellow; the open windows let in the grime of the city and, more recently, the exhaust fumes of many vehicles. The storm of controversy which was provoked by the cleaning and restoration work undertaken by the Vatican in the 1980s and 1990s was completely misplaced. The work was done by some of the world's finest art restorers and was closely supervised by an international team of art experts and historians. People had become accustomed to seeing the frescoes with their colours dulled by soot, varnish and general grime and just didn't realise how much brighter and more vivid the originals were.

16 Perhaps the most common response to the call for the elimination of animals from toxicity testing is the benefits argument. It runs thus: human beings and animals have benefited from toxicity tests on animals; therefore these tests are justified. But those who

support this argument would have to establish that the benefits to animals outweighed the distress and indignity caused to them in the process, and if it did we might be on our way to receiving an interesting defence of toxicity testing. Unfortunately for those who countenance these tests, however, the benefit to animals cannot be established and therefore the benefits argument fails. (*OCR 1999, Paper 1*)

17 Some people succeed against all the odds. Even though they are not born to wealth or don't have parents who can help them achieve their ambitions, they still make it. This just shows what determination and hard work can achieve, so if you really want something, go for it.

18 Recent research suggests that our understanding of how clouds interact with sunlight might be wrong: new measurements suggest that clouds absorb four times as much energy as previously thought. Since existing models of how the climate functions are based on the original measurements, if the new measurements are shown to be accurate, models of how the climate works will need to be completely overhauled. Climate models are used in our attempts to measure global warming so, if these climate models are shown to be inaccurate, we will have to completely revise our understanding of global warming. (*OCR 2000, Paper 1*)

19 According to the World Health Organization, thousands of people around the world die prematurely from heart disease triggered by long-term exposure to excessive noise. Noise pollution can cause disturbed sleep and stress – and high levels of stress can increase the levels of stress hormones such as cortisol, adrenaline and noradrenalin in the body, even during sleep. The longer these hormones stay in circulation around the blood stream, the more likely they are to cause high blood pressure, strokes and heart failure. (Cf. article in the *Guardian* by Alok Jha, science correspondent, 23 August 2007, p. 12)

20 Many people believe that they are at risk from some kind of violent attack but the crime statistics clearly show that very few are really at risk. Those most at risk are young men, since most violence – by a long way – is committed by young men against other young men. Television news and drama and films greatly exaggerate how common violence is. It is high time the media

gave a realistic impression of the real risks and in particular that the elderly really have little to fear.

21 The history of the world is full of many examples of the extinction of species, and we should not regard such extinction as a serious environmental problem. 245 million years ago, 90 per cent of all species vanished; 65 million years ago, 50 per cent of all living things died out, including the dinosaurs. Much more recently, the influence of humans has led to many species becoming extinct: in Hawaii, for example, this influence led to huge losses of species of plants, insects, and animals. But in each example of extinction we have seen the lost species replaced by new ones (for instance, mammals replaced dinosaurs). (*OCR 2000, Paper 1*)

22 There is a simple argument against an economic system which leaves most people poor. It is inefficient. An efficient economic system allocates resources where they will maximise utility – will produce the greatest benefit. Though some economists will dispute this, everyone knows that the average poor man can make better use of a dollar than the average rich man. A system which gives great wealth to those who have no real use for it, like prodigal 'billion heirs', diverting resources from those who could use them better, is plain inefficient.

23 *Decanter* magazine organised a 'water tasting' in 2007. The tasting panel consisted of Masters of Wine, top sommeliers and the like; they voted London tap water third equal in a blind tasting of 24 different waters. The most expensive water, at £21 a bottle, came 18th and Bling H_2O, from California, the second most expensive, came 22nd in the tasting. London water is free out of the tap.

24 If Neil Armstrong didn't really walk on the Moon in 1969 there must have been the most extraordinary conspiracy, involving thousands of people. Such a conspiracy could not possibly have happened. Someone would have spilled the beans by now. So Neil Armstrong did walk on the Moon in 1969.

25 Some people say that admission to college or university should be based only on objective measures of academic achievement – such as high school grades and college admissions exams. But this ignores the fact that students from different backgrounds and different schools may have had very different educational opportunities. Success at university depends a great deal on your

ability to think for yourself and a willingness to work hard – which can be true of students who have had poor educational opportunities.

26 In the film *Breakfast at Tiffany's* the actress Audrey Hepburn wore a beautiful black dress, designed by Givenchy. In December 2006, the dress was sold at a Christie's auction for an amazing £467,200. The proceeds from the sale of the dress were given to the Calcutta-based City of Joy Foundation, a charity established by the French author Dominique Lapierre to help India's poor. The money was used to build 15 schools across West Bengal, one of the poorest parts of India. Audrey Hepburn spent the last years of her life working to help the poor and destitute. (Cf. the *Independent*, 2 March 2007)

27 There has been a decline in the rate of many illnesses of old age. For example, arthritis, dementia, and strokes are all declining year by year. The causes of this decline include such medical advances as beta-blockers to control high blood pressure and the fitting of hip replacements. There is, however, another factor. The present generation of 60- and 70-year-olds had much better nutrition as children than did their parents. Good nutrition in childhood is important in laying the foundations of good health in adulthood. Since improvements in nutrition have continued over the past 60 years, we can expect that many of the illnesses of old age will continue to decline. (*OCR 2000, Paper 1*)

28 A recent study of 8,000 white-collar workers in America found that men who had considerable control over the way they carried out their jobs had a low rate of heart disease, a stress-related illness. Thus the most stressful jobs are those in which employees have little control over the pace of their work and how it is organised. If employers want to reduce the level of stress-related illness, they need to give their workers high levels of control over their work.

29 If a Roman Catholic priest administered arsenic to the philosopher Descartes in communion wafers, that would have killed Descartes. A Roman Catholic priest did administer arsenic to the philosopher Descartes in communion wafers. So that must have been what killed Descartes. (Cf. story in the *Guardian*, 15 February 2010)

30 'If you drink from a bottle marked "poison" it will harm you. This bottle is not marked "poison", so it won't harm you.' (Cf. Lewis Carroll's *Alice in Wonderland*)

31 One group of theories about the origin of human language suggests that the first words were imitative of sounds in the natural world. The apparent survival of such words in modern languages is offered as evidence for this explanation. For instance, a Welsh term for owl, *gwdihŵ,* pronounced 'goody-hoo', seems to mimic the sound an owl makes. But if some words really did resemble natural sounds, you would expect them to be the same or similar in every language. Interestingly enough, they are not. Dogs go *ouah-ouah* in France, *bu-bu* in Italy, *mung-mung* in Korea. A purring cat goes *ron-ron* in France, *schnurr* in Germany – and so on. These so-called imitative words are no more alike than any other synonymous words from different languages. (*OCR 2000, Paper 1*)

32 If your husband has disappeared, leaving traces of his blood near a 'crocodile slide' where he was walking, he has probably been taken by a crocodile. The police have found traces of his blood by the crocodile slide and he has disappeared. So he has probably been taken by a crocodile. (Cf. story in the *Daily Telegraph*, 6 October 2008)

33 Most cars currently travel at about 80 mph on motorways in Britain, although the speed limit is only 70 mph. Flouting the law in this way encourages disrespect for the law. The government should increase the speed limit on motorways to 80 mph. This would allow drivers to drive at a reasonable speed without breaking the law.

34 The idea that we should cease to aim for economic growth is crazy. When should growth have stopped? Presumably the Stone Age – surely the most 'environmentally friendly' period ever. Think of all the wildlife and the rainforests. Or perhaps the Victorian Age? But in those days, before the availability of affordable transport, you could travel to beauty spots only if you were rich. Now all those who wish to see these places can do so. How can this be bad? One can still find such beautiful places – all the more so as economic growth makes so many people better off. It would be a complete tragedy if our worries about the environment had made us stop economic growth in earlier periods. (*TLR*)

35 Despite the appalling mortality rate from polio in the past, some parents choose not to have their children vaccinated against it, because they think there is now only a low risk that their children

will become infected with the disease. Moreover, some believe that there is a more than negligible risk that the vaccine will have harmful side-effects. In their eyes, a decision to avoid vaccination may appear entirely rational. But what they do not realise is that if a substantial percentage of a population is not vaccinated against polio, there will be regular outbreaks of the disease every few years as the number of non-immune people increases. (*TLR*)

36 The police force should ban their officers from driving at high speed in pursuit of young joyriders who steal cars. Many deaths, both of joyriders and of innocent bystanders, have occurred as a result of such chases. The police say that they have policies which are aimed at preventing danger to the public during car chases, by requiring police drivers to abandon the chase when speeds become too high for safety. But the excitement of the chase inevitably makes the police drivers forget the policy, and disregard public safety. No stolen car is worth a human life. (*TLR*)

37 If you don't vote, it won't make any difference to the election result. But if everyone like you doesn't vote, it will make a difference to the election result. So you should vote. (*TLR*)

38 Imagine a bucket of water, say three-quarters full, suspended by a long cord which has been twisted many times. Let the water settle so it is still; then the surface of the water will be flat. Now release the bucket so that the cord untwists and makes the bucket spin. What happens to the surface of the water? Initially, it remains flat – although the water is spinning fast *relative to the bucket*. Gradually, as the spinning bucket imparts its motion to the water, the surface of the water will become concave as centrifugal force makes the water 'climb up' the inside of the bucket. At a certain point the water will be revolving at the same speed as the bucket, so, *relative to the bucket* the water is stationary; but the surface of the water will be concave. This shows that there is a great difference between relative and absolute motion. At first the 'motion' of the water is only relative – but then it is absolute, as is shown by its concave surface. (See Isaac Newton *Principia*, Book 1: Scholium [1687]. This is his famous 'bucket argument'; see Wikipedia.)

39 When prisoners under sentence of death are given the choice between life in prison and execution, 99 per cent of them choose life imprisonment. This shows that they fear death more than

they fear life imprisonment. Since one is most deterred by what one most fears, it is evident that the threat of the death penalty is more likely to deter most potential murderers than is the threat of life imprisonment. (*TLR*)

40 Is there intelligent life anywhere in the Universe besides here on Earth? In a famous experiment, conducted in 1952, two Chicago scientists, Stanley Miller and Harold Urey, attempted to simulate the Earth's early environment within a sealed, sterile container. There was water (for the oceans), methane, ammonia and hydrogen (like the Earth's early atmosphere) and an electric arc (to simulate lightning). Some of the water was heated to produce water vapour in the atmosphere and sparks were passed through the resulting atmosphere for a week. At the end of this time they found amino acids, which are the building blocks of protein, some fatty acids and urea – all of which are important chemicals for life processes. Repetitions of the experiment using different combinations of initial chemicals have shown similar results. These experiments suggest that life should not be uncommon in the Universe, since the important chemicals are not uncommon and there are over 100 billion stars in our own galaxy alone.

However, the surprising thing is that scientists who have searched for intelligent life elsewhere in the Universe – by listening for radio signals – have found no solid evidence to suggest that it exists anywhere other than here.

41 On 6 April 2009 an earthquake hit the city of L'Aquila in Italy, 95 kilometres northeast of Rome, killing many people. A laboratory technician named Giuliani who lived in L'Aquila had predicted the earthquake because he had felt tremors and had detected increased emissions of the gas radon in the area: over a number of years he had become convinced that in seismically active areas these factors signalled that an earthquake was imminent. However, Italian seismologists did not agree with him, pointing out that the possibility of using increased radon emissions to predict earthquakes had been studied and shown not to work because many quakes are not preceded by radon increases and many radon increases are not followed by quakes. Giuliani tried to warn everyone by touring the town with a loudspeaker and putting a video on YouTube, but the mayor got the police to stop him spreading alarm. Despite the disaster, the mayor was right

to do what he did; it was just chance that Giuliani's prediction turned out true. (Cf. article in the *Guardian* by Ben Goldacre entitled 'Predictions are fine, but there are better ways to protect a population', 19 June 2010)

42 In the following passage you are given some information about a situation in which there is a dispute as to what happened. Using this information, write out a reasoned case for judging who was to blame. In your answer you should make clear what assumptions you are making about what the participants could have seen, what motives they might have for saying what they do, what expertise they have and any other relevant factors.

One of the employees at Dovetail Joinery has had an accident with a circular saw. Whilst pushing a large piece of wood through the saw, his hand was seriously injured by the blade. A dispute has now arisen over who was to blame for the accident.

The injured employee, Ashworth (A), claims that he followed all the company's safety procedures but that the owner of the company, Bell (B), had not ensured that the machine was as safe as possible. (B) is adamant that the machine was in a perfectly safe condition. If it had not been, he argues, he would have been told by the foreman (F).

(F) also insists that the machine was always maintained satisfactorily and, to emphasise his point, he has given the written maintenance record to the tribunal. In addition, (F) claims that just before the accident he saw (A) 'laughing and joking and messing about with his workmates'.

One of these workmates, Chandra (C), agrees with (A) that, despite its regular maintenance, the saw was not as safe as it should have been because its safety guard was poorly designed and did not function well. Furthermore, they had told (F) about it.

A health and safety inspector (I), who has inspected the machine, reports that the safety guard is poorly designed to protect operators in a number of circumstances which are familiar to operators of circular saws. (*OCR 1999, Paper 2*)

43 A classic paper by Lord, Ross and Lepper (of Stanford University), published in 1979, reports an experiment to test how ready we are to change our minds in the face of evidence. They took two groups of people, one in favour of the death penalty, the other against

it; they then presented everyone with two studies, one of which supported their pre-existing view whilst the other challenged it (for example, showing that murder rates went up or down after the abolition of capital punishment in a state). Each group found that their pre-existing beliefs were reinforced; in particular, they found methodological holes in the 'evidence' they disagreed with, but ignored the same holes in evidence that supported their views. More recent evidence suggests that some people go further when presented with scientific evidence which tells against their beliefs and insist that the subject (homosexuality or whatever) cannot be studied by scientific methods. (Cf. paper by Professor Geoffrey Munro in the *Journal of Applied Psychology*, 2010)

44 In the following passage you are given some information about a situation in which there is a dispute as to what happened. Using this information, write out a reasoned case for judging who was to blame. In your answer you should make clear what assumptions you are making about what the participants could have seen, what motives they might have for saying what they do, what expertise they have and any other relevant factors.

The appointment of a new ambassador from a South American country provoked considerable controversy because the ambassador had been accused by many human rights groups of having been personally involved in the torture of political opponents some years before. A demonstration against his appointment had been organised by the Anti-Fascist Alliance (AFA). However, this had provoked the group Rebirth-1933 (known otherwise as R33) to organise a counter-demonstration, welcoming the ambassador as 'a fighter against moral decline'.

The police had decided to let both demonstrations go ahead, but provided a very heavy presence in order to keep the two sides apart. Unfortunately, the number of demonstrators on both sides was greater than had been expected, and it became difficult to prevent violence breaking out. One of those hurt was the leader of the AFA, Fran Lee (F), who suffered serious head injuries, and had to be taken to the nearest hospital, where she remains in a coma.

The situation surrounding (F)'s injuries remains unclear. At the time that she was injured, she was being restrained by a policeman (P) who claims she was hit by a 'rock' which also

struck his helmet and which was thrown from a group of R33 demonstrators. Furthermore, (P) insists that he was one of a group of police who were trying to ensure that she did not get attacked by some R33 members who had got very close to her. On the other hand, a well-known lawyer and human-rights activist (A), who was with her at the time, claims that (F) was injured by the police, especially (P), using their batons with unreasonable force against her and other AFA members. A TV news crew (N), whose camera was damaged in the melee, said that they had filmed several police using their batons heavily on (F) and other AFA members just before she was grabbed by (P).

Though R33 leaders deny that she was hit by something thrown by one of their supporters, an R33 member (M), who refused to be identified, boasted in an interview to the BBC of having 'split Lee's head with a brick'. Another witness, a Dutch tourist (D), who had taken shelter from the violence in a doorway, says she certainly saw police using their batons strongly to try to separate people, but she did not see any missiles being thrown. A hospital spokesperson (H) said, 'Frances Lee sustained a fractured skull which appears to have been caused by at least one very severe blow to the head.' (*OCR 2000, Paper 2*)

45 The following is a difficult piece of reasoning about credibility considerations – and miracles:

. . . there is no species of reasoning more common, more useful, and even necessary to human life, than that which is derived from the testimony of men, and the reports of eye-witnesses and spectators. . . . [But]

Suppose . . . that the fact, which the testimony endeavours to establish, partakes of the extraordinary and the marvellous; in that case, the evidence, resulting from the testimony, admits of a diminution, greater or less, in proportion as the fact is more or less unusual. . . .

[Now] suppose that the fact which [witnesses] affirm, instead of being only marvellous, is really miraculous; . . .

Nothing is esteemed a miracle, if it ever happen in the common course of nature. It is no miracle that a man, seemingly in good health, should die on a sudden: because such a kind of death, though more unusual than any other, yet has been frequently observed to

happen. But it is a miracle, that a dead man should come to life; because that has never been observed in any age or country. . . .

The plain consequence is . . ., 'That no testimony is sufficient to establish a miracle, unless the testimony be of such a kind, that its falsehood would be more miraculous, than the fact, which it endeavours to establish.' . . . When anyone tells me, that he saw a dead man restored to life, I immediately consider with myself, whether it be more probable, that this person should either deceive or be deceived, or that the fact, which he relates, should really have happened. I weigh the one miracle against the other; and according to the superiority, which I discover, I pronounce my decision, and always reject the greater miracle. If the falsehood of his testimony would be the more miraculous, than the event which he relates; then, and not till then, can he pretend to command my belief or opinion. (David Hume, *Enquiries Concerning Human Understanding*, section X, part 1, para. 91)

46 Neuroscientists from Cambridge University and colleagues at the US National Institute on Ageing in Maryland claim to have shown that running stimulates the brain to grow thousands of new cells and this has a big effect on memory and the ability to learn. They studied two groups of mice, one which ran a lot and one which didn't. After plenty of running, the 'runners' scored nearly twice as well on memory tests as the non-runners, were better at distinguishing similar memories and learned more quickly when there were changes to their environment. Brain tissue taken from the runners showed an average of 6,000 new brain cells in every cubic centimetre of the dentate gyrus – part of the hippocampus. The author of the study was Timothy Bussey, a behavioural neuroscientist at Cambridge University. (Cf. the *Guardian*, 19 January 2010, p. 3)

47 A study carried out by Sheffield University and the Scottish Crop Research Institute, and published in the journal *Food Chemistry*, suggests that eating rhubarb crumble may help fight cancer. The researchers found that rhubarb, like many red vegetables, contains polyphenols – chemicals which kill or prevent the growth of cancer cells – and baking rhubarb for 20 minutes in a crumble greatly increases the chemicals' concentration. It is hoped that extracting the chemicals could produce new anti-cancer

drugs which are less toxic than those in use at present. (Cf. *Daily Telegraph*, 12 February 2010)

48 A report in the respected medical journal the *Lancet* showed as far back as 1972 that 100 doctors' stethoscopes collected from various departments in a London teaching hospital *all* carried infectious bacteria. The authors, Gerken et al., also reported that the risk of infection could easily be eliminated by wiping the bell and diaphragm of a stethoscope with disinfectant after use on each patient. More than two decades later, a paper in the *British Medical Journal* reported that of 29 doctors questioned, only three had ever cleaned their stethoscopes – two of them intermittently and one only once! Yet again, in December 2008, the *Journal of Infection* contained a report from researchers at Sandwell General Hospital, West Bromwich, England, which found that of 40 stethoscopes sampled, 37 harboured bugs. What is likely to be the case in 10 or 20 years' time? (Cf. *Daily Telegraph*, 15 December 2008)

49 Tim Entwisle, chief of Sydney's Royal Botanic Gardens, believes the European system of four seasons is wrong for Australia. He wants to introduce 'sprummer' between spring and summer and 'sprinter' between winter and spring. He says that having four three-month seasons doesn't make any sense in Australia, that Australian aborigines use up to eight seasons in different parts of the country and that what is needed are seasons which reflect what's going on around us – so different regions of the country might have different numbers of seasons. (Cf. *Daily Telegraph*, 24 August 2009)

50 I dare the strongest supporter of King George to show a single thing the colonies can gain by being connected with Great Britain. We do not need to trade with Britain alone. We can sell our corn in any Market in Europe.

The harm we suffer by our ties to Great Britain are without number. Thus, our duty to the world as well as to ourselves tells us to break this tie. Now our link with Britain draws us into European wars and quarrels. It makes enemies of countries who might otherwise seek our friendship. And whenever a war breaks out between Britain and another country, the trade of America will be ruined.

Everything that is right or natural begs for separation. The blood of the slain, the weeping voice of nature cries, "Tis time to

part.' Even the distance at which God has placed England and America is proof that the rule of one over the other was never the plan of heaven. (Thomas Paine, *Common Sense*, 1776)

51 Suppose (as Aristotle believed) that the heavier a body is, the faster it falls to the ground and suppose that we have two bodies, a heavy one called M and a light one called m. Under our initial assumption M will fall faster than m. Now suppose that M and m are joined together, thus $M + m$. Now what happens? Well, $M + m$ is heavier than M so by our initial assumption it should fall *faster* than M alone. But in the joined body $M + m$, m and M will each tend to fall just as fast as before they were joined, so m will act as a 'brake' on M and $M + m$ will fall *slower* than M alone. Hence it follows from our initial assumption that $M + m$ will fall both *faster* and *slower* than M alone. Since this is absurd our initial assumption must be false. (From Galileo's *Dialogues Concerning Two New Sciences*, 1638)

52 James Lovelock, author of the Gaia hypothesis (that the Earth is a self-regulating biosphere geared to preserving life) has recently argued that the way to power economies without the damage being done to the biosphere by polluting greenhouse gases is to adopt nuclear power: 'The real dangers to humanity and the ecosystems of the earth from nuclear power are almost negligible. You get things like Chernobyl but what happens? Thirty-odd brave firemen died who needn't have died but its general effect on the world population is almost negligible.

'What has it done to wildlife? All around Chernobyl, where people are not allowed to go because the ground is too radioactive, well, the wildlife doesn't care about radiation. It has come flooding in. It is one of the richest ecosystems in the region. And then they say: what shall we do with nuclear waste?' Lovelock has an answer for that, too. Stick it in some precious wilderness, he says. If you wanted to preserve the biodiversity of the rainforest, drop pockets of nuclear waste deep into it to keep the developers out. The lifespans of the wild things might be shortened a bit, but the animals wouldn't know, or care. Natural selection would take care of the mutations. Life would go on.

'I have told [British Nuclear Fuels] that I would happily take the full output of one of their big power stations. I think the

high-level waste is a stainless steel cube of about a metre in size and I would be very happy to have a concrete pit . . . that they would put it in.' He says he would use the waste for two purposes. 'One would be home heating. You would get free home heat from it. And the other would be to sterilise the stuff from the supermarket, the chicken and whatnot, full of salmonella. . . . Just drop it down through a hole. [And] they would be welcome to take pictures of my grandchildren sitting on top of it.' (Cf. the *Guardian: Saturday Review*, 16 September 2000, p. 1)

53 The highly respected *British Medical Journal* (BMJ) recently reported that acupuncture could increase IVF (in vitro fertilisation) success rates by 65 per cent, based on analysis of seven separate trials involving 1,366 women. . . .

So how accurate were the trials analysed in the BMJ? The problem is that four out of the seven trials did not include a 'sham' acupuncture group [a 'control' group who thought they were having acupuncture but were not really doing so] but merely compared the effect of acupuncture with no acupuncture at all; any benefit could be due to the placebo effect and therefore these trials should be ignored. When focusing on the remaining three trials which had included such a sham group, the results are less than impressive. Two out of the three failed to show that real acupuncture offers any significant benefit (in terms of the likelihood of pregnancy) beyond the fake treatment.

The sensible conclusion is that acupuncture is still unproven in terms of increasing IVF success rates. So it is worth avoiding acupuncture in the context of IVF, since 10 per cent of patients complain of pain, bleeding or bruising, and some even experience fainting, dizziness, nausea or vomiting. These adverse effects are not serious, but the known risks outweigh the unproven benefits. (Cf. article by Simon Singh in the *Daily Telegraph*, 21 April 2008)

54 [I have engaged in practical] research, covering some thirty years, into Leonardo da Vinci's painting methods. By successfully attempting to recreate the distinctive effects that Leonardo achieved in the *Mona Lisa*, I have come to realise that when depicting flesh areas, he departed from the specific rules then laid down for producing resistant oil paint films. In these zones, his method was more akin to the techniques of the watercolourist

than to those of the painter in oils. It consisted of a very gradual, deliberate build-up of successive layers of ultra-thin, overdiluted glazes. (I call this technique 'micro-divisionism'.)

For this reason, the glazes in areas of paint where Leonardo was depicting flesh are extremely thin and, in all probability, porous, friable and powdery. Therefore, the varnish lying on these areas acts as a useful reinforcement. Any attempt to remove this integrated varnish by solvents and scalpels could seriously endanger Leonardo's own work. Since the last stages of artistic 'finessing' form the upper layer, the most crucial effects are also the most vulnerable ones.

In 1994, the Louvre in Paris considered cleaning Leonardo's *The Virgin and Child with St Anne.* As a consultant to the museum, I was able to point out in advance these physical and aesthetic vulnerabilities. My testimony was listened to and accepted by the Louvre's committee for restoration. Happily, the project was dropped, but other experts challenged my views – in particular David Bull, head of painting conservation at the National Gallery of Art in Washington.

Although Bull agreed that the surfaces produced by Leonardo are 'paper-thin' and finely manipulated, he doubted that they would be fragile structurally. He recommended that the varnish be removed. Bull contends that Leonardo achieved his impalpably vaporous effects not by delicate brush-work, but much more robustly by spreading paint with his fingertips and, even, with the palm of his hand. I reject this hypothesis. It might, at first glance, seem to rest on hard physical evidence – namely the traces of fingerprints found on certain of Leonardo's paintings. In reality, it reveals a misreading of evidence and a misconstruing of technical practice.

I am familiar with the method of work that Bull suggests, and in my attempts to 'reconstruct' Leonardo's art I have experimented with it over several years. It is a process which I abandoned when I realised that no matter how skilled or practised the hand, the master's famous sfumato effects, characterised by an extremely gradual transition between areas of different colour or shade, could not be so obtained. It might be argued that the failure was my own. But, having tried to create the effects with both brush and hand, and comparing them at length, I can justifiably claim sufficient expertise on the issue.

Those who continue to champion 'fingerwork' have yet to demonstrate the practice. My 'practical' position is widely supported by historical evidence and scientific pointers, but nowhere in Leonardo's extensive theoretical writings on painting does he mention anything like the method which Bull adduced.

In fact, the fingerprints are only encountered in unfinished sections – which is to say those that have not received the delicate multilayered treatment described above. The 'fingerwork', self-evidently, served as a quick method of laying down an intermediate stage, and not as a final effect. The more fragile of these two distinct types of work is the more complex and subtle one (that is to say, 'micro-divisionism'), which means that restoration must be equally delicate and cautious. (From 'Let's keep the Mona Lisa smiling', *New Scientist*, 11 May 1996, p. 48 by Jacques Franck, an art historian specialising in the techniques of old masters, in Sucy-en-Brie, France)

55 It is important to note, however, that thinking is always thinking *about* something. To think about nothing is a conceptual impossibility. The importance of this simple point is that it raises serious questions about the meaning of such commonly heard claims as 'I teach thinking', or 'I teach students to think'. One may well ask 'About what?' Nor would the claim that one taught 'thinking in general' or 'thinking about everything' be any more helpful. For to think about nothing in particular is equivalent to not thinking at all. And to think of 'everything in general' is incoherent.

. . . it is a matter of conceptual truth that thinking is always *thinking about* X, and that X can never be 'everything in general' but must always be something in particular. Thus the claim 'I teach my students to think' is at worst false and at best misleading.

Thinking, then, is logically connected to an X. Since this fundamental point is reasonably easy to grasp, it is surprising that critical thinking should have become reified into a curriculum subject and the teaching of it an area of expertise on its own. . . .

In isolation from a particular subject, the phrase 'critical thinking' neither refers to nor denotes any particular skill. It follows from this that it makes no sense to talk about critical thinking as a distinct subject and that it therefore cannot be profitably taught as such. To the extent that critical thinking is not

about a specific subject X, it is both conceptually and practically empty. The statement 'I teach critical thinking', *simpliciter*, is vacuous because there is no generalised skill properly called critical thinking. (McPeck, 1981, pp. 3, 4)

56 **Pascal's Wager**

Either there is a Christian God or there isn't. If you believe in Him and live a Christian life, then if He exists you will enjoy eternal bliss and if He doesn't exist you will lose very little. On the other hand, if you don't believe in Him and don't live a Christian life, then if He doesn't exist you will lose nothing, but if He does exist, you will suffer eternal damnation! So it is rational to believe in God's existence and live a Christian life.

57 Richard Dawkins, 'The more you understand evolution, the more you move towards atheism'. (This is an edited version of Dawkins' speech at the Edinburgh International Science Festival on 15 April 1992, reprinted from the *Independent* with his permission.)

As a Darwinian, something strikes me when I look at religion. Religion shows a pattern of heredity which I think is similar to genetic heredity. The vast majority of people have an allegiance to one particular religion. There are hundreds of different religious sects, and every religious person is loyal to just one of these.

Out of all the sects in the world, we notice an uncanny coincidence: the overwhelming majority just happen to choose the one their parents belonged to. Not the sect that has the best evidence in its favour, the best miracles, the best moral code, the best cathedral, the best stained-glass, the best music: when it comes to choosing from the smorgasbord of available religions, their potential virtues seem to count for nothing compared to the matter of heredity.

This is an unmistakeable fact; nobody could seriously deny it. Yet people with full knowledge of the arbitrary nature of this heredity somehow manage to go on believing in their religion, often with such fanaticism that they are prepared to murder people who follow a different one.

Truths about the cosmos are true all around the universe. They don't differ in Pakistan, Afghanistan, Poland or Norway. Yet we are apparently prepared to accept that the religion we adopt is a matter of an accident of geography.

If you ask people why they are convinced of the truth of their religion, they don't appeal to heredity. Put like that it sounds too obviously stupid. Nor do they appeal to evidence. There isn't any, and nowadays the better educated admit it. No, they appeal to faith. Faith is the great cop-out, the great excuse to evade the need to think and evaluate evidence. Faith is belief in spite of, even perhaps because of, the lack of evidence. The worst thing is that the rest of us are supposed to respect it: to treat it with kid gloves.

If a slaughterman doesn't comply with the law in respect of cruelty to animals, he is rightly prosecuted and punished. But if he complains that his cruel practices are necessitated by religious faith, we back off apologetically and allow him to get on with it. Any other position that someone takes up can expect to be defended with reasoned argument. But faith is immune. Faith is allowed not to justify itself by argument. Faith must be respected: and if you don't respect it, you are accused of violating basic human rights.

Even those with no faith have been brainwashed into respecting the faith of others. When so-called Muslim community leaders go on the radio and advocate the killing of Salman Rushdie, they are clearly committing incitement to murder – a crime for which they would ordinarily be prosecuted and possibly imprisoned. But are they arrested? They are not, because our secular society respects their faith, and sympathizes with the deep hurt and insult to it.

Well I don't. I will respect your views if you can justify them. But if you justify your views only by saying you have faith in them, I shall not respect them.

I want to end by returning to science. It is often said . . . that although there is no positive evidence for the existence of a God, nor is there evidence against His existence. So it is best to keep an open mind and be agnostic.

At first sight that seems an unassailable position, at least in the weak sense of Pascal's wager. But on second thoughts it seems a cop-out, because the same could be said of Father Christmas and tooth fairies. There may be fairies at the bottom of the garden. There is no evidence of it, but you can't *prove* that there aren't any, so shouldn't we be agnostic with respect to fairies?

The trouble with the agnostic argument is that it can be applied to anything. There is an infinite number of hypothetical beliefs we could hold which we can't positively disprove. On the whole,

people don't believe in most of them, such as fairies, unicorns, dragons, Father Christmas, and so on. But on the whole they do believe in a creator God, together with whatever particular baggage goes with the religion of their parents.

I suspect the reason is that most people . . . nevertheless have a residue of feeling that Darwinian evolution isn't quite big enough to explain everything about life. All I can say as a biologist is that the feeling disappears progressively the more you read about and study what is known about life and evolution.

I want to add one thing more. The more you understand the significance of evolution, the more you are pushed away from the agnostic position and towards atheism. Complex, statistically improbable things are by their nature more difficult to explain than simple, statistically probable things.

The great beauty of Darwin's theory of evolution is that it explains how complex, difficult to understand things could have arisen step by plausible step, from simple, easy to understand beginnings. We start our explanation from almost infinitely simple beginnings: pure hydrogen and a huge amount of energy. Our scientific, Darwinian explanations carry us through a series of well-understood gradual steps to all the spectacular beauty and complexity of life.

The alternative hypothesis, that it was all started by a supernatural creator, is not only superfluous; it is also highly improbable. It falls foul of the very argument that was originally put forward in its favour. This is because any god worthy of the name must have been a being of colossal intelligence, a supermind, an entity of enormous sophistication and complexity. In other words, an entity of extremely low statistical probability – a very improbable being.

Even if the postulation of such an entity explained anything (and we don't need it to), it still wouldn't help because it raises a bigger mystery than it solves.

Science offers us an explanation of how complexity (the difficult) arose out of simplicity (the easy). The hypothesis of God offers no worthwhile explanation for anything, for it simply postulates the difficult to explain and leaves it at that. We cannot prove that there is no God, but we can safely conclude that He is very, very improbable indeed.

58 Mine's a placebo and tonic

Could I really get the same lovely feeling from drinking alcohol-free vodka as from drinking the real thing? About half of subjects given what they were told was an alcoholic drink reported feeling slightly drunk. The placebo effect is a very powerful one and the history of medical treatment is essentially the history of the placebo effect.

Ancient remedies used in China involved over 2,000 drugs and some 16,000 prescriptions, and Galenic medicine, from Greek till quite recent times, prescribed some 800 remedies. It is extremely unlikely that these had the effect claimed for them and the Graeco-Roman physician Claudius Galen himself showed significant insight when he observed that the doctor who cures most successfully is the one in whom people have the most confidence.

The most reliable way to assess any medical treatment is by random clinical trials, in which treatment is given to some patients and not to others. The patients are assigned to one or other treatment randomly, and neither the patients nor the doctors should know who has been given which treatment. This anonymity is essential since if either group knows what is going on, this can significantly affect the outcome in all sorts of subtle ways.

Such trials are a relatively recent procedure, only having started in this country in the 1940s. And trials of any sort to find out if a medical treatment works were virtually unknown until the pioneering work of Pierre Louis in Paris in the 1830s, when he showed, at last, that bloodletting, based on the ancient belief that too much of that humour was the cause of the illness, did patients more harm than good.

It is such trials that reveal the placebo effect. In many trials, patients are given a completely neutral treatment such as a solution of sugar and, typically, this placebo treatment works for about a quarter to half of the patients. This effect is particularly strong with treatment of a mental illness like depression where there is substantial evidence that about a third of patients in trials of antidepressants improve due to a placebo treatment. Confidence in one's doctor can itself have a positive effect and a diagnosis alone can act as a placebo, which is better than being told that the illness is not understood.

Just how effective placebos are with physical illnesses like cancer or heart disease is less clear; but they can work with pain.

Patients told that ultra-sound would prevent pain when their teeth were being extracted did better even when the machine was turned off. Probably the very presence of the machine was important, for the placebo effect is intimately associated with desires and expectations of patients.

Further support for this is that a placebo given by injection is more effective than, for example, pills. Again, if patients are first given a painkiller and then given a placebo, the effect of the placebo is greater because of the earlier effect since it increased the belief that the treatment would work. However, this may reflect a type of learning and conditioning. Evidence for this comes from animal studies. Rats fed saccharin with an immuno-suppressive drug continued to respond to saccharin alone, with their immune systems continuing to be suppressed.

The other, and unpleasant, side to the placebo effect is termed the nocebo. The nocebo involves getting ill or having symptoms because of the expectation that this will happen. For example, 80% of patients given sugar as an emetic vomited, and asthmatics have had an attack caused by a neutral inhalant they were told would cause one, and been cured when told it would help.

'Medical students' disease' is well known – students get the symptoms of the disease they are studying. Also, depressed patients have a greater probability of heart disease because, perhaps, of their negative expectations of their health.

The nocebo effect might also explain 'voodoo death', due to a spell being put on a victim by someone in whose powers they have strong beliefs. Could this be what hypochondriacs like myself do to themselves? Would that I could believe that malt whisky will cure me. (The *Independent*, 12 January 2001; used with permission of the author, Professor Lewis Wolpert, Professor of Biology as Applied to Medicine at University College London)

59 The following passage was written by Sydney Smith in 1824 as a kind of summary review of Jeremy Bentham's *The Handbook of Political Fallacies*. The text is an imaginary speech by a Member of Parliament (the Noodle) which commits a large number of the fallacies Bentham discussed in his book. The exercise for you is to find all the mistaken lines of argument you can. This will require some interpretation of the text because its language is rather antiquated, but express the fallacies in modern language and you will then see how common they are today!

The Noodle's Oration

What would our ancestors say to this, Sir? How does this measure tally with their institutions? How does it agree with their experience? Are we to put the wisdom of yesterday in competition with the wisdom of centuries? (*Hear, hear.*) Is beardless youth to show no respect for the decisions of mature age? (Loud cries of *Hear! hear!*) If this measure is right, would it have escaped the wisdom of those Saxon progenitors to whom we are indebted for so many of our best political institutions? Would the Dane have passed it over? Would the Norman have rejected it? Would such a notable discovery have been reserved for these modern and degenerate times?

Besides, Sir, if the measure is good, I ask the honourable gentleman if this is the time for carrying it into execution – whether, in fact, a more unfortunate period could have been selected than that which he has chosen? If this were an ordinary measure, I should not oppose it with so much vehemence; but, Sir, it calls in question the wisdom of an irrevocable law – of a law passed at the memorable period of the Revolution. What right have we, Sir, to break down this firm column, on which the great men of that day stamped a character of eternity? Are not all authorities against this measure: Pitt, Fox, Cicero, and the Attorney- and Solicitor-General?

The proposition is new, Sir; it is the first time it was ever heard in this house. I am not prepared, Sir – this house is not prepared, to receive it. The measure implies a distrust of His Majesty's government; their disapproval is sufficient to warrant opposition. Precaution only is requisite where danger is apprehended. Here the high character of the individuals in question is a sufficient guarantee against any ground of alarm.

Give not then your sanction to this measure; for, whatever be its character, if you do give your sanction to it, the same man by whom this is proposed, will propose to you others to which it will be impossible to give your consent. I care very little, Sir, for the ostensible measure; but what is there behind it? What are the honourable gentleman's future schemes? If we pass this bill, what fresh concessions may he not require? What further degradation is he planning for his country?

Talk of evil and inconvenience, Sir! Look to other countries – study other aggregations and societies of men, and then see whether the laws of this country demand a remedy, or deserve

a panegyric. Was the honourable gentleman (let me ask him) always of this way of thinking? Do I not remember when he was the advocate in this house of very opposite opinions? I not only quarrel with his present sentiments, Sir, but I declare very frankly, I do not like the party with which he acts. If his own motives were as pure as possible, they cannot but suffer contamination from those with whom he is politically associated. This measure may be a boon to the constitution, but I will accept no favour to the constitution from such hands. (Loud cries of *Hear! hear!*)

I profess myself, Sir, an honest and upright member of the British Parliament, and I am not afraid to profess myself an enemy to all change, and all innovation, I am satisfied with things as they are; and it will be my pride and pleasure to hand down this country to my children as I received it from those who preceded us.

The honourable gentleman pretends to justify the severity with which he has attacked the noble Lord who presides in the Court of Chancery. But I say such attacks are pregnant with mischief to government itself. Oppose ministers, you oppose government; disgrace ministers, you disgrace government; bring ministers into contempt, you bring government into contempt; and anarchy and civil war are the consequences.

Besides, Sir, the measure is unnecessary. Nobody complains of disorder in that shape in which it is the aim of your measure to propose a remedy to it. The business is one of the greatest importance; there is need of the greatest caution and circumspection. Do not let us be precipitate, Sir; it is impossible to foresee all the consequences. Everything should be gradual; the example of a neighbouring nation should fill us with alarm!

The honourable gentleman has taxed me with illiberality, Sir. I defy the charge. I hate innovation, but I love improvement. I am an enemy to the corruption of government, but I defend its influence. I dread reform, but I dread it only when it is intemperate. I consider the liberty of the press as the great palladium of the constitution; but, at the same time, I hold the licentiousness of the press in the greatest abhorrence. Nobody is more conscious than I am of the splendid abilities of the honourable mover, but I tell him at once, his scheme is too good to be practicable. It savours of utopia. It looks well in theory, but it won't do in practice. It will not do, I repeat, Sir, in practice; and so the advocates of the

measure will find, if unfortunately, it should find its way through Parliament. (*Cheers.*)

The source of that corruption to which the honourable member alludes is in the minds of the people; so rank and extensive is that corruption, Sir, that no political reform can have any effect in removing it. Instead of reforming others, instead of reforming the state, the constitution, and everything that is most excellent, let each man reform himself! Let him look at home, he will find there enough to do, without looking abroad, and aiming at what is out of his power: (*Loud cheers.*) And now, Sir, as it is frequently the custom in this house to end with a quotation, and since the gentleman who preceded me in the debate has anticipated me in my favourite quotation of the 'Strong pull and the long pull', I shall end with the memorable words of the assembled barons – '*Nolumus leges Angliae muturi.*'

Answers to questions

Chapter 1

1.1 This is entirely your own answer.

1.2 On Dewey's account, critical thinking requires 'active, persistent, and careful consideration' (etc.) and it is clear this passage displays these qualities, whether you agree with its claims or not.

1.3 Your new definition should pick up on all or most of the points made so far, that critical thinking (i) is 'active' (you construct your own answers, you do not just accept what others tell you), (ii) is 'persistent' (you weigh alternatives and take time to consider the issues, you do not just decide quickly without thinking), (iii) centrally involves giving and evaluating reasons, (iv) concerns both what we believe and what we do, and (v) involves dispositions as well as skills.

1.4 'Great. Now we'll play basketball again, but this time, guard your opponents well in the way we just practised – and when you get a chance to pass, try to do it in the way we just practised – and if you get a chance to shoot, don't forget what we practised there too.'

1.5.1 You reflect on your current way of doing something; watch while a good model is shown; then practise doing it yourself, trying to do it in the way shown in the good model. The fourth stage is when you deploy the skill in real situations, monitoring what you are doing and trying to do it well.

1.5.2 This is for you to say.

1.6.1 This depends roughly on whether you have to work things out or just let it 'wash over you'.

1.6.2 Even if you are reasoning your way through to a conclusion, if this is a 'mechanical' process requiring little judgement, interpretation and so on, it involves very little critical thinking.

1.6.3 This depends on how much 'figuring out' he is doing, trying to work out what his opponents will do and how to outwit them; this could involve a lot of critical thinking.

1.6.4 Provided you do not just make a snap decision, but investigate the alternatives, get the information you need to judge which will suit you best and so on, this could involve a great deal of critical thinking (see chapter 11).

1.6.5 If you have to 'figure things out' from the guidance you are given, this might involve a lot of critical thinking, but if you are just blindly following the instructions, it does not.

1.7.1 The differences between the two imagined cases are that in one case Andy questions evidence from a source which has a vested interest, looks for relevant evidence from independent sources, and weighs the pros and cons reasonably skilfully (allowing for the costs of doing this and the time available); in case 2 Andy displays 'reflective thinking' before buying his car, but not in case 1.

1.7.2 In case 1 Andy does none of these things and in case 2, arguably, he does all of them.

1.7.3 Only in case 2.

1.8.1 In this case, the issue is how to interpret a TV news report (about the reported accuracy of certain US weapons). On the face of it, Bertha is simply swallowing what is presented to her, and does not even want to question it when her friend raises doubts. The question then is whether Cheryl has reasonable grounds for being sceptical and the evidence is that she has if what she says is true. In this case, Bertha is not thinking critically, but Cheryl is.

1.8.2 Broadly 'yes' for Cheryl, but broadly 'no' for Bertha.

1.8.3 Same as 1.8.2.

1.9.1 See chapter 12 'How to get reliable information from the internet'.

1.9.2 None.

1.9.3 Left to you.

1.9.4 This could require very little critical thinking, although if you found an error in the recipe or if you decided to adapt the recipe, this might require quite a bit of 'careful consideration' and reasoning.

Chapter 2

2.1.1 This is simply a descriptive passage. It does not give reasons for a conclusion (though we naturally make various inferences as we read it).

2.1.2 This passage does give reasons for a conclusion; it gives reasons for thinking we may have to revise completely our understanding of global warming.

2.1.3 This does not give reasons for a conclusion. It describes a possible 'solution' to a problem but there is no reasoning.

2.1.4 This passage gives reasons for the conclusion that certain problems can only be addressed by international action.

2.1.5 This passage is largely just descriptive though it does report Gall's basic reason for his basic conclusion.

2.1.6 This passage gives reasons for the conclusion that the only way to deliver 'thinking schools' is to assess thinking skills and dispositions directly.

2.1.7 This passage does not contain reasoning to a conclusion. As the sketch goes on to say, this exchange is not an argument but is simply abuse!

2.2 Answered in the text immediately following the question.

2.3 Answered in the text immediately following the question.

2.4 Answered in section 2.4, but give your own answer before reading that answer.

2.5 We mark the argument indicators in bold below. To show which parts are reasons and which are conclusions, let us use the notation R1<. . .>, R2<. . .>, and C1[. . .], C2[. . .], etc. (and notice that a sentence can be both a reason and a conclusion). Given this, a natural way (though not the only possible way) to construe the reasoning presented for the various conclusions is as follows:

2.5.1 R1<During the football game he committed a serious foul>, **so** C1[he deserved to be sent off].

2.5.2 R1<Women's brains are on average smaller than men's>, **therefore** C1[women are less intelligent than men].

2.5.3 R1<The butler was in the pantry>. R2<In that case he couldn't have shot the master, who was in his study>. **Hence** C1[the butler couldn't have done it]!

2.5.4 C1[The sovereignty of Parliament is open to abuse by any government] **as** R1<power in Britain is too centralised>.

2.5.5 C1[The Green movement is mistaken in thinking we should recycle materials like paper and glass] **because** R1<paper comes from trees, an easily renewable resource, and glass is made from sand, which is plentiful and cheap>. Furthermore, R2<in some American cities recycling schemes have been abandoned because they are too expensive>.

Note that R2 is an explanation (see section 3.7) and it is plausible to say that both R1 and R2 are being presented as reasons for C1.

2.5.6 In 2000 Dr Eleanor McGuire published a groundbreaking study of the brains of London taxi drivers. R1<Brain scans showed that they had a much larger hippocampus (the part of the brain used for navigation) than most people>. R2<More importantly, it grew larger the longer they spent doing their job>. **So** C1[it is a mistake to think that we continually lose brain cells from an early age].

2.5.7 R1<Some people succeed against all the odds. Even though they are not born to wealth or don't have parents who can help them achieve their ambitions, they still make it>. **This just shows** C1[what determination and hard work can achieve], **so** C2[if you really want something, go for it].

2.5.8 R1<There has been a decline in the rate of many illnesses of old age. For example, arthritis, dementia, and strokes are all declining year by year>. R2<The causes of this decline include such medical advances as beta-blockers to control high blood pressure and the fitting of hip replacements>. R3<There is, however, another factor. The present generation of 60- and 70-year-olds had much better nutrition as children than did their parents>. R4<Good nutrition in childhood is important in laying the foundations of good health in adulthood>. **Since** R5<improvements in nutrition have continued over the past 60 years>, C1[we can expect that many of the illnesses of old age will continue to decline].

2.6 Again, using the notation introduced in the answer to question 2.5, a natural way (though not the only possible way) to construe the reasoning presented for the various conclusions is as follows:

(a) R1<Most parents want their children to have successful careers> and R2<education is essential to success>, **so** C1[it is the duty of parents to give children the best possible education].

(b) R3<It is the duty of parents to give children the best possible education> and R4<it is also in the country's economic interest to have a highly educated population>, **so** C2[the government should help parents to provide for their children's education].

(c) R5<The government should help parents to provide for their children's education> **therefore** C3[all parents should receive financial help towards the cost of their children's education].

(d) R6<All parents should receive financial help towards the cost of their children's education>, **so** C4[the low paid should receive tax credits and those who are better off should receive tax relief].

2.7.1 R1<The traditional British approach to food safety has been one where local health officials only intervene at the level of food retailing, for instance inspecting premises where food is prepared or sold>. However, R2<many of the dangers to our health resulting from the food we eat arise from the way it is produced in the first place, that is, the modern intensive farming practices involved, rather than small-scale organic farming>. **Therefore** C1[a national food safety agency that fails to address the question of food production will be unlikely to protect us effectively from damaging our health through the food we eat] **so** C2[a much broader approach to the question is needed].

2.7.2 R1<Many species are in danger of becoming extinct from the destruction of their habitats>. R2<Sometimes this is caused directly by human activity, as when forests are cleared for agriculture>, but R3<global warming is also having a devastating effect on many habitats>. R4<Of course, human activity is the main cause of this too> **therefore** C1[human beings are the main threat to the existence of many species].

2.7.3 R1<If a government bans cigarette advertising within its territory then cigarette manufacturers save their advertising budgets in that country> **so** C1[in order to compete with each other, they are then likely to lower their prices]. **Therefore** C2[banning cigarette advertising could easily lead to an increase in smoking].

2.7.4 R1<The famous frescoes on the ceiling of the Sistine Chapel were painted by Michelangelo between 1508 and 1512. Every time there was a Mass, the candle flames sent up wax and soot; after some years, varnish was applied which gradually became yellow; the open windows let in the grime of the city and more recently, the exhaust fumes of many vehicles> **therefore** C1[in the following 500 years they must have lost their original colours]. C2[The storm of controversy which was provoked by the cleaning and restoration work undertaken by the Vatican in the 1980s and 1990s was completely misplaced] **because** R2<the work was done by some of the world's finest art restorers and was closely supervised by an international team of art experts and historians> and R3<people had become accustomed to seeing the frescoes with their colours dulled by soot, varnish and general grime and just didn't realise how much brighter and more vivid the originals were>.

2.7.5 R1<245 million years ago, 90 per cent of all species vanished; 65 million years ago, 50 per cent of all living things died out, including the dinosaurs>. R2<Much more recently, the influence of humans has led to many species becoming extinct: in Hawaii, for example, this influence led to huge losses of species of plants, insects, and animals>. R3<But in each example of extinction we have seen the lost species replaced by new ones (for instance, mammals replaced dinosaurs)>. **Therefore** C1[the history of the world is full of many examples of the extinction of species, and we should not regard such extinction as a serious environmental problem].

2.8.1 'do not demonstrate' is a strong claim that the evidence does *not* support the widely accepted conclusion. 'So' is a straightforward conclusion indicator.

2.8.2 'Since' and 'so' are straightforward argument indicators; 'suggests' is tentative.

2.8.3 'This is an unmistakeable fact; nobody could seriously deny it' shows that Dawkins is very confident of his claim.

2.8.4 Numerous argument indicator words, 'because', 'so', etc., are used straightforwardly. 'At first sight that seems an unassailable position . . . But on second thoughts it seems a cop-out' shows a considerable measure of confidence in his position.

2.9 These are for you to do.

Chapter 3

3.1.1 Questions appendix, passage 3 is side-by-side reasoning.

3.1.2 Questions appendix, passage 11 is side-by-side reasoning.

3.1.3 Questions appendix, passage 17 has a chain structure.

3.1.4 Questions appendix, passage 29 is side-by-side reasoning.

3.1.5 This argument has a chain structure.

3.2.1 Questions appendix, passage 21 is side-by-side reasoning.

3.2.2 Questions appendix, passage 28 has a chain structure.

3.2.3 Questions appendix, passage 36 is side-by-side reasoning.

3.2.4 Questions appendix, passage 38 is complex, but is mainly side-by-side reasoning to the conclusion that the two lines of argument show 'a great difference between relative and absolute motion'.

3.3.1 Joint.

3.3.2 Not joint.

3.3.3 Joint.

3.4.1 (a) R1<Recent research suggests that our understanding of how clouds interact with sunlight might be wrong: new measurements suggest that clouds absorb four times as much energy as previously thought> and R2<existing models of how the climate functions are based on the original measurements>, **therefore** C1[if the new measurements are shown to be accurate, models of how the climate works will need to be completely overhauled].

(b) C1 = R3<if the new measurements are shown to be accurate, models of how the climate works will need to be completely overhauled> and R4<climate models are used in our attempts to measure global warming> **so**, C2[if these

climate models are shown to be inaccurate, we will have to completely revise our understanding of global warming].

3.4.2 (a) C1[According to the World Health Organization, thousands of people around the world die prematurely from heart disease triggered by long-term exposure to excessive noise] **because** R1<noise pollution can cause disturbed sleep and stress> – and R2<high levels of stress can increase the levels of stress hormones such as cortisol, adrenaline and noradrenalin in the body, even during sleep> and R3<the longer these hormones stay in circulation around the blood stream, the more likely they are to cause high blood pressure, strokes and heart failure>.

3.4.3 (a) R1<If Neil Armstrong didn't really walk on the Moon in 1969 there must have been the most extraordinary conspiracy, involving thousands of people> and R2<such a conspiracy could not possibly have happened>. **So** C2[Neil Armstrong did walk on the Moon in 1969].

(b) R2 = C1[Such a conspiracy could not possibly have happened] **because** R3<someone would have spilled the beans by now>.

3.4.4 (a) R1<When prisoners under sentence of death are given the choice between life in prison and execution, 99 per cent of them choose life imprisonment>. **This shows that** C1[they fear death more than they fear life imprisonment].

(b) C1 = R2<{prisoners under sentence of death} fear death more than they fear life imprisonment> and R3<one is most deterred by what one most fears>, **therefore** C2[it is evident that the threat of the death penalty is more likely to deter most potential murderers than is the threat of life imprisonment].

3.5.1 (a) R1<Radioactive elements disintegrate and eventually turn into lead> **therefore** C1[if matter has always existed, there should be no radioactive elements left].

(b) C1 = R2<If matter has always existed, there should be no radioactive elements left> but R3<there are still uranium and other radioactive elements around> **therefore this is scientific proof that** C2[matter has not always existed].

3.5.2 R1<If the civil population cannot be defended in the event of nuclear war, we do not need a civil defence policy>. But

R2<we do need a civil defence policy if 'deterrence' is to be a convincing strategy>. **Therefore** C1[deterrence is not a convincing strategy].

3.5.3 Left to you.

3.5.4 Left to you.

3.6.1 The councillor is presumably arguing for the conclusion: 'we should get new, bright high-level sodium lights'. It is not quite clear whether 'Because our street lights are too dim, we have more accidents and more crime than we should' is an explanation or an argument; if it is accepted that 'our street lights are too dim' it is an argument. If it is accepted that 'we have more accidents and more crime than we should' then it is probably an explanation. Similarly with 'they are so low that they are easily and often damaged by vandals'.

3.6.2 Explanation.

3.6.3 Explanation.

3.6.4 Explanation.

3.6.5 Argument (but you could read it as an explanation).

3.6.6 The seismologist is *explaining* the 1906 San Francisco earthquake and *arguing* that another is due soon.

3.7.1 (i) If extinction of species is threatened, we need to prevent the destruction of their habitats wherever we can. (ii) We need to minimise global warming if we are not to lose many species. (Perhaps others too.)

3.7.2 (i) If enough parents refuse vaccination, they will put their children at real risk of contracting polio. (ii) The medical authorities should publicise these risks and urge parents to have their children vaccinated.

Chapter 4

4.1.1 The only possible (or 'the most plausible') explanation for the rise in sea level is that the world's climate is getting warmer.

4.1.2 Better education on health issues would lead people to improve their lifestyles.

4.1.3 If a bottle isn't marked 'poison' it won't harm you.

4.1.4 Your husband hasn't fabricated the evidence to make it look as though he is dead.

4.1.5 Drivers would still travel at about 80 mph if the speed limit were 80 mph.

4.2.1 Yale is a very prestigious university in the US so most people would assume its report was reliable. The American Federal Drugs Administration has a reasonably good reputation in the US for being careful about the safety of drugs, so Americans will tend to assume the FDA's judgement is sound. (There is no comparable government agency, with a comparable reputation, in the UK, where BSE and other scares have given rise to quite different assumptions about food and drug safety.) One would need to investigate these matters – and perhaps the 'culture of litigation' which exists in the US – to know what other assumptions could be attributed in virtue of context.

4.2.2 This argument has to assume that unmarried men and women (of comparable ages to married people) do *not* put on weight to the same extent. Also, many women have children during the early years of marriage and this could be a cause of weight increase rather than marriage itself.

4.2.3 Investigation will show you that this argument was written not long after the French Revolution, when many 'upper class' and wealthy people in Europe were afraid revolution might come to them. It assumes that what happens in the plant and animal worlds also happens to human beings (so far as population growth is concerned), that there is not much scope for science and technology to increase food production etc., that redistributive policies could not work, and that birth control could not have a significant impact on population growth (see Fisher, 2004, chapter 3).

4.3 The reasons are plausible in the UK context and the argument seems reasonable. Attitudes to cars and railways are very different in the US. Investigation would probably show that nearly everyone in America expects to have a car and to travel freely by road, that the Mid-West is not very crowded and its roads are not generally crowded, so the reasons do not apply there and the argument will probably seem irrelevant.

4.4.1 The structure of the reasoning and the argument's conclusion are clear. The argument assumes that drivers would still drive at about 80 mph if the speed limit were raised to 80 mph.

This seems doubtful because most drivers in Britain believe they will only be prosecuted for speeding on a motorway if they travel at more than 10 mph higher than the speed limit and that it is rare to get caught anyway. Such attitudes suggest people would drive faster if the speed limit were higher. Furthermore, there are arguments for lower speed limits – on grounds of safety and fuel economy. So overall, this argument doesn't have much merit.

4.4.2 Left to you.

4.4.3 Left to you.

4.4.4 This argument clearly assumes that there are no compensating benefits to the paintings – such as earning extra revenue which can enable their owners to provide better for their care and normal keeping. Nothing is said about that possibility, but we really need to quantify the risks and benefits to decide this issue. It is a weak argument at present.

4.5 Left to you.

Chapter 5

5.1.1– 5.1.3 Left to you, but you might find the internet helpful. Try your answers on fellow students. Discussed in section 5.1.

5.2.1 In this case the problem is to explain a term which has a *precise* mathematical meaning to a child. If you do not know what a polygon is, you could look in a dictionary or a mathematics book, or search the internet or ask someone you would expect to know (like a mathematics teacher). If you do know, the best way to explain this idea to a child would probably be to say something like, 'It is a figure which has many straight sides ("poly" means "many" and "gon" means "angle"), like these,' and then draw a pentagon (a five-sided figure), a hexagon (a six-sided figure) and so on, and explain that the word 'polygon' enables us to talk about lots of different figures, with different numbers of sides. More or less help might be needed depending on the ease with which the child grasped the idea.

5.2.2 A natural resource like oil or coal is used up and does not get renewed, whereas the water that flows in a river gets renewed. Curiously I find that people in different parts of the world view this example differently; those with plentiful

supplies of water (like the UK) think it is different from other natural resources, whereas those with limited water supplies (like Singapore) think it is very similar.

5.3.1 Someone who had never heard the phrase; the members of a jury; a university law seminar discussing recent decisions in some criminal cases. The rest is left to you.

5.3.2 Left to you.

5.4 My *New Shorter Oxford English Dictionary* tells me that circumstantial evidence 'tends to establish a conclusion by inference from known facts which are otherwise hard to explain'. So the circumstantial evidence would suggest Jones had stolen the paintings if we discover that Jones is not a wealthy man, that many of the paintings were reported stolen years ago, that Jones has no proof of purchase, that his friends did not know he had these paintings, and that his explanations of how he came by them are vague, uncheckable or provide no proof that he got them legitimately. This would probably suffice for everyday purposes though, of course, in a court of law greater precision would be needed.

5.5.1 To say that the conclusion of an argument *necessarily follows* from its reasons means that if the reasons are *true* then the conclusion *must* also be true; or, to put it another way, it is *impossible* for the reasons to be true and the conclusion false. For example, if someone argues as follows: 'All students work hard and all those who work hard deserve to succeed, therefore all students deserve to succeed', the conclusion *necessarily follows* from the reasons. This claim says nothing about whether the reasons *are* true (do all students work hard?); it simply says *if* they are true then the conclusion *must be*. As such the argument may fail to convince you of its conclusion even though it necessarily follows from its reasons.

To say that a conclusion *does not necessarily follow* from its reasons means that the reasons could be true and the conclusion false; or, to put it another way, it is possible for the reasons to be true and the conclusion false. For example, if someone argues, 'The weapon which killed Smith was found in Jones's house and had his fingerprints on it; Jones hated Smith and cannot provide an alibi for the time of the murder;

therefore he must have killed Smith', the conclusion *does not necessarily follow* (it may be a very compelling argument but it does not necessarily follow).

5.5.2 A dictionary or encyclopaedia will give you a phrase like, 'Democracy is government *of* the people, *by* the people, *for* the people. The essential thing is that the government is *elected* by the people and can be replaced by others if the people are dissatisfied.' This simply reports common usage (it is a reported definition); it may still be too vague to make it easy to decide if certain countries are democracies, though some examples might help. Presumably, if any country is a 'democracy', the United States is, as are various western European countries; certainly Hitler's Germany was not, Franco's Spain was not, and modern Saudi Arabia is not. A politics text would no doubt give you much more detail.

5.6.1 Left to you.

5.6.2 In a famous study of poverty in the United Kingdom, Peter Townsend gave the following definition: 'Individuals, families and groups in the population can be said to be in poverty when they lack the resources to obtain the types of diet, participate in the activities and have the living conditions and amenities which are customary, or at least widely encouraged or approved, in the societies to which they belong' (*Poverty in the United Kingdom*, 1979, p. 31). This criterion was defended in terms of general usage, but also as usable for empirical research, and has been widely acknowledged for its usefulness. No doubt there is a history of other usages before and since which could help you.

5.6.3 Left to you.

5.7.1 This may seem plausible at first, but the student evaluation form may be measuring how much the students *like* the teacher and surely someone could be a good teacher without being greatly liked or could be liked without being a good teacher. They might be popular, friendly and give high grades but might be relatively ineffective at getting students to learn what the course is about; surely we should not count such a person as a good teacher. Thus, in the terminology favoured by philosophers, a good score is neither a *necessary condition* for being a good teacher nor a *sufficient condition*

either. (For an instructive discussion of this example see Scriven, 1976, p. 127.)

5.7.2 You might think games must be 'played by several players' but what about the card game of patience? You might think they should be 'fun' but what about poker? You might think they need 'rules' but what about when children simply throw a ball to each other? And so on. In fact it turns out to be impossible to find necessary and sufficient conditions for something to be a game.

5.8 Here is a possible response:

It is easy to see what is being argued here, but what do 'sensitive' and 'fully developed human being' mean? Most of us probably feel we could describe situations in which individuals were sensitive or insensitive, human or inhuman. The dictionary tells me that 'being sensitive' means 'being aware of and responsive to other people's feelings'. This is obviously a matter of degree and even though it may be true that many men have difficulty in being sensitive to others, many are reasonably sensitive. Whatever the author means by 'fully developed human being' this would probably include a reasonable degree of sensitivity to others (if someone was utterly insensitive this would surely make them less than fully human) but you do not need to be enormously sensitive to be human. For example, is the employer who makes employees redundant being 'insensitive' or failing as a 'fully developed human being'? And how sensitive do you want your soldiers to be? Are tough, thick-skinned political leaders less than fully developed human beings? Perhaps women do generally find it easier to be sensitive to others but surely the responsibility for being sensitive must be placed squarely on the person concerned. Yes, perhaps others can help but the inference that women have a special responsibility to help men is very weak. I am assuming that this whole argument relates to Britain at the present time, where there is much debate about the relationship between men and women, including their respective rights and responsibilities; perhaps it has more force in other societies.

5.9 Left to you.

5.10 Left to you.

5.11.1 Toxicity testing is the testing of drugs or other substances to see if they are harmful (toxic) to animals and human beings. Such tests – of medicines, pesticides, food additives, household cleaners and so on – are usually carried out on animals initially.

5.11.2 Technically, it means that, of all those who are diagnosed with mesothelioma, half will die within eight months of diagnosis (i.e. very quickly) and the other half will die after that (but perhaps also quite quickly?). The next thing Gould wanted to know was 'What is my chance of being in the half of people who live longer than eight months?' (i.e. what determines whether I will die quickly or live longer than eight months?) and 'What does the graph look like for those who live longer than eight months?' (and what determines where I might be on that graph?). In fact the right-hand side of the graph has a long 'tail' and Gould lived for another 20 years.

5.11.3–5.11.5 Left to you.

5.11.6 Oil reserves are the *estimated* quantities of *crude oil* which are believed to be recoverable under *existing economic and operating conditions*. The total amount of oil in an oil field includes much oil which will not be recoverable; how much is recoverable depends on economics and on technology, so the oil reserves in a field can change considerably over time. (Cf. Wikipedia)

5.11.7 Though this is a famous argument which was influential in its day, it is surely clear now that you *can* teach people how to argue, judge whom to believe, make decisions and so on better than they did before. (Consider your own experience of working through this book.) Furthermore, you might as well say you can't teach writing in general.

Chapter 6

6.1 Discussed in the text immediately following the question.

6.2 Discussed in section 9.3 but do not look ahead yet.

6.3 Discussed in the text immediately following the question.

6.4 It is difficult to say what evidence would show that 'TV violence affects people's behaviour' because so many factors come into play. However, if the claim is that watching TV violence makes more people more violent than they would otherwise be, it might be possible to measure levels of violence in otherwise similar groups who have very different TV habits (with some groups tending to watch more violence than others). It might also be relevant to find people who say they are influenced by what they see on TV, some being made more violent and some being put off violence. One only has to suggest these possibilities to think of complications, so perhaps this claim cannot really be shown to be true or false. 'TV advertising works' is left to you.

6.5 'People should be free to watch what they choose.' We live in a society which puts a high value on allowing people to make their own choices about how to live their lives – provided they do not harm others – so this principle will give strong support to the claim made here. However, by that same principle, if we could show that watching certain kinds of TV programmes had undesirable effects – inclining some people to harm others – this would tell against this claim. But we just saw that this could be very difficult.

6.6 Factual claim: 'acupuncture could increase IVF (in vitro fertilisation) success rates by 65 per cent'; trial reports should show this claim true or false.

Recommendation: 'it is worth avoiding acupuncture in the context of IVF'; to evaluate it look at the pros and cons (Singh makes a strong case for this recommendation).

Definition: '"sham" acupuncture group [a "control" group who thought they were having acupuncture but were not really doing so]'; this seems a clear idea; we could check with doctors whether it is.

Causal claim: 'acupuncture is still unproven in terms of increasing IVF success rates'; the figures support this negative causal claim.

6.7.1 Dawkins is claiming with great certainty that 'religion shows a pattern of heredity which . . . is similar to genetic heredity'. He made his claim originally in a public lecture

which was subsequently published in the *Independent*. It does seem true that most people remain in the faith in which they are brought up, at least in broad terms of remaining Christian, Muslim, Hindu, Buddhist and so on. Perhaps people change sects more within their broad religious category, say changing from Church of England to Roman Catholic, but the broad claim seems true within my experience of people's religious beliefs and seems to fit well with everything else I know. The rough analogy with heredity also captures something of the unreflective nature of most people's religious commitment in my experience. So, for these reasons, I regard it as acceptable for the purpose of this argument.

6.7.2 This piece reports a famous experiment which had remarkable results – which were replicated in similar experiments. It is easy to check the report on the internet and there seem to be no doubts about the results. The inference about there being life elsewhere in the universe is tentative and evidence is even presented against that (no radio signals). It is a typical scientific report, definite about what happened and tentative about inferences from the evidence. It fits easily with my other beliefs so I have no difficulty in accepting what it says.

6.7.3 This report was in *New Scientist*, which lends it credibility. The marine scientists who reviewed the plan are likely to have the relevant expertise and were probably independent, so that lends credibility to their views. So I accept that Norsk Hydro's plan was probably flawed. This exercise is really all about credibility – with which we are about to deal.

6.8 Examples 1 and 2 are discussed in the text immediately following the question. Examples 3 and 4 are left to you.

6.9.1 Photographs are easy to fabricate. It remains incredible for the reasons already given.

6.9.2 Left to you.

Chapter 7

7.1 Example 3: the driving instructor is very likely to have the relevant expertise to be highly credible.

Example 4: the reporter will probably have the expertise to pass on what was presented. This may depend on the TV channel.

7.2 Partly answered in the text immediately following the question, but you may have your own questions.

7.3 Most people say at this point that the red and white car drivers cannot be given much credence because they have too much to lose or gain (even though they can see well, etc.). The policeman and mother seem to be credible witnesses at this stage (though we may need to know a bit more) and the child is not.

7.4 Left to you.

7.5 Rufus probably has much to lose if he is convicted, so (although he *may* be telling the truth) we generally treat such testimony rather sceptically in the absence of other evidence – because he has a vested interest. The doctor presumably has nothing to lose from reporting accurately what she found – she has nothing to gain or lose, no vested interest, in giving evidence one way rather than another. She also has the relevant expertise and the court context adds weight, so her evidence is highly credible. Rufus is in trouble!

7.6 Left to you.

7.7 Direct evidence would come from someone saying, for example, that they had seen little green men walking out of a spaceship onto the lawn in their garden. Circumstantial evidence for the same claim might be indentations and scorch marks where the spaceship was alleged to have landed and taken off, what looked like footprints of small men on the lawn surface and forensic evidence from the grass that unusual chemical compounds had come into contact with it. Perhaps you can think of more.

7.8 The mother might say, 'I wanted to cross the road safely with my child so, although he was having a tantrum, I was holding his hand tightly and attending carefully to the traffic and the traffic lights. I have good eyesight, visibility was excellent, I drive myself so I know what to look for and

I don't know the two drivers or the policeman.' The policeman might say, 'I was on duty. I was attending to the junction; I have good eyesight and visibility was excellent. I have also received extensive training in how to observe and report accurately what I observe and on my responsibilities when giving evidence in court.'

7.9 'The Sun crosses the sky from east to west' versus 'The Sun orbits the Earth', or 'The gate is open and the sheep are on the road' versus 'The sheep have escaped from their field'.

7.10 I assume everyone had good visibility, was not under the influence of drugs of any kind and was attending to what happened, except the child, who was having a tantrum. I also assume that none of the witnesses knew any of the other witnesses (except of course the mother and child). I also assume that the traffic lights were working correctly and normally.

We cannot attach much credibility to either of the drivers because they have a vested interest. Nor can we attach any weight to what the child says (he was not attending to what happened). I assume the mother was a driver and knew what to look for; given everything else we assumed about her, her evidence is highly credible. I assume the policeman was under no particular pressure to get convictions. Then, given his expertise and everything else we have assumed, his evidence is highly credible too. Furthermore, we have two independent witnesses saying the same thing so their evidence corroborates each other's. This makes a very strong case for believing that the red car *did* jump the red light.

7.11.1 The passage reports direct evidence – of what I saw. I was young and had good eyesight so the description is probably reasonably accurate. To claim that it was a ghost would be an inferred judgement – about which most of us would normally be very sceptical. In fact, my bicycle light was shining down into the strangely shaped puddle, which was rippling in the wind, and my light was reflected onto the tree trunk, the angle of reflection increasing as I approached the puddle, making the image move up the tree (as I was able to corroborate, by doing it again)!

7.11.2 The tobacco industry stands to gain from such results, so we need to know how independent the scientists were. If they were reputable, independent scientists who were free to publish whatever they found and were under no pressure to find results helpful to the tobacco industry, their results are credible. If they were tobacco industry scientists or scientists who had some other kind of vested interest, their results deserve less credence.

7.11.3 We know relatively little about Giuliani; he is described as a laboratory technician, so we might assume he knows about scientific methods (and therefore has relevant expertise). However, the clear experts in this case, seismologists, had checked the basis of his claims and disagreed with him. They have no vested interest, have the relevant expertise, and so on – consequently they deserve more credence.

7.11.4 Here is a possible answer – based on particular assumptions:

The owner of the company (B) probably has much to lose (he may have to pay compensation or suffer in other ways) if he is held to be responsible for (A)'s injury, so this reduces the credibility of his testimony. For the same reasons, the foreman's (F) credibility is threatened – however, he is able to produce a maintenance schedule which corroborates his claim that the machine was always maintained satisfactorily, so this point seems well proven.

(A) has much to lose (or gain) so the credibility of his claim that he followed all the safety procedures is not high, because of his vested interest. (C) is a workmate of (A), so his credibility is reduced since he is probably biased in favour of his workmate.

We assume that the health and safety inspector (I) does not know any of those involved, or at least has no relationships with them which would bias him for or against anyone (otherwise he should not be giving his professional judgement). Thus he is independent and has no vested interest in arriving at one judgement rather than another. We also assume he has the relevant training and expertise to judge this issue (otherwise why is he involved?). Furthermore, he must be able to justify his judgement professionally so his reputation is at risk too. On these assumptions, (I) is a

highly credible witness and his judgement suggests that the company which designed the saw and its guard is ultimately responsible for the accident. Interestingly, (C) claims to have told (F) that the saw guard 'was poorly designed and did not function well'. If (C) had the relevant expertise (was familiar with the problems of operating circular saws) and if (F) knew this, it would suggest that (F) failed to act on reliable advice and that would make (F) to blame.

7.11.5 Left to you.

7.11.6 Here is a possible answer – based on particular assumptions:

I assume from the context that this is set in London. The policeman (P) probably has much to lose if he is found to have harmed Fran Lee (F) with a baton blow, so this greatly reduces his credibility. The 'well-known lawyer and human-rights activist' (A) was probably sympathetic to the AFA but we do not know his views about the police, so it is hard to estimate if there was any bias regarding them. He will obviously know the law and how it works and his professional reputation is at risk if he says something which turns out to be false, so this lends credibility to his testimony. I shall assume the TV news crew (N) were from a reputable news company (BBC or ITV, say) and this lends credibility to their evidence. Furthermore, if there is any possibility of rescuing some of their film, this could provide evidence concerning the accuracy of what they say, so this puts their reputation at risk, which again increases their credibility. No doubt the R33 leaders would deny blame because they have much to lose if they are found to be partly to blame. Equally the R33 member (M) who boasts that he threw the brick refuses to be identified and therefore is not very credible. I shall assume the Dutch tourist (D) could see much of what was happening; presumably she has no vested interest or bias and was merely an impartial observer so all that lends credibility to her testimony. No doubt the hospital spokesperson (H) is independent and speaks with the relevant expertise behind her, but her testimony does not help us decide who struck 'at least one very severe blow' to Fran Lee's head. On balance and given the assumptions I have made, several credible witnesses, (A), (N) and (D), corroborate one another and their evidence suggests fairly strongly that

(P) – and perhaps other police officers – were responsible for the head injuries to Fran Lee.

7.11.7 Left to you.

Chapter 8

8.1.1 The inference is simply the move from the first two sentences, taken as side-by-side reasoning, to 'a suitable job for anyone who doesn't go to university is to become an entrepreneur'.

8.1.2 The inference is the move from everything before 'so', taken as joint reasoning, to 'the world's climate must be getting warmer'.

8.1.3 The inference is the move from the first three sentences, taken as side-by-side reasoning, to 'findings of the analysis are likely to be incorrect'.

8.1.4 The inference is the move from 'If a Roman Catholic priest administered arsenic to the philosopher Descartes in communion wafers, that would have killed Descartes' and 'A Roman Catholic priest did administer arsenic to the philosopher Descartes in communion wafers' to 'that must have been what killed Descartes'.

8.1.5 The inference is the move from everything else (probably best construed as joint reasoning) to 'The police force should ban their officers from driving at high speed in pursuit of young joyriders who steal cars'.

8.1.6 This contains two inferences. The first is from 'When prisoners under sentence of death are given the choice between life in prison and execution, 99 per cent of them choose life imprisonment' to 'they fear death more than they fear life imprisonment'. The second is from 'they fear death more than they fear life imprisonment' and 'one is most deterred by what one most fears' taken jointly to 'the threat of the death penalty is more likely to deter most potential murderers than is the threat of life imprisonment'.

8.2 Discussed in the text immediately following the question.

8.3.1 Clearly a poor inference by our test.

8.3.2 As it stands, a weak inference by our test. If we add the assumption that 'there is no other possible explanation' it becomes a good inference by our test.

8.3.3 A weak inference by our test.

8.3.4 It is *possible* that Descartes died of an unrelated heart attack before the arsenic had time to take effect, so the argument as it stands is not a good inference. However, if the conclusion had read, 'So that would have killed Descartes', it would have been a good inference.

8.3.5 Not so easy to judge by our test; depends on whether there are any alternatives. See discussion in section 9.5, example 2.

8.3.6 Left to you.

8.4.1 Deductively valid. If Tom really hates *everyone* Mary loves, then if she loves him he *must* hate himself. Of course, if it means Tom hates everyone Mary loves other than himself, then it is not deductively valid.

8.4.2 If 'in that case he couldn't have shot the master' means 'if the butler was in the pantry he couldn't have shot the master' then the argument is deductively valid; if the reasons are true the conclusion must be.

8.4.3 Clearly not deductively valid. However, with the added assumption 'there is no other possible explanation', which means in effect 'if the sea level is rising, the world's climate must be getting warmer', it is deductively valid.

8.4.4 Not deductively valid. See the answer to 8.3.4. Again, if the conclusion had read, 'So that would have killed Descartes', the argument would have been deductively valid.

8.5.1 Deductively valid.

8.5.2 Not deductively valid.

8.5.3 Not deductively valid.

8.5.4 Not deductively valid.

8.6 Perhaps Daley and Thompson were both able to give details about the murder which no one could know except the murderer and the police. Perhaps the prosecution was able to show that Daley and Thompson could not have discussed their own evidence with each other so it was given independently and thus corroborated what the other said (see section 7.5 on corroboration). Perhaps they were able to convince the jury that they had no 'motive' other than to tell the truth in this case because they had nothing to gain or lose by giving or not giving their evidence against Stone. These – and perhaps

other facts you thought of – could surely convince a jury that the case was established 'beyond a reasonable doubt'.

8.7.1 Left to you.

8.7.2 Left to you.

8.8.1 You probably do not know whether the figures are correct and so cannot comment on the acceptability of the reasons, but the inference is clearly weak; there could be all sorts of social factors causing all sorts of differences, so the reasons could be true and the conclusion quite unwarranted.

8.8.2 Left to you.

8.9.1 Dawkins begins by considering the following inference: 'although there is no positive evidence for the existence of a God, nor is there evidence against His existence. **So** it is best to keep an open mind and be agnostic.' This has seemed a very reasonable inference to many people, but Dawkins argues that there are many similar inferences which no one accepts: for example, 'although there is no positive evidence for there being fairies at the bottom of the garden, nor is there evidence against their existence, **so** it is best to keep an open mind and be agnostic'. He contends that since the second (fairies) inference impresses almost no one, the first shouldn't either. Although it is clear that neither inference is deductively valid, the question is whether they nonetheless have some force. Though almost no one suggests that the 'fairies inference' has any weight, some people think the one about God's existence is very strong – because the consequence of being mistaken (and wrongly believing God does not exist) might be so serious – perhaps even involving eternal damnation! (For closely related inferences, see Pascal's Wager, Questions appendix, passage 56.)

8.9.2 Argument indicator words make the structure of this argument very clear. Ayer's basic reason is that 'all claims about the nature of God are meaningless'. From this he infers that the sentence 'God exists' is meaningless. From this and the belief that 'only meaningful claims can be meaningfully denied' he infers that the sentence 'God does not exist' must be meaningless too. He then infers from the fact that the agnostic thinks 'Does God exist?' is a meaningful question,

that the agnostic must think 'God exists' and 'God does not exist' are meaningful sentences. Thus, he infers, his view conflicts both with atheism and agnosticism.

These are all hard inferences to fault; if it is true that 'all claims about the nature of God are meaningless' it is hard to see how his other conclusions could be false. Thus, if you dispute Ayer's conclusions you have to challenge his basic reason. Philosophers and theologians have been doing this for a long time!

Chapter 9

9.1 Discussed in the text immediately following the question.

9.2.1 This appears to assume that only the benefits to *animals* from toxicity testing will count towards the benefits argument. The benefits argument also counts the benefits to humans in weighing the case for toxicity testing, so the author is mistaken in inferring from 'the benefit to animals cannot be established' to 'the benefits argument fails'.

9.2.2 This assumes that it is better to have the original, vivid colours than those dulled by soot, varnish and general grime. It is not difficult to find 'before' and 'after' images on the internet; if you think the more vivid colours are better, this strengthens the argument, but if you think the duller colours are better, that weakens the argument.

9.2.3 On the face of it, this inference commits a classic fallacy, since it has the pattern 'if A then B and B is true, so A is true' which is a common mistake. However, in this case, it is surely reasonable to attribute the assumption 'there is no other plausible explanation for the pile of ashes and rubble' and this is very likely to be true, so adding this assumption shows the inference to be a good one (by any reasonable standard) and not a fallacy at all. (For a more extensive discussion of the significance of this example see Everitt and Fisher, 1995, p. 171.)

9.2.4 It is not enough to add the assumption 'if Jones has worked hard he will pass the exam'; this makes the argument deductively valid, but any questions about the original inference will now have to be asked of the assumption. For the inference to succeed the speaker must be assuming that

Jones is clever enough, that he has been well taught, that the exam is no more difficult than usual, that Jones will not collapse with exam nerves, and the like (since these are all ways in which the inference could misfire). So if all or most of these assumptions are true the inference is a good one, but if some are false it is weak.

9.2.5 The inference 'Mars has long been too dry and cold for life to flourish on its surface' seems to be based on the assumption that 'life requires warmth and water'. Since (I think) this is true of every form of life we know, this seems a reasonable inference. Some people may wish to insist that other forms of life are 'possible' and they will think this is a weak inference.

9.2.6 In the first paragraph the Noodle is surely assuming that if our ancestors did not think of some 'measure' it cannot be a good one (or at least that policies and practices which have been in place for some time are generally better than new ones). Identifying this underlying assumption shows how weak his inference is. Perhaps the measure he is discussing is a poor measure, perhaps it is not; but the inference from the fact that it is *new* to the conclusion that it is *poor* is hopeless. Apart from anything else, circumstances change over time and the measure has to be weighed on its merits. (However, note that it is surprisingly common for older people to say to younger people, 'It was good enough for us, so it should be good enough for you'!)

In the fourth paragraph the Noodle is assuming that if the measure is supported, then further and less reasonable measures will be proposed and these will have to be accepted. This is often called the 'thin end of the wedge' argument or the 'slippery slope' argument. The Noodle would almost certainly have argued that women should not be given the vote because they would then demand even more – and these additional demands would have to be accepted! Again the inference is hopeless and the measure has to be weighed on its merits. This is another very common pattern of inference which has little merit.

9.3 Left to you.

9.4.1 Left to you.

9.4.2 Discussed in the text immediately following the question.

9.5 Answer for yourself now, but the argument is discussed in section 11.2.2.

9.6.1 Here are some 'other relevant considerations' which students have suggested: (i) Perhaps the extinction of species is simply a 'natural' process and we should not interfere with it.

(ii) If we accept the argument, many species might quickly disappear. For example, pandas and tigers are threatened with extinction because of the loss of habitat and some species of whales are threatened because of over-hunting. Surely we simply do not want to lose these species and perhaps thousands or even millions of others?

(iii) We might like to eliminate some species, like mosquitoes, which are responsible for spreading disease.

(iv) Losing some species might produce great changes in ecosystems (for example, losing whales might have a big impact on oceans) – and some such losses might harm us. (Other such losses might benefit us.)

(v) Genetic material might be lost forever and we do not yet know how significant that loss might be.

(vi) It usually costs money and resources to keep species in existence (because, for example, habitats have to be preserved and guarded) so we may face some hard choices about the use of scarce resources.

(vii) The loss of some species might particularly impoverish human life. Surely we have much to gain from a species-rich environment, including being able to extract useful genetic materials and simply being able to wonder at its variety and beauty.

(viii) It would be wonderful if there were still some dinosaurs around – in somewhere like Jurassic Park!

Perhaps you thought of other considerations too and you might feel you need to do some research about some of these before being able to resolve this question to your own satisfaction.

9.6.2 Here are some 'other relevant considerations':

(i) It must surely be possible nowadays to make containers which, like aircraft 'black boxes', are almost indestructible in the event of a crash, thus reducing the risk to paintings being transported by air.

(ii) It must also be possible to construct such containers so that they can properly regulate the atmosphere in which the painting is transported.

(iii) Paintings are often at risk in their 'home' galleries – from excessive humidity, poor security (some famous paintings have been attacked by vandals), over-zealous cleaning and even from flooding.

(iv) 'Home' galleries are often too impoverished to maintain paintings in the best possible atmosphere and security.

(v) Many people attend these big art exhibitions – which brings in a great deal of money – so this money could help impoverished 'home' galleries protect and look after the paintings.

(vi) Many people get enormous pleasure from these big art exhibitions and they increase interest in such paintings throughout the world and this is good for the paintings.

Again, perhaps you thought of other considerations too.

9.6.3 Left to you.

9.7.1 Largely left to you, but if you skim the piece looking for argument indicator words, you will quickly find the following, and these will help you locate quite a few of Dawkins' arguments: '**So** it is best to keep an open mind and be agnostic [about the existence of God]. . . . **because** the same could be said of Father Christmas and tooth fairies. . . . **so** shouldn't we be agnostic with respect to fairies? . . . **the reason is** that most people . . . have a residue of feeling that Darwinian evolution isn't quite big enough to explain everything about life. . . . **because** any god worthy of the name must have been a being of colossal intelligence, a supermind, an entity of enormous sophistication and complexity. . . . **because** it raises a bigger mystery than it solves. . . . **for** it simply postulates the difficult to explain. . . . **we can safely conclude** that He [God] is very, very improbable indeed.'

Applying other techniques, like the 'therefore' test, will then help you understand the rest of Dawkins' arguments (which he mostly displays very clearly). Doing these things and applying the other lessons we have practised should help you to write a response which genuinely focuses on the issues.

9.7.2 This is entirely for you to answer.

Chapter 10

10.1 Perhaps they (i) use landmarks and knowledge of features on the ground; (ii) use the Sun, Moon and stars; (iii) simply have inbuilt instinct; (iv) learn by following adult birds. Other explanations may have occurred to you.

10.2 Perhaps you could look for magnetite in birds' brains. Perhaps you could attach tiny magnets to the birds which would obscure the Earth's magnetic field – so they would not be able to find their way. Perhaps you could show that some other mechanism was more plausible.

10.3 It is clear that this example about the death of Napoleon exhibits the right structure; it considers two alternatives and provides evidence which tells *against* the cancer option and provides quite a lot of evidence *for* the arsenic poisoning option. Whether this is a good and persuasive argument depends on whether there are other possibilities which should have been considered and other evidence which should have been taken into account or looked for. For example, perhaps we need to check whether there are other ways the arsenic could have got into Napoleon's hair (say through a hair dressing that was commonly used then?). Or is there any other evidence about his behaviour before his death which might suggest other possible causes of death? For most of us the argument will be convincing enough if we have good reason to believe that the experts are right about what his symptoms signify and no other experts have had good evidence for other possible causes of his death. Of course, someone who researches Napoleon's death for a PhD thesis will need to investigate the evidence and what other experts have said very carefully, but most of us have to rely on experts in a situation like this. Incidentally, this explanation fits well with something else I know, namely that horsemen used to give their horses small doses of arsenic before they entered a show to make their coats shine!

10.4.1 Perhaps there was colossal volcanic activity on Earth at that time which threw enormous quantities of dust and poisonous fumes into the air. Perhaps some disease killed off

the dinosaurs. You may be able to think of other possible explanations.

10.4.2 Evidence of colossal volcanic activity could be shown by huge lava flows of the correct date over a huge part of the Earth (as is the case in India). I do not know enough about how disease might be revealed as the cause; perhaps experts could say what to look for in fossils.

10.4.3 Presumably, there are no dinosaur fossils later than 65 million years old. What is said fits well with everything else I know and believe, so the reasoning seems quite strong. You may know more and have a different view.

10.4.4 Left to you.

10.4.5 Maybe some dinosaurs could have survived on these plants; we would need to know how extensive the plants were and whether at least some species of dinosaurs ate them to know whether this evidence conflicts with the explanation we have been offered for the death of the dinosaurs.

10.4.6 This seems to render the meteor explanation less likely (unless we can find evidence of other colossal meteors arriving at the right times). Perhaps we would need to look for different explanations for different extinctions.

10.4.7 Left to you.

10.5 It is reasonably easy to mark this passage up so that you can see what is going on: An underwater survey of the Witch Ground, in the North Sea off Aberdeen, has discovered C1[a trawler which was probably sunk by a sudden burst of methane gas escaping from a vent in the sea floor – known as the Witch's Hole]. R1<When methane – or natural gas – bubbles through the sea in big enough volumes, it lowers the density of the water around it to a point at which objects, including ships, will no longer float>. R2<'Any ship caught above [such a blow-out] would sink as if it were in a lift shaft,'> said Alan Judd, a marine geologist from the University of Sunderland who led the Witch Ground survey expedition. R3<In this case the trawler had sunk 'flat' with its hull sitting horizontally on the sea bed exactly over the Witch's Hole vent>. R4<This was consistent with the vessel being sunk by a methane blow-out>; R5<if AH [she had been holed] [she would have sunk with the holed end

lowest]>. Even sailors who jumped overboard wearing life-jackets would sink like stones. R6<The Witch Ground, and the Witch's Hole in particular, have long been known among fishermen as treacherous waters>. R7<Methane blow-outs are thought to have destroyed about 40 oil platforms around the world; the Witch Ground is just 22 miles from the Forties oil field>.

Does the report consider other reasonable possibilities? Could there simply have been a great storm? Could a submarine have caught in the trawler's nets and dragged it down (as happened in another famous case)? What evidence would show for or against these possibilities? A storm would probably have allowed the trawler time to send a Mayday message and might have allowed the men time to put on safety clothing or launch a safety raft; a methane blow-out might be too sudden to allow any of these. We should look for evidence on these. On the submarine theory, the nets would have to be out; were they? Presumably there is no evidence of the trawler being holed. If the area is known to be treacherous, presumably this is because other ships have been lost there, so shouldn't we expect to find other ships nearby similarly unholed, sitting 'flat' on the bottom, if this explanation is to be completely convincing?

10.6 Marking this up is easy so we leave that to you. This piece provides three pieces of evidence and two other reasons *against* believing that 'consumption of cholesterol affects the level of blood cholesterol' (let's call this claim the Hypothesis). Assuming the evidence is from reliable sources (and the MRC *is* reputable) they tell against the Hypothesis. Assuming the other reasons are true too, they also raise questions about the Hypothesis. If what this says is true, it is interesting and persuasive to me. Back to eating butter!

10.7 The piece explains why the number of twin births has increased and why the increase is entirely due to increased numbers of 'fraternal twins'. It then argues that these will continue to increase 'because more women will seek fertility treatments' but will stabilise – with most births still being single births. Given the figures, it seems a plausible argument.

10.8 Discussed in the text immediately following the question.

10.9 Left to you.

10.10 Left to you.

Chapter 11

11.1 Left to you.

11.2.1 Hans has complained that your test was unfair. That gives you a problem. Three options are: (i) re-mark the paper then discuss the result with Hans, (ii) refer the test paper to a colleague for scrutiny and also refer Hans's paper for re-marking, (iii) do nothing.

11.2.2 Left to you.

11.2.3 Here are some options: (i) threaten the Japanese government with the use of the bomb, (ii) try to negotiate peace, (iii) demonstrate the power of the bomb to the Japanese on an uninhabited area, (iv) don't use the bomb but continue the war with conventional arms, (v) drop the bomb on military targets, (vi) drop the bomb on cities, (vii) abandon the war.

11.3.1 (i) If you re-mark the paper and discuss the result with Hans, you may change your mark or confirm it; if you discuss it with Hans he may be persuaded of the correctness of your final mark or not; if he is not, he may appeal; other students may be affected by the reasonableness of what you do.

(ii) Suppose you refer the test paper to a colleague for scrutiny and Hans's paper for re-marking. Assuming the colleague has the relevant expertise, doing this will either confirm the fairness of the paper or raise questions about it. It will also either confirm the fairness of Hans's mark or raise questions about it. A second pair of (expert) eyes is often helpful to both the teacher and the student in such cases and may be seen as fair by other students.

(ii) If you do nothing you will be saved some work initially, but Hans may appeal and other students may be disaffected.

11.3.2 Left to you.

11.3.3 If you demonstrate the power of the bomb to the Japanese on an uninhabited area (option iii), this will avoid killing thousands of innocent civilians (which will happen if it is

dropped on cities); may persuade them to surrender (but they have fought fiercely despite huge losses so maybe it won't); will limit your scope for further attacks (because you have only two of these bombs); may cost the lives of further US soldiers as the conventional war drags on (and this may cost you votes in the next election).

If you don't use the bomb but continue the war with conventional arms (option iv), this will certainly cost you far more US servicemen's lives (the Japanese fight very fiercely and don't yield easily); it will be costly in conventional arms too; the war will probably drag on for much longer than it would if you use the bomb; all this may cost you votes in the next election.

11.4 Left to you.

11.5.1 The obvious moral principle here is that you should not break promises. Most people act on this principle much of the time (otherwise trust would soon break down), but it might depend on the case.

11.5.2 This is a moral dilemma; on the one hand, most people believe that you should not break promises but, on the other, most people also believe you should save friends from harm if you can. It will depend on the seriousness of the case.

11.5.3 Left to you.

11.6 Discussed in the text immediately following the question.

11.7 Left to you.

11.8.1 Here is a possible brief response: Bullfighting countries could (i) leave things as they are, (ii) try to educate people or (iii) ban bullfighting. Option (i) causes suffering to some animals, gives pleasure to those who watch and disgusts some other people. If enough people in a country are opposed, option (iii) would produce most happiness, but some people would feel very aggrieved. It is important to be tolerant of others and let people live their lives as they choose but cruelty to animals should not be tolerated. Trying to educate people against this activity is a slow process; many animals would suffer in the meantime and such a policy might not succeed in eliminating this activity.

11.8.2 Left to you.

11.8.3 Left to you.

11.8.4 Here is a possible brief response: We could leave things as they are, try to educate people or compel parents to have their children vaccinated. Because more and more parents are worried about the risks from vaccination, the first course carries a substantial risk of polio outbreaks, with resulting injury to children, perhaps even death. Perhaps education could succeed in persuading more parents to have their children vaccinated, but many are increasingly suspicious of government claims about health. Compulsion would generate passionate hostility from some parents, is contrary to our traditions and would probably produce martyrs. We need reliable research on the risks of all options.

11.9.1 You have been made an offer so you have to decide. There are two options; you take the bet or you don't (maybe you could try to negotiate too!). What are the likely consequences? If you win, you get £1,000; if you lose you lose £100, so you have a 50–50 chance of winning ten times what you risk losing. If you value the possibility of winning £1,000 more than you hate the idea of losing £100 it is rational to accept the bet, but the outcome might be that you lose £100. That's life! (Cf. Sutherland, 1992, p. 5f.)

11.9.2 It is *very* likely that the coin will land heads at least once, so the chance of losing any money is tiny and the chance of gaining quite a lot is very considerable!

11.10 Left to you.

11.11.1 Kessler has the right background and expertise for us to take his claims seriously. So there is a problem about how 'foods have been engineered to be tasty [and to] stimulate us to want more of them' thus contributing to current obesity problems. We could do nothing and people would eat more and more – and obesity problems would increase. Or we could try to educate people to eat more healthily, but the food industry has huge resources so this course of action would be unlikely to be effective unless it was given huge resources too – which is unlikely. Or we could try to control or ban the techniques Kessler is referring to. This would almost certainly

encounter huge opposition from the food producers; many people would argue that people should be free to eat what they want – and calls for 'freedom' work powerfully in this context. Maybe we could put health warnings on unhealthy food or tax it heavily – as governments do with cigarettes. On the face of it the simplest thing to do, and the one most likely to be effective, is to ban the chemicals and so on which stimulate us to want more of the foods that contain them. That would no doubt require a fierce political campaign.

11.11.2 Left to you.

11.11.3 Left to you.

11.11.4 This piece addresses a major contemporary issue. Roughly speaking, we need enormous amounts of energy to power homes and economies throughout the world. The options are (i) to use coal, oil and gas – with adverse consequences for the environment, (ii) to use renewable sources, like solar power, wind power and tidal power – an option which is developing rapidly, (iii) to use nuclear power – which can produce the enormous amounts of energy needed – but which is seen to be risky, (iv) to settle for consuming less energy worldwide – which has very few supporters, or (v) some combination of the above. You might like to develop Lovelock's argument about using nuclear power – perhaps with a little help from the internet, trying to work out the likely consequences of 'going nuclear', how likely accidents are, how serious a problem nuclear waste is, and so on, to see whether or not you agree with Lovelock.

Chapter 12

It is not always easy to give answers to the questions in this chapter because what is on the internet changes so rapidly, but help is given where possible.

12.1.1 This is about the causes, symptoms, diagnosis (and so on) of obesity. It was published by the National Health Service (author unknown), in the UK, on 25 February 2010.

12.1.2 This is about obesity and is published in Wikipedia ('en' means the English language Wikipedia), but is undated and the author is unknown.

12.1.3 This is about the fragility of the ozone layer, published in the *Guardian* newspaper, 27 April 2005, in the UK; the author is probably their science correspondent.

12.2.1 Bats good luck. (These are the key words Google suggests.)

12.2.2 Van Gogh digitalis. (You should find a good site at www.psych.ucalgary.ca/pace/va-lab/avde-website/vangogh.html)

12.2.3 Napoleon arsenic.

12.2.4 Global warming (man-made OR anthropogenic).

12.2.5 Finding reliable information internet. (You should find a good site at www.library.jhu.edu/researchhelp/general/evaluating)

12.2.6 Critical thinking about internet. (Many universities publish information about using the internet; high on your list is likely to be www.vts.intute.ac.uk/detective)

12.3 The National Health Service in the UK is staffed by medical experts, has a good reputation for medical information and has no relevant vested interests; it is a reliable source of medical information.

Wikipedia is discussed at length later in this chapter; its entries are sometimes written by experts and sometimes not; its reputation is that it is sometimes reliable and sometimes less so, but it often cites reliable sources.

The *Guardian* newspaper has a good reputation for reliability and if this piece was written by their science correspondent, it is likely to be reliable (though it may be dated now).

12.4 All three sources are likely to be reliable. The BBC and the *New York Times* have a good reputation for reliability, which is at risk if they get things wrong, so they check their facts. The IWC has experts on its staff to ensure reasonable reliability.

12.5.1 Left to you.

12.5.2 This is guidance from the library of John Hopkins University, in the USA, and clicking on 'similar' produces similar guidance from a number of other university libraries.

12.5.3 Left to you.

12.6–12.8 Left to you.

12.9 Using the methods described earlier, you will find this at: en.wikipedia.org/wiki/Wikipedia:Press_releases/Nature_compares_Wikipedia_and_Britannica (accessed December 2010).

12.10.1 There is a section on the 'cause of death' in the Wikipedia article about Napoleon. It refers to a 1961 paper in *Nature*, by Sten Forshufvud, which discusses various possible causes of death, including deliberate arsenic poisoning. There are also several inline citations to more recent publications which discuss Forshufvud's paper and which look reputable. So this looks like reliable information.

12.10.2 Under the heading 'Dinosaurs' there is a section called 'Extinction' which includes numerous inline citations to what are obviously reputable sources – from *Science*, Princeton University Press, and various scholarly publications. So this is likely to be a reliable source.

12.10.3 An article called 'Global warming' discusses different models and contains numerous inline citations to scholarly publications, so is likely to be reliable – certainly a good place to start one's investigations. There are numerous other relevant items, including 'Global warming controversy' and 'Global warming conspiracy theory'; you may check these yourself.

12.10.4 There are several relevant entries. One, entitled 'Debate over the atomic bombings of Hiroshima and Nagasaki', has many inline citations to scholarly sources so is likely to be reliable.

12.10.5–12.10.7 Left to you.

12.10.8 An initial note on the 'War on Terror' item (accessed December 2010) says it needs improvements from an expert, needs verification and has problems with neutrality. So, be wary if this note or something similar is still there.

12.10.9–12.10.10 Left to you.

12.11–12.13 Left to you.

Glossary

Critical thinking uses ordinary English. Virtually all the language which is used in this book can be found quite easily in a good English dictionary. That is deliberate. Critical thinking simply requires us to use ideas and distinctions which are already available to most sixth-form students who understand ordinary English. There is no need for jargon and although a few semi-technical terms (such as *ad hominem*, *hypothetical*, *converse*) can be helpful, most of the language we need in order to engage in even high-level critical thinking is ordinary language. However, it *is* necessary to be clear-headed about the use of this language, hence the need for this glossary. The glossary explains the meanings of terms and phrases as an aid to keeping students' thinking properly focused, but there is no intention that the usage explained here should differ from (clear-headed) ordinary usage.

Analogy: Sometimes it is helpful to think about one matter by recognising that it is like another, more familiar one. For example, people sometimes explain what the movement of molecules in a gas is like by saying that they move like a large number of snooker balls bouncing off one another (but in three dimensions rather than just two). If someone reasons 'by analogy', they say, 'A is very like B in a number of respects so A is (probably) like B in another respect which interests us' – for example, in chapter 1 I say that critical thinking is like basketball in that it comprises various skills, so you can (probably) improve your critical thinking ability in the same way that coaches teach basketball.

Argument: In the context of critical thinking, the term 'argument' refers to a set of claims, some of which are presented as reasons for accepting some further claim – the conclusion. The reasons are presented with the aim of persuading the listener or reader to accept the conclusion. Many clear examples are provided in the book; in each of these some basic claims are put forward as reasons which support further claims – the

conclusions. Reasons may be factual claims, value judgements, definitions or interpretations (for example, see question 6.3) and there is a similar variety among conclusions. Arguments can vary enormously in their structure and content, but they always contain a set of claims which are presented as reasons for accepting some further claim – a conclusion – and they are always intended to persuade their audience.

There is an everyday sense of 'argument' which means roughly 'quarrel', but this is a quite different usage.

Argument indicators: These are the words we commonly use to show that reasons are being presented in support of a conclusion; they are words like *therefore . . ., so . . ., thus . . ., hence . . ., consequently . . ., which proves that . . ., I conclude that . . ., it follows that . . .*(where the dots are the conclusion) and *because . . ., since . . ., the reasons are . . ., the evidence is . . .* (where the dots indicate reasons). Of course, there are many other phrases in English which play a similar role (cf. section 2.3).

Assumption: We commonly call a belief an assumption when it is clearly accepted or 'taken for granted' by a speaker or writer but is *not stated or made explicit* by them; for example, someone engaging in a discussion about miracles may fail to mention that he believes in the existence of an omnipotent Christian God, but this may be obvious from other things he says. This is the most important usage of assumption in our context (see section 4.1).

Note also that in ordinary usage we sometimes call an *explicit* claim made by a speaker or writer an assumption, either (i) because we wish to note that the speaker or writer *has given no reasons for accepting it* or (ii) because *we wish to challenge the claim*. Thus, if someone was arguing the case for believing in miracles and clearly and explicitly based their case on the claim that there is an omnipotent Christian God, one might say (i) 'but this is only an assumption; why should I accept it?' or (ii) 'but this is only an assumption; I don't believe it at all'.

Belief: This term is used in an entirely everyday sense; it refers broadly to the claims/sentences that an individual person holds to be true or right. There are many different kinds of beliefs, including scientific claims ('the Earth goes around the Sun'), religious beliefs ('God created man in His own image'), moral principles ('it is wrong to break promises'), prudential beliefs ('it is better to buy a house than to rent in the UK') – and many others. If an individual A believes a claim P, P

may be true or false, scientific, religious, moral, mystical or whatever; it may be harmless or pernicious, plausible or not, and much more – it may even be meaningless or vacuous. Many arguments throughout the text express the beliefs of their authors.

Conclusion: If reasons are presented for accepting some other belief, C, then C is called a conclusion (of that argument). Thus the conclusion of example 2 in section 4.1 is 'all the unemployed could solve their unemployment problem by great ingenuity in searching for a job or by willingness to work for less'. If reasons are given for one conclusion, which is then used as a reason for a further conclusion and so on – in a chain of reasoning – the last is called the **main conclusion** and the earlier ones in the chain are sometimes called **intermediate conclusions**. Thus, in the first example in section 3.5, the main conclusion is 'we should prohibit the use of techniques which enable people to choose the sex of their children' and an intermediate conclusion is 'if people can choose the sex of their child, it is likely that there will eventually be more males than females in the population'.

Notice that a conclusion does not necessarily come at the end of a piece of reasoning. It may be stated at the beginning, then argued for (there are many examples in the text). The conclusion of a piece of reasoning might be a factual claim ('so he must be the murderer'), a recommendation ('so you ought to buy this car'), an interpretation ('so Iago should be seen as a treacherous villain'), a decision ('so I shall take the job') and so on.

Consistent/Inconsistent: Two claims are consistent provided they could both be true or correct at the same time. They are inconsistent if they *cannot* both be true or correct at the same time. Thus, 'Mount Everest is 29,000 feet high' is consistent with 'Mount Everest is in Nepal' but is inconsistent with 'Mount Everest is 15,000 metres high'. The Ten Commandments are consistent with each other but not with 'Do whatever you want to do regardless of others'. Someone is consistent provided the things they say could be true or correct together; they are inconsistent if this is not the case. Sometimes inconsistency is fairly obvious, as in the Mount Everest example above, but sometimes it is more deeply buried in what people say and believe. Here is a less obvious example of inconsistency: 'People strive only to obtain praise. When certain behaviour brings approval, a baby will try to continue in that way. But what if your baby's behaviour disappoints you? The less

attention you give this behaviour the better; children love drama and may repeat their misbehaviour just to witness your explosion.'

Contradiction: Strictly speaking, someone contradicts themselves if they say or believe both A and not-A at the same time (both A and its 'opposite'): for example, 'Mount Everest is 29,000 feet high' and 'Mount Everest is *not* 29,000 feet high'. Sometimes contradictions are very evident in what a person says or believes (as in the Everest example) but often they are deeply buried. If someone contradicts themselves they are clearly **inconsistent** (since their claims cannot both be true) but the inconsistent claims could both be false, so inconsistency does not necessarily imply a contradiction. However, these two terms are commonly used to mean much the same (see the *Oxford English Dictionary*).

Converse: The term 'converse' is usually used to refer to the 'opposite' (in a certain sense) of a **hypothetical**. The converse of any hypothetical 'If A then B' is simply the hypothetical 'If B then A'. Thus the converse of 'If fire is burning then oxygen is present' is 'If oxygen is present then fire is burning'. Notice that in this example although the initial hypothetical is true, its converse is not. Sometimes the converse of a hypothetical is true and sometimes not.

Counter-example: General claims may be challenged by finding counter-examples. Suppose someone claims that 'all politicians are dishonest'; if we can find one or more politicians who are honest then these are counter-examples to the general claim and it is shown to be mistaken. Whether counter-examples greatly weaken a general claim depends on the case; in this case, the proponent of the original claim might just say, 'OK, *nearly* all politicians are dishonest,' and this may serve his purpose just as well. To challenge this further claim would require quite a few more counter-examples. In another example, consider the general principle that 'it is always wrong to break a promise'; if you can then describe a situation in which it seems right to break a promise, this will be a counter-example to the general principle. For example, if breaking your promise will save the lives of many innocent people in a war situation, this might well be the right thing to do and this would be a counter-example to the general principle.

Disposition: If you say that someone has a 'sunny disposition' you mean that they are inclined to be cheerful; confronted with situations which might worry or displease other people, such a person is more likely to react cheerfully. So dispositions are behaviour patterns; they

are a propensity to behave in a particular way; a kind of habit. In our context this idea matters because we do not want students just to be good at doing critical thinking tasks when they are set, but we hope that they will become *disposed* to use the techniques explained and practised here.

Entails: In contexts where a case is being argued or discussed, this term is often used to mean something like 'is a logically inescapable consequence'. For example, a company's finance director might say of his company's finances, 'The planned investment programme will entail heavy borrowing,' meaning, 'We don't have sufficient cash to finance this investment ourselves and the only way to finance it will be by heavy borrowing.' In everyday contexts, 'entails' is often used more loosely – so that there is a substantial overlap between the common usage of 'entails' and 'implies' (see **Imply/Implication**).

Fallacy: The central usage of this term is to refer to a pattern of reasoning which is mistaken and which people commonly use. For example, it is surprisingly common for people to argue 'B came after A so A caused B' (as in 'I got my cold after Mary had hers, so I must have caught it from her'). Another such pattern is what is called the 'ad hominem fallacy', where you argue against what someone says not by criticising their claims but by making a personal attack (as in 'We should not listen to the complaints of prisoners because they are convicted criminals'). It can be quite difficult to recognise when a piece of reasoning is genuinely a fallacy. The term 'fallacy' is also more loosely used to refer to any error in reasoning.

Hypothesis: This is a claim which is put forward 'for consideration' or to be investigated, rather than being presented as true. For example, scientists investigating the death of the dinosaurs put forward the 'hypothesis' that an enormous meteor crashed into the Earth at about the time the dinosaurs became extinct. Subsequent investigation has confirmed this hypothesis to a large extent – there is strong evidence that such a meteor landed in the Gulf of Mexico – so the hypothesis has moved from being a hypothesis to being widely believed to be true (though whether it caused the extinction of the dinosaurs is more open to question).

Hypothetical: This is a sentence which has the general form 'if A then B' where A and B are sentences. For example, 'If you work hard at this course then you will get a good grade' is a hypothetical; so is the

sentence 'If people can choose the sex of their child, it is likely that there will eventually be more males than females in the population.' There are many equivalent ways of expressing hypotheticals in English, for example 'suppose we continue to damage the ozone layer; in that case there will be far more cases of skin cancer' or 'unless we stop damaging the ozone layer, there will be far more cases of skin cancer'; there are many others too (see section 3.6).

Imply/Implication: There is an everyday sense of these words which means roughly 'suggests' or 'leads me to believe', as when the detective says, 'The evidence implies that Smith was present at the murder'. However, the word 'imply' is often used more strictly to mean 'if . . . then . . .' For example, if someone says, 'The presence of fire implies that there must be oxygen present', they could equally well say, 'If there is fire then oxygen must be present'. In general, to say that 'A implies B' is to say something like 'if A then B', either strictly as in the oxygen example or more loosely as in the detective example. To say that 'A entails B' is usually to say 'A implies B' in this strict usage of 'implies' (see **Entails**).

Inference: In everyday usage, when people speak of an inference they sometimes mean a 'guess' based on the information they have, or the conclusion they have reached – again based on other things they believe. For example, after reviewing the evidence a detective might say, 'The inference is that we should question Jones again about this case.' In our context, the usage is derived from logic, where it means the 'step' from reasons to conclusion, the 'move' from one to the other. Thus, if someone argues, 'Some people have solved their own unemployment problem by great ingenuity in searching for a job or by willingness to work for less so all the unemployed could do this', the inference is the move from 'some people have . . .' to 'all the unemployed could do this'. Though the reason presented in this argument has often been true, the inference to the argument's conclusion is much more questionable. Other instructive examples are given in several chapters, especially chapters 8 and 9.

Knowledge: In its everyday usage this term is used in several ways. One usage refers to knowing some fact (as in 'John knows that chemotherapy will make his hair fall out' or 'John knows that the Earth is 93 million miles from the Sun'); another refers to knowing *how* to do something (as in 'John knows *how* to ride a bicycle'); another refers to *being acquainted* with an object, place or person (as in 'John knows New York').

If someone claims to know the fact, for example, that 'the Earth is 93 million miles from the Sun', then this is something they take to be true. If they are not sure about its truth they will tend not to say they 'know' it, but will probably say they 'suspect it is true' or it is their 'opinion' or some such alternative to 'know'.

Of course, people *claim* to know things which are in fact *false* and in that case the rest of us would say they do not really *know* it. For a person to *know*, for example, that Aids is caused by a virus, they would have to accept this, they would have to have good reason to accept it and it would have to be *true*.

Metacognition: This simply means 'thinking about your thinking'. For example, if you reflect on the thinking you went through in arriving at some decision, you are engaged in metacognition – in thinking about your own thinking. Sometimes you just work out the solution to some problem without thinking about how you are doing it or how you should do it; then you are just thinking, without engaging in metacognition. If you are not very skilled at, say, decision-making, many people believe that the way to improve it is through thinking about how you usually do it and then trying to correct the weaknesses by thinking self-consciously about how you make the next decision.

Necessary and sufficient conditions: To say that 'A is a necessary condition for B' is to say that 'if A is not the case, B will not be the case either'. Thus, if a teacher says, 'A good musical ear is a necessary condition for learning to play the violin', this is the same as saying, 'If you do not have a good musical ear, you will not be able to learn to play the violin'.

To say that 'A is a sufficient condition for B' is to say that 'if A is the case, then B will be the case also'. Thus, if a lawyer says, 'Being born in the UK of British parents is a sufficient condition for obtaining a British passport', this is the same as saying, 'If you are born in Britain of British parents you can have a British passport'. Clearly, although a good musical ear is necessary for learning to play the violin, it is not sufficient (you will need lots of practice too). Equally clearly, although being born in the UK of British parents is sufficient to get a British passport, it is not necessary, since other people can qualify for British passports too.

Reasons: Arguing a case consists in giving reasons for a conclusion. The reasons are presented as supporting the conclusion and are intended to persuade an audience to accept the conclusion. Thus, in the first

example in section 3.5, 'Most prospective parents would prefer to have sons' is given as a reason for believing that 'if people can choose the sex of their child, it is likely that there will eventually be more males than females in the population' – which in turn is one of the reasons for the main conclusion. If someone offers reasons for a conclusion they present themselves as both believing the reasons and believing that the reasons support the conclusion.

Sufficient conditions: (see **Necessary and sufficient conditions**).

Supposition (and **Suppositional reasoning**)**:** A supposition is a sentence which begins with the word 'suppose' or some synonym. For example, someone who is thinking about current experiments on genetically modified crops might say, 'Suppose these experiments do risk dangerous contamination of other crops . . .' Such a sentence does not commit the speaker to the view that these experiments do carry a dangerous risk; he or she is simply speculating about what would be the case *if* this were so. Reasoning from such starting points is often called 'what if' reasoning, or hypothetical reasoning or suppositional reasoning. This kind of reasoning is very common in theoretical contexts. The police have to use it a good deal too, for example asking, 'What if Smith really was in Amsterdam at the time of the murder . . .?' For a famous example see the Questions appendix, passage 51.

Truth: This term is used in an ordinary, everyday sense. Physical scientists aim to establish truths about the world, like Newton's laws of motion. Biologists try to establish truths about living things, like which genes determine which human characteristics. Psychologists aim to establish truths about the human mind, for example whether it works in some important respects like a computer. When these questions are answered with sufficient evidence then we tend to speak of the truth of the matter. Before we are sure, we tend to speak of hypotheses, opinions, beliefs, conjectures, guesses and so on. Truths are often called facts. It is disputed whether the term 'true' should be applied to moral, political, religious and other beliefs.

Valid/Invalid: 'Valid' is often used as a very general term of approval, as in 'the headteacher's policy on truancy is perfectly valid' but it also has a very specific meaning in the context of argument appraisal where it is short for 'deductively valid'. Consider the argument 'If Smith's fingerprints are on the gun then he is the murderer. His fingerprints

are clearly detectable on the gun. So he must be the murderer.' In this argument, *if* the two reasons are true, then the conclusion *must* be true; it is *impossible* for the reasons to be true and the conclusion false. Such an argument is called 'deductively valid' or 'valid' for short. Validity in this sense is a matter of the *relationship* between the reasons and their conclusion. In this usage, reasons and conclusion may be true or false (but are not said to be valid) and arguments as a whole are valid or not (but not true or false). An argument is said to be **invalid** if it is not valid (cf. chapter 8).

Bibliography and further reading

Dewey, J. (1998) *How We Think*. Dover Publications. (The beginnings of the modern tradition of critical thinking; first published 1909.)

Ennis, R.H. (1996) *Critical Thinking*. Pearson Education. (A good critical thinking text.)

Everitt, N. and Fisher, A. (1995) *Modern Epistemology: A New Introduction*. McGraw-Hill.

Facione, P. (2010) 'Critical Thinking: What It Is and Why It Counts'. California Academic Press. (Easy to find on the internet.)

Fisher, A. (2004) *The Logic of Real Arguments* (2nd edn). Cambridge University Press. (A way of handling complex reasoning.)

Fisher, A. and Scriven, M. (1997) *Critical Thinking: Its Definition and Assessment*. Edgepress and Centre for Research in Critical Thinking, University of East Anglia.

Fisher, A. and Thomson, A. (1993) *Test of Logical Reasoning*. Centre for Research in Critical Thinking, University of East Anglia.

Gardner, Martin (2000) *Fads and Fallacies in the Name of Science* (2nd edn). Dover Publications. (First published 1957. A classic and entertaining critique of pseudoscience and irrationality.)

Glaser, E. (1941) *An Experiment in the Development of Critical Thinking*. Advanced School of Education at Teacher's College, Columbia University.

Goldacre, Ben (2008) *Bad Science*. Fourth Estate. (Well written and clear headed against widely believed nonsense.)

MacCormick, N. (1978) *Legal Reasoning and Legal Theory*. Oxford University Press. (New edition Clarendon Press 1994.)

McPeck, J.E. (1981) *Critical Thinking and Education*. Martin Robertson. (The classic text which argues that critical thinking cannot be taught; now published by Palgrave Macmillan.)

Morton, A. (1988) 'Making arguments explicit: the theoretical interest of practical difficulties', in A. Fisher (ed.), *Critical Thinking: Proceedings of the First British Conference on Informal Logic and Critical Thinking*. University of East Anglia.

Norris, S. and Ennis, R. (1989) *Evaluating Critical Thinking*. Lawrence Erlbaum.

Passmore, J. (2009) 'On teaching to be critical', in R.S. Peters (ed.), *The Concept of Education*, pp. 192–211. Routledge. (Originally published 1967.)

Paul, R., Fisher, A. and Nosich, G. (1993) *Workshop on Critical Thinking Strategies*. Foundation for Critical Thinking, Sonoma State University, California.

Phipson, S.L. and Elliott, D.W. (1980) *Manual of the Law of Evidence* (11th edn). Sweet and Maxwell.

Schick, T. and Vaughn, L. (2010) *How to Think About Weird Things: Critical Thinking for a New Age* (6th edn). McGraw-Hill. (A very engaging critique of pseudoscience and the paranormal, based on sound epistemology, clearly explained.)

Scriven, M. (1976) *Reasoning*. McGraw-Hill. (A classic text on how to improve reasoning skills.)

Sutherland, S. (1992) *Irrationality: The Enemy Within*. Penguin Books. (2nd edn Pinter and Martin Ltd 2007.) (Full of surprises – and mistakes we are all inclined to make!)

Swartz, R.J., Costa, A. et al. (2010) *Thinking-Based Learning*. Teachers College Press, Columbia University.

Index

academic internet searches, 206–10
acceptability of claims, 83–5, 88–91
  thinking map for judging, 92
  see also credibility of sources, judging
acupuncture and IVF argument, 141–4
advertising, 78–9, 80, 87, 188
'affirming the antecedent', 123
alternative decisions, 170–3
alternative hypotheses, 151–2
ambiguity, 78–9
analogy
  definition of term, 276
  from basketball, 5–7
analysis of arguments, 59, 60
argument indicators, 24–5, 277
arguments, 276–7
  and context, 54–9
  evaluation of, 141–7
  expressing clearly, 31–2
  standards for evaluating, 118–20
  versus explanations, 45–7, 49
assumptions, 50–4
  assumption, definition of, 277
  hypothetical, 137–8
  implicit, 133–7
audience, providing clarification to, 67–78
authoritative definition/explanation, 69–70

balance of evidence/probabilities, 120, 126–9
belief, 277–8
  and claims, 90–1
  see also credibility
bias of sources, 103–5
brainstorming, 140, 170

causal claims, 86, 254
causal explanations
  example of, 151–2
  language of, 159–62
  pattern of reasoning in, 149–51
  and scientific evidence, 162–5
  thinking map, 153–4
  versus arguments, 45–7, 49
  weaknesses of, 152–3
certainty of claims, level of, 88–9
chain of reasoning, 37
cholesterol levels, 158–9
circumstantial evidence, 77–8, 107–8
civil courts, standard of proof required in, 126–9

claims
  acceptability of, 88–91, 92
  based on instrument readings, 102
  context of, 55–6, 105–6
  justifying using credibility considerations, 108
  nature of, 94–5, 109–10
  thinking map, 92
  types of, 86–8
clarification, 64–5
  and ambiguous expressions, 78–9
  problem identification, 66–7
  sources of, 68–72
  and target audience, 67–8, 77–8
  of terms and ideas, 72–7
competencies of critical thinking, 7–9
conclusion indicators, 23–4
conclusions, 278
  conclusion indicators, 23–4
  drawing more than one, 47–8
  examples of, 19–23
  identifying, 16–18, 61
  and inferences, 114–22
  jumping to, 75–6, 152
consequence, 46
consequences of decisions, 172–7
consistency, 278–9
context, 54–8
  and acceptability of claims, 83–6, 89
contradiction, 279
contrasts, drawing, 75–6
converse, 279
corroboration, 94, 95, 98, 99
  from independent sources, 110–11, 196
counter-example, definition, 279
courts
  direct vs. circumstantial evidence, 77–8, 107–8
  standards of proof required by, 124–9
creative thinking, 14, 138–40, 170
credibility
  and context, 55–6
  versus truth, 95–6
credibility of sources, judging, 92–5, 98–9
  circumstances/context, 105–6
  corroborative evidence, 110
  justification offered by source, 106–9
  nature of claim, 109–10
  person/source, 99–105
  thinking map, 111
criminal courts, standard of proof required in, 124–6
critical thinking
  classic definitions of, 2–5
  disposition to apply, 12–13
  Scriven's definition, 11–12
  skills underlying, 7–8
critico-creative thinking, 14, 138–40

decision making, 4, 167–8
  common flaws in, 168–9
  model for good, 169–77
  procedures, 180–2
  thinking map for, 177–9
deductive validity, 119, 120–2
  and patterns of argument, 122–4
definitions, 86–8
  of critical thinking, 2–5, 11–12
  dictionary, 68–9

of technical terms, 69–70
using internet to find, 196–7
'denying the consequent', 123
Dewey, John, 2–3
dictionary definitions, 68–9
dinosaurs, extinction of, 156–7
direct evidence, 77–8, 107–8
disjunctions, 42–3
dispositions, 12–13, 279–80

Encyclopædia Britannica, 199, 200–1
Ennis, Robert, 4
'entails', meaning of, 280
ethics, 175
evaluation
of arguments, 59, 60, 141–7
critical thinking as, 12
evidence
in a court of law, 69, 71, 77–8
balance of, 120, 126–9
direct vs. circumstantial, 77–8, 107–8
first-hand vs. reported, 106–7
scientific explanations, 162–5
and witness credibility, 101–2
examples, providing good, 75
expertise
of a source, and credibility, 99–100, 186–7
and acceptability of claims, 89–90
explanations
alternative plausible, 135
versus arguments, 45–7, 49
see also causal explanations
expressions
clarification of, 72–7
understanding, 66–7

factual claims, 86–8
fallacy of equivocation, 78–9
fallacy, definition of, 280
first-hand vs. reported evidence, 106–7

Gaia hypothesis, 137, 227
Gall, Francis Joseph, 17
Glaser, Edward, 3–4, 8
God's existence, 131–2, 146, 231–3
Google ranking algorithm, 187–8
Google Scholar, 207–10
Gosse, Philip Henry, 17

hearsay evidence, 77, 78, 107
historical context
and argument evaluation, 56
of critical thinking, 2–5
of ideas, explaining, 76–7
Holmes, Sherlock, 114, 126
Hume, David, 109
hunches versus certainty, 88–9
hypothesis
alternative, 151–2
definition of, 280
hypothetical assumptions, 137–8
hypotheticals, 42–4, 280–1

ideas, clarification of, 72–7
'if and only if' definitions, 73–4
imagination, 138–40
implicit assumptions, 51–2, 53–4, 133–7
implies, definition of, 281
inconsistency, 278–9
inferences, 115–16
deductive validity, 120–4
definition of, 281

initial test for good, 115–18
standards for evaluating, 118–20
information sources *see* sources
instrument readings, accuracy of, 102
internet *see* search engines; websites
interpretation *see* clarification
interpretation skills, 12, 15
invalid, definition of, 283–4
IVF and acupuncture argument, 141–4

jargon, obtaining explanation of, 69–70
joint reasoning, 38–40
joyriders, argument evaluation, 144–6

key words, search engines, 191–3
Kimmel Admiral, 181–2
knowledge
definition of, 281–2
expertise, 89–90, 94, 99–100
first-hand vs. reported, 106–7
philosophers' definition of, 73, 74

language of reasoning, 23–32
lottery, likelihood of winning, 173–4

Malthus, Thomas, 56
meaning(s), 64–7
context aiding, 56
and fallacy of equivocation, 78–9
stipulation of, 70–1
of terms, ways of clarifying, 72–7
of words, sources of definitions, 68–9, 196–7
metacognition, 12, 282
basketball analogy, 5–7
Mill, John Stuart, 78
modus ponens, 123
modus tollens, 123
Monty Python argument clinic sketch, 18
morality, 175
'more likely than not on balance of evidence', 120, 126–9

Napoleon's death, causal explanation, 151–2
necessary and sufficient conditions, 73–4, 282
neutral point of view (NPOV), 203–4
nocebo effect, 235

observation
and accuracy of our senses, 100–2
in Scriven's definition of critical thinking, 11, 12
original evidence, 78
original research, 203–4

paradigm examples, 75
Pascal's Wager, 231
Paul, Richard, 4–5
peer-reviews as sources, 206–7
per genus et differentiam, 75–6
placebo effect, 142–4, 234–5
probabilities, 173–4
'proof beyond reasonable doubt', 119–20, 124–6

questionable sources, 202–3

reason indicators, 24
reasoning, 35
  arguments vs. explanations, 45–7
  assumptions, 50–4
  chain structure of, 37
  complex patterns of, 40–2
  and context, 54–8
  drawing more than one conclusion, 47–8
  examples of, 19–23
  hypotheticals, 42–4
  identifying, 16–18
  language of, 23–31
  patterns of, 149–51
  structure of, 32–3
  the 'therefore' test, 26–8
  thinking maps, 59–62
reasons
  acceptability of, 83–96
  definition of term, 282–3
  joint reasons, 38–40
  side-by-side reasons, 35–7
recommendations, 254
  thinking map, 177
  see also decision making
reflective thinking, Dewey, 2–3
reliability
  of proof, 124–9
  and reputation, 102–3
  of websites, 193–6
  of Wikipedia, 204–6
reported evidence, 106–7
reputable sources, 102–3, 186–7, 199

scholarly internet searches, 206–10
Scriven, Michael, 11–12
search engines, 187
  Google ranking, 187–8
  key words, 191–3
  search results, reading, 188–9
  special searches, 196–8
side-by-side reasoning, 35–7
Singh, Simon, 141–4
skills underlying critical thinking, 7–8
sources
  Britannica and Wikipedia, 199–205
  of clarification, 68–72
  credibility of, 91–5, 98–111
  peer-reviewed, 206–7
  primary versus secondary, 107
  questionable, 202–3
  reliability of internet, 185–7, 193–6
standards of proof
  in civil courts, 126–9
  in criminal courts, 124–6
structure of reasoning, 32–3
sufficient and necessary conditions, 73–4, 282
supposition, definition of, 283
synonymous expressions, 72–3

technical words, checking meaning of, 69–70
terminology of reasoning, 23–33
terms, clarification of, 72–7
'therefore'
  conclusion indicator word, 23–4
  the 'therefore' test, 26–8
thinking maps, 59–62
  analysis and evaluation of arguments, 60–2
  causal explanations, 153–4

clarifying ideas, 80–1
decisions/recommendations, 177–8
judging acceptability of claims, 92
judging credibility of sources, 111
judging inferences, 147
thinking skills, teaching, 1
truth
definition of term, 283
versus credibility, 95–6

unreflective thinking, 2–3, 15, 76
URLs (uniform resource locators), 190–1, 194–5

vagueness of language, 66–7
valid
deductive validity, 120–4, 130–1
definition of, 283–4
value claims/judgements, 86–8
values of the critical thinker, 12–13
verifiability of Wikipedia, 201–3
vested interest, 103–5, 194

websites
addresses (URLs), 190–1, 194–5
reliability of, 193–6
Wikipedia
neutrality of, 203–4
reliability of, 199–201, 204–5
verifiability of, 201–3
Witch Ground survey, 157–8
witnesses *see* evidence
Wittgenstein, Ludwig, 74

Acknowledgements

The authors and publishers acknowledge the following sources of copyright material and are grateful for the permissions granted. While every effort has been made, it has not always been possible to identify the sources of all the material used, or to trace all copyright holders. If any omissions are brought to our notice, we will be happy to include the appropriate acknowledgements on reprinting.

p. 77–78 extract from *Manual of the Law of Evidence*, Phipson & Elliott, 1980, Thomson Reuters (Legal) Ltd trading as Sweet & Maxwell;

p. 112 adapted from 'Nun tells of her 'miracle' cure by John Paul II' by Henry Samuel for The Telegraph, March 2007, © Telegraph Media Group 2007;

pp. 141, 228 extracts from 'Acupuncture: the lie of the needle' by Simon Singh for The Telegraph, April 2008, © Telegraph Media Group 2008;

p. 157 Mark Henderson/The Times/NI Syndication;

pp. 158, 181 from *Irrationality* by Stuart Sutherland by permission of Constable & Robinson Ltd and the Estate of Stuart Sutherland;

p. 162 from 'When North flies South' by Lou Bergeron in the New Scientist, March 1996;

pp. 200, 202–203 material from Wikipedia available under the Creative Commons Attribution-ShareAlike License;

pp. 213–224 various excerpts from OCR May 1999 and May 2000 Critical Thinking examination papers, by permission of OCR;

p. 227 from 'The whole world in our hands' by Tim Radford for The Guardian, September 2000 © Guardian News and Media Ltd, 2000;

p. 228 from 'Let's keep the Mona Lisa smiling' in the New Scientist, May 1996;

p. 231 edited version of Richard Dawkins' speech 'The more you understand evolution, the more you move towards atheism' given at the Edinburgh International Science Festival, 1992, used with permission of Professor Dawkins;

p. 234 'Mine's a placebo and tonic' by Lewis Wolpert from The Independent, January 2001 © Independent News & Media Ltd;

The painting shown on the cover is a detail from *The Disquieting Muses* by Giorgio de Chirico, 1918 © DACS 2011.

MARYMOUNT MANHATTAN COLLEGE
3 3203 00107 530 9

Critical Thinking

An Introduction Second Edition

Alec Fisher

This engaging coursebook aims to teach critical thinking skills – the ability to interpret, analyse and evaluate ideas and arguments. Alec Fisher's approach is based on a widely shared conception of critical thinking and covers many of the basic skills and competencies displayed by good critical thinkers.

This second edition has been extensively revised with updated examples and a brand new chapter on how to obtain reliable information from the internet. It is suitable for a wide range of students in schools, colleges and universities and may also be used for self-study.

The coursebook contains:

- a wide variety of materials on topical issues to stimulate critical thinking and discussion
- over 220 questions with answers to three-quarters of them providing students with feedback and opportunities for self-evaluation
- 'thinking maps' that enable students to monitor and improve their critical thinking skills
- a comprehensive glossary.

Dr Alec Fisher is an experienced author and a distinguished researcher and lecturer in critical thinking. He designed the AS level examination in Critical Thinking for the OCR Examinations Board and was its chief examiner for some years.



Other titles available:
The Logic of Real Arguments ISBN 978-0-521-65481-4
Thinking Skills ISBN 978-0-521-52149-9

CAMBRIDGE UNIVERSITY PRESS
www.cambridge.org
ISBN 978-1-10
9 781107 4